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Units, Dimensions and Measurement

SOLUTIONS

Beginner Box - 1

1. Dimension of gravitational constant

$= [M^{-1}L^3T^{-2}]$

$$n_2 = n_1\left[\frac{M_1}{M_2}\right]^{-1}\left[\frac{L_1}{L_2}\right]^{3}\left[\frac{T_1}{T_2}\right]^{-2}$$

here,

MKS	CGS
$n_1 = 6.67 \times 10^{-11}$	$n_2 = ?$
$M_1 = kg$	$M_2 = g$
$L_1 = m$	$L_2 = cm$
$T_1 = sec$	$T_2 = sec$

$$n_2 = 6.67 \times 10^{-11}\left[\frac{kg}{g}\right]^{-1}\left[\frac{m}{cm}\right]^{3}\left[\frac{sec}{sec}\right]^{-2}$$

$= 6.67 \times 10^{-11} \times 10^{-3} \times 10^{6} \times 1$

$= 6.67 \times 10^{-8}$

Beginner Box - 2

2. (a) [Temperature] = [K]

(b) $KE = \frac{1}{2}mv^2$

$\Rightarrow [KE] = [m]\,[v]^2 = M(LT^{-1})^2 = [ML^2T^{-2}]$

(c) $P = \frac{F}{A} \Rightarrow [P] = \frac{[F]}{[A]} = \frac{MLT^{-2}}{L^2} = [ML^{-1}T^{-2}]$

(d) $\omega = \frac{v}{r} \Rightarrow [\omega] = \frac{[v]}{[r]} = \frac{LT^{-1}}{L} = [T^{-1}]$

3. $E = hf \Rightarrow h = \frac{E}{f} \Rightarrow [h] = \frac{[E]}{[f]} = \frac{ML^2T^{-2}}{T^{-1}} = [ML^2T^{-1}]$

4. $F \propto m^a v^b r^c \Rightarrow F = km^a v^b r^c$

$\Rightarrow [M^1L^1T^{-2}] = [M]^a[L^1T^{-1}]^b[L]^c = [M]^a[L]^{b+c}[T]^{-b}$

by comparison a = 1, b = 2, c = –1

So $F = k\frac{mv^2}{r}$

Beginner Box - 4

2. Radius of circle $= \frac{1.06}{2} = 0.53$ m

Area of circle $= \pi r^2$

$= 3.14 \times (0.53)^2 = 0.882\ m^2$ (3 SF)

3. $3.9 \times 10^5 - 2.5 \times 10^4 = 3.9 \times 10^5 - 0.25 \times 10^5$

$= 3.65 \times 10^5 = 3.6 \times 10^5$ (2 SF)

4. (a) Total mass of the box

= 2.3 kg + 20.15 g + 20.17 g = 2.34017 kg

= 2.3 kg

(b) Difference in masses of gold pieces

= 20.17 g – 20.15 g = 0.02 g

Beginner Box - 6

1. $L = L_1 + L_2 = (1.8 \pm 0.2) + (2.3 \pm 0.1)$

$= 4.1 \pm 0.3$ m

2. Elongation $= L_2 - L_1$

$= (155.3 \pm 0.2) - (153.7 \pm 0.6)$

$= (1.6 \pm 0.8)$ cm

3. Let y = AB

$$\frac{\Delta y}{y} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$y = 2.5 \times 0.1 = 0.25$

$$\frac{\Delta y}{0.25} = \frac{0.5}{2.5} + \frac{0.01}{0.10}$$

$\Rightarrow \Delta y = 0.075 = 0.08$ (2SF)

4. Surface area of sphere $= 4\pi r^2$

$= 4 \times 3.14 \times (2.1)^2 = 55.3896 = 55.39$

$$\Delta A = 2\frac{\Delta r}{r} \times A = 2 \times \frac{0.5}{2.1} \times 55.39$$

$= 2.6376 = 2.64$

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5. $x = \dfrac{a^3b^2}{\sqrt{cd}}$

$$\frac{\Delta x}{x} = 3\frac{\Delta a}{a} + 2\frac{\Delta b}{b} + \frac{1}{2}\frac{\Delta c}{c} + \frac{1}{2}\frac{\Delta d}{d}$$

$$\frac{\Delta x}{x} \times 100 = 3 \times 1 + 2 \times 3 + 4 \times 1/2 + 2 \times 1/2$$

$= 12\%$

6. $V = \dfrac{16.0}{4.0} = 4.0$ m/s

$\because \quad \dfrac{\Delta V}{V} = \dfrac{\Delta s}{s} + \dfrac{\Delta t}{t}$

$\because \quad \Delta V = (4.0)\left[\dfrac{0.4}{16.0} + \dfrac{0.2}{4.0}\right] = 4.0[0.2 + 0.5]$

$= 0.28$ after (rounding off) 0.3

$\Rightarrow$ Speed (4.0 ± 0.3) m/s

Beginner Box - 7

1. L.C. $= \left(\dfrac{b-a}{b}\right)$ M.S.D. $= \left(\dfrac{20-19}{20}\right) \times \dfrac{1}{10}$ cm

$= 0.005$ cm

2. Pitch $= \dfrac{4}{8}$ mm $= 0.5$ mm

L.C. $= \dfrac{0.5}{100}$ mm $= 0.005$ mm

3. L.C. $= \dfrac{0.0625}{250}$ cm $= 2.5 \times 10^{-4}$ cm

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Wave Motion

SOLUTIONS

BEGINNER'S BOX-1

1. Wave velocity $v = n\lambda$

$\Rightarrow \lambda = \frac{v}{n} = \frac{300}{500} = 0.6m$

Relation between phase difference & path difference $\frac{\Delta\phi}{2\pi} = \frac{\Delta\lambda}{\lambda} \Rightarrow \Delta\lambda = \frac{\pi/3}{2\pi} \times 0.60 =$ 0.10m = 10 cm.

2. Time taken by the stone to reach the water of pond $t_1 = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times300}{9.8}} = 7.825$ s

Also time taken by splash of sound $t_2 = \frac{\text{distance}}{\text{velocity}} = \frac{300}{340} = 0.882$ s

$\therefore$ Total time $t = t_1 + t_2 = 7.825 + 0.882 = 8.707$ s

BEGINNER'S BOX-2

1. Maximum speed of particles = $A\omega = 40 \times 5 = 200$ m/s

2. Compare it with $y = a \sin(\omega t - kx)$

$a = \frac{10}{\pi}$ cm, $\omega = 200\,\pi$ rad/s

so $T = \frac{2\pi}{\omega} = \frac{1}{100}$ s

Maximum velocity of particle

$v_{max.} = a\omega = \frac{10}{\pi} \times 10^{-2} \times 200\,\pi = 20$ m/s

3. No converse is not true, for representing a travelling wave, function must be finite at everywhere and at all times. Here no one function satisfies this requirement.

4. Speed of transverse wave in string $v = \sqrt{\frac{T}{m}}$

where $T = 2g = 20N$ & $m = \frac{4.5\times10^{-3}}{2.25}$

$= 2 \times 10^{-3}$ kg/m $\Rightarrow v = \sqrt{\frac{20}{2\times10^{-3}}} = 100$ ms^{-1}

Therefore required time $= \frac{\ell}{v} = \frac{2}{100} = 0.02$ s

5. Speed of transverse wave on a wire,

$v = \sqrt{\frac{T}{m}}$

$\therefore \quad 300 = \sqrt{\frac{T}{m}}$ & $360 = \sqrt{\frac{T'}{m}}$

$\Rightarrow \quad \frac{300}{360} = \frac{300}{360} \Rightarrow \frac{T}{T'} = \left(\frac{5}{6}\right)^2$

$\Rightarrow \quad T' = \frac{36}{25}\,T = \left(1+\frac{11}{25}\right)T$

$\Rightarrow \quad \Delta T = \frac{11}{25} \times T = \frac{11}{25} \times 500 = 220$ N

6. Velocity of transverse wave

$v = \sqrt{\frac{T}{m}} = \sqrt{\frac{200\times20}{2.5}} = 40$ ms^{-1}

Time (t) $= \frac{\text{distance}}{\text{speed}} = \frac{20}{40} = 0.5$ s

7. rope of mass = 0.1 kg

length = 2.45 m

(a) speed of transverse wave in the rope at a point 0.5m distance from the lower end

$v = \sqrt{\frac{T}{\mu}}$ and $T = m'g$

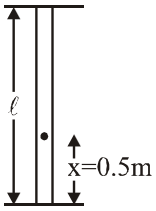
$m' = \frac{M}{\ell}x \quad \because \mu = \frac{M}{\ell}$

$T = \frac{M}{\ell}xg$

$v = \sqrt{\frac{\frac{M}{\ell}xg}{M/\ell}}$

$v = \sqrt{xg}$

$v = \sqrt{0.5\times9.8} = 2.21$ m/s



(b) time taken by a transverse wave to travel the full length of rope

$$\text{speed} = \frac{\text{distance}}{\text{time}} \quad \because dt = \frac{dx}{dv}$$

$$\int_0^t dt = \int_0^{\ell} \frac{dx}{\sqrt{xg}} \Rightarrow t = \frac{1}{\sqrt{g}} \int_0^{\ell} x^{-1/2} dx$$

$$t = \frac{1}{\sqrt{g}} \left[\frac{x^{-1/2+1}}{-\frac{1}{2}+1} \right]_0^{\ell} = \frac{1}{\sqrt{g}} \frac{\ell^{1/2}}{1/2}$$

$$t = 2\sqrt{\frac{\ell}{g}} \Rightarrow t = 2\sqrt{\frac{2.45}{9.8}}$$

$$t = 2\sqrt{0.25} \Rightarrow t = 2 \times 0.5 = 1 \text{ sec}$$

$$a = \frac{v\,dv}{dx} = \sqrt{xg}\,\frac{d}{dx}\left(\sqrt{xg}\right)$$

OR

$$= \sqrt{xg}\,\frac{\sqrt{g}}{2\sqrt{x}} = \frac{g}{2} \text{ (constant acceleration)}$$

so $s = \frac{1}{2}at^2 \Rightarrow \ell = \frac{1}{2}\left(\frac{g}{2}\right)t^2 \Rightarrow t = 2\sqrt{\frac{\ell}{g}}$

BEGINNER'S BOX-3

1. Velocity of longitudinal wave pulse

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{10^{11}}{10 \times 10^4}} = 10^3 \text{ ms}^{-1}$$

Required time $\frac{2\ell}{v} = \frac{2 \times 1000}{10^3} = 2$ s

2. Speed of sound $v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{4000 \times 10^6}{1000}}$

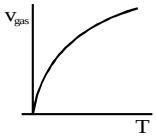
$= 2000 \text{ ms}^{-1}$

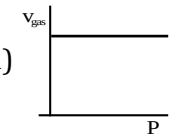
3. Intensity level in decibel $= 10 \log_{10}\left(\frac{I}{I_0}\right)$

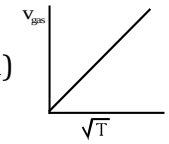
$= 10 \log_{10}\left(\frac{10^{-9}}{10^{-12}}\right) = 10 \log_{10} 10^3 = 30\text{dB}$

4. After rain fall, the humidity of air increases. This lowers the density of air hence there is increase in velocity of sound $\left(\because v \propto \frac{1}{\sqrt{\rho}}\right)$

5. $v_{gas} = \sqrt{\frac{\gamma RT}{M_W}}$

(i) 

(ii) 

(iii) 

6. Due to this loudness of sound increases.

7. Velocity of sound wave in air

$$v_{air} = \sqrt{\frac{\gamma RT}{M_w}} \propto \sqrt{T} \qquad \text{[for a given gas]}$$

so sound travels faster in warm air than in cool air.

8. Sound wave require material medium for their propagation. Since there is vacuum in the space in between the earth and the planets, so the sound produced in the explosion on other planets can not reach our ear.

9. Velocity of sound in gaseous mixture

$$v_s = \sqrt{\frac{\gamma_{mix.} RT}{M_{W_{mix.}}}}$$

$$M_{W_{mix}} = \frac{\mu_1 M_{w_1} + \mu_2 M_{w_2}}{\mu_1 + \mu_2} = \frac{1 \times 4 + 1 \times 32}{2}$$

$$= \frac{36}{2} = 18 \text{ g/mole}$$

$$\gamma_{mix.} = \frac{\mu_1 C_{P_1} + \mu_2 C_{P_2}}{\mu_1 C_{V_1} + \mu_2 C_{V_2}} = \frac{1 \times \frac{5}{2}R + 1 \times \frac{7}{2}R}{1 \times \frac{3}{2}R + 1 \times \frac{5}{2}R}$$

$$= \frac{12}{8} = \frac{3}{2} \text{ and } T = 127 + 273 = 400\text{K}$$

$$v_s = \sqrt{\frac{3 \times 8{\cdot}31 \times 400}{2 \times 18 \times 10^{-3}}} = \sqrt{27{\cdot}7} \times 100 \text{ m/s}$$

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10. Assume ideal gas law : $P = \frac{\rho RT}{M}$, where ρ is the density, M is the molecular mass, and T is the temperature of the gas. This gives $v = \sqrt{\frac{\gamma RT}{M}}$. This shows that v is :

(a) Independent of pressure.

(b) Increases as $\sqrt{T}$.

(c) The molecular mass of water (18) is less than that of N_2 (28) and O_2 (32.) Therefore as humidity increases, the effective molecular mass of air decreases and hence v increases.

BEGINNER'S BOX-4

1. 0°, 90°, 120°, 180°[By properties of vector]

2. given $\frac{a_{max}}{a_{min}} = \frac{9}{7} \Rightarrow \frac{a_1 + a_2}{a_1 - a_2} = \frac{9}{7}$

$\Rightarrow \frac{a_1}{a_2} = \frac{8}{1}$

So, $\frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2 = 64:1$

3. given $\frac{a_1}{a_2} = \frac{5}{4}$

then $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{5+4}{5-4}\right)^2 = \left(\frac{9}{1}\right)^2$

= 81 : 1

4. Silents points means minima

min – min = $\Delta\lambda = \lambda$

λ = 60 – 36

λ = 24 cm & v = 330 m/s

$n = \frac{v}{\lambda} = \frac{330}{24} \times 100 = 1375$ Hz

5. Sliding distance x = 16.6 cm

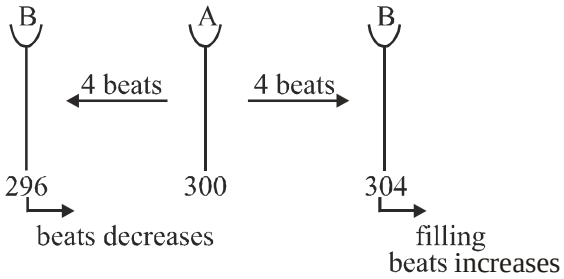
$\lambda = 2x$ n = ?

$\lambda = 2 \times 16.6$ cm

$n = \frac{v}{\lambda} = \frac{332 \times 100}{2 \times 16.6} = \frac{332 \times 1000}{332}$

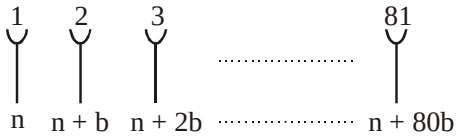
n = 1000 Hz

6. Let tuning fork A having frequency =300 Hz produce 4 beats/sec with unknown B then frequency of B is



So answer is 296 Hz

7.



according to question

n + 80b = 2n (octave = 2 times)

n = 80 × 4 = 320 Hz (b = 4 beats)

$n_{last} = n + 80b$

= 320 + 320

n_{last} = 640 hz

8. $y_1 = 3\sin 400\pi t$ $y_1 = 4\sin 402\pi t$

(a) Frequency of resultant wave

$n_{avg} = \frac{n_1 + n_2}{2}$

$\omega_1 = 2\pi n_1 = 400\pi$ $n_1 = 200$ Hz

$\omega_2 = 2\pi n_2 = 402\pi$ $n_2 = 201$ Hz

$n_{avg} = \frac{200+201}{2} = 200.5$ Hz

(b) Beat frequency $\Delta n = n_2 - n_1$

= 201 – 200

= 1 Hz

(c) Intensity ratio of wave $I \propto a^2$

$\frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16} = 9:16$

9. Frequency of standard tunning fork = n_0

So $n_A = n_0 + \frac{n_0 \times 20}{100} = 1.2\, n_0$

$n_B = n_0 + \frac{n_0 \times 30}{100} = 1.3\, n_0$

then $n_B - n_A = 6$ Hz

$1.3\, n_0 - 1.2\, n_0 = 6$

$n_0 = 60$

$n_A = 1.2 \times 60 = 72$ Hz

$n_B = 1.3 \times 60 = 78$ Hz

10. (a) Beat time period $T_b = \frac{1}{\text{beats}} = \frac{1}{4}$ s

(b) It is time taken to complete half beat

$= \frac{T_b}{2} = \frac{1}{2\times4}$ s $= \frac{1}{8}$ s

(c) Beat per three second = 3 × 4 = 12 beats

11. Path difference = (6.4 – 6.0) = 0.4 m

For destructive interference 0.4 m = $(2N + 1)\frac{\lambda}{2} \Rightarrow \lambda = \frac{0.8}{2N+1}$ where N = 0, 1, 2, 3, ...

The corresponding frequency

$n = \frac{v}{\lambda} = \frac{320\times(2N+1)}{0.8} = 400\ (2N+1)$

Therefore the frequency in the 500 Hz to 5000Hz for which listener would hear a minimum intensity = 1200 Hz, 2000 Hz, 2800 Hz, 3600 Hz, 4400Hz

12. Let frequency of first fork $n_1 = n$

According to question $n_{51} = n_1 + 50 \times b$

where b = 5, $n_1 = n$, $n_{51} = 6n$

$6n = n + 50 \times 5 \Rightarrow n = n_1 = 50$ Hz

$n_{11} = n + 10\times 5 = 50 + 50 = 100$ Hz

$n_{17} = n + 16\times 5 = 50 + 80 = 130$ Hz

$n_{27} = n + 26\times 5 = 50 + 130 = 180$ Hz

$n_{33} = n + 32 \times 5 = 50 + 160 = 210$ Hz

$n_{51} = 6n = 6 \times 50 = 300$ Hz

BEGINNER'S BOX-5

1. (i) $y = 0.06 \sin\left[8\pi\left(t + \frac{x}{20}\right) + \pi\right]$

(ii) at x = 0

$y = 0.06 \sin(8\pi t + \pi)$

$y = -0.06 \sin(8\pi t)$

2. (a) The function represents a stationary wave.

(b) The given equation is

$y(x, t) = 0.06 \sin\left(\frac{2\pi}{3}x\right) \cos(120\pi t)$ ...(1)

and a stationary wave is given by

$y = 2a \sin\frac{2\pi}{\lambda}x \cos\frac{2\pi}{\lambda}vt$ ...(2)

On comparing equation (1) and (2) we get

wave length λ = 3m and also $\frac{2\pi v}{\lambda} = 120\pi$

$\Rightarrow$ $v = 60\lambda = 60 \times 3 = 180\ ms^{-1}$ and

frequency $n = \frac{v}{\lambda} = \frac{180}{3} = 60$ Hz

(c) For tension in string $v = \sqrt{\frac{T}{m}}$

$\Rightarrow T = v^2m = (180)^2 \times \frac{3\times10^{-2}}{1.5} = 648$ N

3. (i) $y = 0.02 \times 0.6 \sin[2\pi(330t + x) + \pi]$

$y = -0.012 \sin 2\pi(330t + x)$

(ii) $y = 0.02 \times 0.75 \sin[2\pi(330t + x)]$

$y = 0.015 \sin[2\pi(330t + x)]$

4. If reflected from rigid end

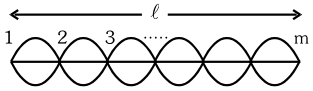
$y = 0.04 \sin[2\pi(200t + x) + \pi]$

$= -0.04 \sin[2\pi(200t + x)]$

If reflected from free end

$y = 0.04 \sin[2\pi(200t + x)]$

5.



Here Number of loops =(m–1) and one loop corresponding to wavelength $\frac{\lambda}{2}$

Therefore $\ell = (m-1)\frac{\lambda}{2}$

6. frequency for sonometer wire is $n = \frac{1}{2\ell}\sqrt{\frac{T}{m}}$

$\Rightarrow \frac{n_1}{n_2} = \frac{\ell_2}{\ell_1}\sqrt{\frac{T_1}{T_2}} = \frac{n_1}{n_2} = \frac{2}{1} \times \sqrt{\frac{9}{16}}$

$\Rightarrow n_2 = \frac{2}{3}\ n_1 = \frac{2}{3} \times 150 = 100$ Hz

7. $b = \frac{5}{3}$ beats/sec

for string $n \propto \sqrt{T}$

Let frequency of TF is N

$\frac{N+5/3}{N-5/3} = \sqrt{\frac{14.4}{10}}$

$\Rightarrow 12N - 20 = 10N + \frac{50}{3}$

$\Rightarrow 2N = \frac{110}{3} = \frac{55}{3}$

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8. $\frac{f_1}{f_2}=\sqrt{\frac{T_1}{T_2}}$

$\frac{2}{3}=\sqrt{\frac{T}{T+50}} \Rightarrow \frac{4}{9}=\frac{T}{T+50}$

$9T = 4T + 200 \Rightarrow T = 40$ N

$\therefore$ T = 4 kg–wt.

9. (i) $n \propto \frac{\sqrt{T}}{L}$

$\frac{n_2}{n_1}=\sqrt{\frac{T_2}{T_1}}\times\frac{L_1}{L_2}=\sqrt{\frac{2T}{T}}\times\frac{L}{2L}=\frac{1}{\sqrt{2}}$

$\therefore \quad n_2 = \frac{n_1}{\sqrt{2}}$

(ii) $n = \frac{1}{2L}\sqrt{\frac{T}{\mu}}=\frac{1}{2L}\sqrt{\frac{T}{M/L}}=\frac{1}{2\sqrt{M}}\sqrt{\frac{T}{L}}$

$n \propto \sqrt{\frac{T}{L}}$

$\frac{n'}{n}=\frac{\sqrt{2T/2L}}{\sqrt{T/L}}=1$

$\therefore \quad n' = n$

(iii) $v = \sqrt{\frac{T}{\mu}}$

μ constant $\therefore v \propto \sqrt{T}$

on doubling tension v will becomes $\sqrt{2}$ times

(iv) $v = \sqrt{\frac{T}{\mu}}=\sqrt{\frac{40}{4\times10^{-3}}}=100\ m/s$

(v) $v=\sqrt{\frac{T}{\mu}}=\sqrt{\frac{80}{\frac{4\times10^{-3}}{2}}}=\sqrt{40\times10^3}=200\ m/s$

(vi) $=\sqrt{\frac{T_2}{T_1}}\times\frac{L_1}{L_2}=\sqrt{\frac{4T}{T}}\times\frac{L}{4L}=\frac{1}{2}$

$n_2 = \frac{n_1}{2}$

$\therefore \quad n_1 = 2n_2$

$\therefore$ No of loop will becomes two.

BEGINNER'S BOX-6

1. According to question frequency ratio – 400 : 800
= 1 : 2 so open organ pipe because even and odd both harmonics are present.

2. For closed organ pipe $n_0 = \frac{v}{4\ell} = \frac{340}{4\times20\times10^{-2}}$

= 425 Hz $\Rightarrow$ first harmonic
If pipe is open, its fundamental frequency
$= \frac{v}{2\ell}$ = 850 Hz so it will not resonant with the given source.

3. Frequency of third overtone of closed organ pipe $n_C = \frac{7v}{4\ell_c}$

Frequency of sixth overtone of open organ pipe $n_0 = \frac{7v}{2\ell_0}$

According to question $\frac{7v}{4\ell_c}=\frac{7v}{2\ell_0} \Rightarrow \frac{\ell_c}{\ell_0}=\frac{1}{2}$

4. Third overtone of closed organ pipe means seventh harmonic.

$(n_7)_{closed} = (n_4)_{open} \Rightarrow 7\left(\frac{v}{4\ell_c}\right) = 4\left(\frac{v}{2\ell_0}\right)$

$\Rightarrow \frac{\ell_c}{\ell_0} = \frac{7}{8}$

5. In open organ pipe overtones are integral multiple of fundamental frequency, so if fundamental
frequency is n_0 then $n_0 = 2592 - 1944 = 648$ Hz

$n_0 = \frac{v}{2\ell} \Rightarrow \ell = \frac{v}{2n_0} = \frac{324}{2\times648} = 0.25m$

6. Frequency at 10th antinode $= \frac{9V}{2L} = 9 \times 300$
$= 2700$

7. $\frac{3V}{4L}+\frac{3V}{2L} = 180$

$\frac{9V}{4L} = 180 \Rightarrow \frac{V}{L} = 80$

$\therefore$ Fundamental frequency of COP
$= \frac{V}{4L} = \frac{80}{4} = 20$
Fundamental frequency of OOP
$= \frac{V}{2L} = \frac{80}{2} = 40$

8. For OOP $n \propto \frac{1}{\ell}$

Let frequency of TF is N

$\frac{N-4}{N+4}=\frac{30}{31}$

$\Rightarrow$ N = 244 Hz

9. $n_1 - n_2 = 2$

$\frac{V}{4\times0.15}-\frac{V}{2\times0.305}=2$

$V\left[\frac{100}{4\times15}-\frac{100}{2\times30.5}\right]=2$

$100V\left[\frac{1}{60}-\frac{1}{61}\right]=2 \Rightarrow 100V\frac{1}{60\times61}=2$

$\therefore V=\frac{60\times61\times2}{100}$

$\therefore n_1=\frac{V}{4\times0.15}=\frac{100}{4\times15}\times\frac{60\times61\times2}{100}=122$ Hz.

$n_2=\frac{V}{2\times0.305}=120$ Hz

10. (a) Fundamental frequency of closed pipe

$=\frac{v}{4\ell}=\frac{330}{4\times1}=82.5$ Hz

(b) At resonance fundamental frequency of stretched wire = fundamental frequency of air column

$\frac{1}{2\ell}\sqrt{\frac{T}{m}}=82.5$ Hz

$\Rightarrow T = m(2\times0.3\times82.5)^2$

$=\frac{0.01}{0.3}(2\times0.3\times82.5)^2$

$= 81.675$ N

11. V = 300 m/sec. n = 500 Hz, L = 125 cm

$\frac{V}{4L}=500 \Rightarrow \frac{300}{4\times L}=500 \Rightarrow L=\frac{3}{20}m=15$ cm

$\therefore$ Possible length of air column below 125 cm are 15 cm, 45 cm, 75 cm, 105 cm

(i) Fourth order (ii) 4

(iii) Maximum water level = 125–15 = 110 cm

Minimum water level = 125–105 = 20 cm

12. (i) $(2\times7+1)\frac{V}{4L}=600$

$\therefore$ Fundamental frequency = $\frac{V}{4L}=40$ Hz

(ii) $(2\times3+1)\frac{V}{4L}=1400$

$\frac{V}{4L}=200$

$\therefore$ frequency of OOP of same length

$=\frac{V}{2L}=400$

(iii) $\frac{8V}{2L}=800$

$\therefore$ Fundamental frequency = $\frac{V}{2L}=100$ Hz

(iv) $(2\times7+1)\frac{V}{4L}=600$

$\therefore \frac{V}{4L}=40$

$\therefore$ Frequency of 3rd overtone

$=(2\times3+1)\frac{V}{4L}=7\times40=280$ Hz

(v) $\frac{8V}{2L}=1600 \Rightarrow \frac{V}{2L}=200$

$\therefore$ Frequency of 3rd overtone = $\frac{4V}{2L}=800$ Hz

(vi) $\frac{8V}{2L}=1600 \Rightarrow \frac{V}{L}=400$

$\therefore$ Fundamental frequency of COP of same length = $\frac{V}{4L}=100$ Hz

(vii) $\frac{8V}{2L}=800 \Rightarrow \frac{V}{2L}=100$

$\therefore$ Frequency of Ist overtone = $\frac{2V}{2L}=200$ Hz

(viii) $\frac{V}{2[L+2x]}=f$

$\frac{340}{2[0.44+2x]}=340 \Rightarrow \frac{1}{2}=0.44+2x$

$0.50 - 0.44 = 2x \Rightarrow 0.06 = 2x$

$\Rightarrow$ x = 0.03 m = 3 cm

(ix) $\frac{V}{2[L+2x]}=f \Rightarrow \frac{340}{2[L+2x]}=340$

$\frac{1}{2}=L+2x \Rightarrow 0.50=0.38+2x$

$\Rightarrow 0.12 = 2\times(0.6r)$

$\Rightarrow 0.06 = 0.6r$

$\Rightarrow r=\frac{0.06}{0.6}=\frac{1}{10}m=10$ cm

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13. (i) $3 \times 25 = 75$ cm

(ii) slightly more than 75 cm

(iii) possible lengths of air = $\frac{\lambda}{4}, \frac{3\lambda}{4}$ .....

= 10, 30, 50, 70, 90, 110, 130 cm.

$\therefore$ Total no of resonance = 7.

(iv) $\frac{v}{4L} = n$

$\Rightarrow L = \frac{1}{5}m = 20cm$

$\therefore$ Possible lengths for resonance = 20, 60, 14, 140 cm

$\therefore$ Possible no of resonance = 4

(v) $\lambda = 80 \Rightarrow \frac{\lambda}{4} = \frac{80}{4}$ = 20 cm.

$\therefore$ Maximum level of liquid = 150 – 20 = 130 cm

(vi) Possible length of air column

= $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$.....

= 20, 60, 100, 140

$\therefore$ Minimum length of liquid

= 150 – 140 = 10 cm.

(vii) $L_2 - L_1 = \frac{\lambda}{2}$

$\Rightarrow \lambda = 2[L_2 - L_1] = 2[55 - 17] = 76$ cm.

(viii) End correction

$e = \frac{L_2 - 3L_1}{2} = \frac{55-51}{2} = 2cm$

$\Rightarrow 0.6r = 2$

$\Rightarrow r = \frac{20}{6} = \frac{10}{3}$cm.

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Oscillation

SOLUTIONS

Beginner's Box - 1

1. (i) $\sin\omega t + \cos\omega t$

$$= \sqrt{2}\left(\frac{1}{\sqrt{2}}\sin\omega t + \frac{1}{\sqrt{2}}\cos\omega t\right)$$

$$= \sqrt{2}\left(\cos\frac{\pi}{4}\sin\omega t + \sin\frac{\pi}{4}\cos\omega t\right)$$

$$= \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$$

This is S.H.M., its time period is $T = \frac{2\pi}{\omega}$

(ii) $\sin\omega t + \cos 2\omega t + \sin 4\omega t$

$\sin\omega t = \sin(\omega t + 2\pi) = \sin\omega\left(t + \frac{2\pi}{\omega}\right)$

$\cos 2\omega t = \cos(2\omega t + 2\pi) = \cos 2\omega\left(t + \frac{2\pi}{2\omega}\right)$

$\sin 4\omega t = \sin(4\omega t + 2\pi) = \sin 4\omega\left(t + \frac{2\pi}{4\omega}\right)$

Each term in the given function represents a periodic function with periods $T = \frac{2\pi}{\omega}$, $\frac{2\pi}{2\omega}$ and $\frac{2\pi}{4\omega}$ respectively. Thus the given function which is sum of three terms is periodic with time period $\frac{2\pi}{\omega}$. During this period $\sin\omega t$ repeats one, $\cos 2\omega t$ repeats twice and $\sin 4\omega t$ repeats four times but this function is not SHM as $\frac{d^2y}{dt^2}$ is not proportional to –y.

(iii) $e^{-\omega t}$, this exponential function never repeats itself (as when $t \to \infty$, $e^{-\omega t} \to 0$) this is a nonperiodic function.

(iv) $\log(\omega t)$, this logarithimic function is also non periodic. It increases as time increases.

(v) $\sin\omega t - \cos\omega t$.

$$= \sqrt{2}\left[\sin\omega t\frac{1}{\sqrt{2}} - \cos\omega t\frac{1}{\sqrt{2}}\right]$$

$$= \sqrt{2}\left[\sin\omega t\cos\frac{\pi}{4} - \cos\omega t\sin\frac{\pi}{4}\right]$$

$$= \sqrt{2}\sin\left(\omega t - \frac{\pi}{4}\right)$$

Given function can be expressed in a single sine or co-sine so function is simple harmonic with time period $T = \frac{2\pi}{\omega}$

(vi) $\sin^3\omega t = [\sin\omega t]^3 = [\sin(\omega t + 2\pi)]^3$

$$= \left[\sin\omega\left(t + \frac{2\pi}{\omega}\right)\right]^3$$

Thus $\sin^3\omega t$ is a periodic motion with period $\frac{2\pi}{\omega}$. Further as $\frac{d^2y}{dt^2}$ is not proportional to (–y).

So this function does not represent SHM.

2. $\because a = \frac{d^2x}{dt^2}$ $\therefore$ $\frac{d^2x}{dt^2} + 16\pi^2 x = 0$

so $\omega^2 = 16\pi^2$ $\Rightarrow \omega = \sqrt{16\pi^2} = 4\pi$ therefore

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4\pi} = \frac{1}{2} = 0.5 \text{ s}$$

3. Compare $y = 0.08\sin\left(3\pi t + \frac{\pi}{4}\right)$

with $y = A\sin(\omega t + \phi)$

(a) time period = 0.67 s

(b) Initial phase $\phi = \frac{\pi}{4}$ rad

(c) Displacement at $t = \frac{7}{36}$ s

$$y = 0.08\sin\left(3\pi \times \frac{7}{36} + \frac{\pi}{4}\right) = 0.08\sin\frac{5\pi}{6}$$

$$= \frac{0.08}{2} = 0.04 \text{ metre}$$

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Beginner's Box - 2

1. The maximum velocity of the particle at the mean position $v_{max} = A\omega = A(2\pi n)$

$\Rightarrow A = \frac{v_{max}}{2\pi n} = \frac{3.14}{2\times 3.14\times 20} = 0.025\text{m}$

If at the instant t = 0, displacement be zero so displacement equation is

$y = A\sin\omega t = A\sin 2\pi nt = 0.025\sin(40\pi t)$ m

2. $\because$ displacement $x = 0.05\sin\left(5t-\frac{\pi}{6}\right)$

$\because$ velocity $v = \frac{dx}{dt} = \left[0.05\cos\left(5t-\frac{\pi}{6}\right)\right](5)$

At t = 0; $v = 0.25\cos\left(-\frac{\pi}{6}\right) = \frac{\sqrt{3}}{8}\text{ms}^{-1}$

3. $x = A\sin(\pi t+\phi)$ at t = 0 x = 1 cm

$\Rightarrow$ $A\sin\phi = 1$ .....(i)

velocity $v = \frac{dx}{dt} = \pi A\cos(\pi t+\phi)$

at t = 0, $\pi = \pi A\cos\phi$

$\Rightarrow$ $A\cos\phi = 1$ .....(ii)

from (i) & (ii) $A^2(\sin^2\phi+\cos^2\phi) = 1+1$

$\Rightarrow$ $A = \sqrt{2}$ cm and $\tan\phi = 1$

$\phi = \frac{\pi}{4}$ rad

4. Velocity $v = \omega\sqrt{a^2-x^2}$

$\therefore \frac{v_2}{v_1} = \frac{\sqrt{a^2-x_2^2}}{\sqrt{a^2-x_1^2}} \Rightarrow \frac{49}{100} = \frac{a^2-16}{a^2-9}$

$\Rightarrow$ a = 4.76 cm

$\Rightarrow$ length of path = 2a = 9.52 cm

5. (a) when particle starts oscillation from mean position.

$\because y = A\sin\omega t \quad \therefore \frac{A}{2} = A\sin\left(\frac{2\pi}{4}t\right)$

$\Rightarrow \sin\left(\frac{\pi}{2}t\right) = \frac{1}{2} \Rightarrow \frac{\pi t}{2} = \frac{\pi}{6} \Rightarrow t = \frac{1}{3}\text{s}$

(b) When particle starts oscillation from extreme

$y = A\cos\omega t$

$\Rightarrow \frac{A}{2} = A\cos\left(\frac{\pi}{2}t\right)$

$\Rightarrow \cos\left(\frac{\pi}{2}t\right) = \cos\frac{\pi}{3} \Rightarrow t = \frac{2}{3}\text{s}$

6. Time to complete $\frac{1}{4}^{\text{th}}$ oscillation is $\frac{T}{4}$, Time to complete $\frac{1}{8}^{\text{th}}$ oscillation from extreme position is

$y = \frac{a}{2} = a\cos\omega t = a\cos\left(\frac{2\pi}{T}.t\right) \Rightarrow t = \frac{T}{6}$

so time to complete $\frac{3}{8}^{\text{th}}$ oscillation.

$= \frac{T}{4}+\frac{T}{6} = \frac{5T}{12}$

7. At t = 0, $y = 10\sin\frac{\pi}{3} = 10\times\frac{\sqrt{3}}{2} = 5\sqrt{3}$ m

$v = \frac{dy}{dt} = 60\cos(6t+\frac{\pi}{3})$ so at t = 0,

$v = 60\cos\frac{\pi}{3} = 60\times\frac{1}{2} = 30$ m/s

8. (a) When particle moves from extreme to mean position then acceleration supports the velocity. Due to this velocity increases up to mean position.

(b) When particle moves from mean to extreme position then acceleration opposes the velocity. Due to this velocity decreases and finally particle stops at extreme position.

9. (a) 0, +, +; (b) 0, –, –;
(c) –, 0, 0; (d) –, –, – ;
(e) +, +, +; (f) –, +, +

10. Stroke twice the amplitude means,
2 × amplitude=1m

$\therefore$ amplitude $a = \frac{1}{2}$ $\omega = 200$

Maximum speed

$v_{max} = a\omega = \frac{1}{2}\times 200 = 100$ m/min

Beginner's Box - 3

1. The equation of SHM will be

$y = A\sin(\omega t+\phi) = 0.1\sin\left[\omega t+\frac{\pi}{4}\right]$

In SHM, kinetic energy at mean position

$= \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2$

$\Rightarrow 8\times 10^{-3} = \frac{1}{2}\times 0.1\times\omega^2\times(0.1)^2$

$\Rightarrow \omega = 4$ rad/s

$\Rightarrow y = 0.1\sin\left[4t+\frac{\pi}{4}\right]$

2. Total energy = $\frac{1}{2}KA^2$

Kinetic energy = $\frac{1}{2}K(A^2 - x^2)$ and

Potential energy = $\frac{1}{2}Kx^2$

When $x = \frac{A}{4}$,

$$\frac{\text{Kinetic energy}}{\text{Total energy}} = \frac{\frac{1}{2}K\left(A^2 - \frac{A^2}{16}\right)}{\frac{1}{2}KA^2} = \frac{15}{16} \text{ and}$$

$$\frac{\text{Potential energy}}{\text{Total energy}} = \frac{\frac{1}{2}K\frac{A^2}{16}}{\frac{1}{2}KA^2} = \frac{1}{16}$$

3. Since each complete oscillation of a body, kinetic energy of oscillations vary from zero to maximum and maximum to zero. Therefore freq. of K.E. will be 2f

4. $\because U = mV = (500x^2 + 1000)$

Force $F = -\frac{dU}{dx} = -1000x = -kx$

$\Rightarrow$ $k = 1000$ dyne/cm

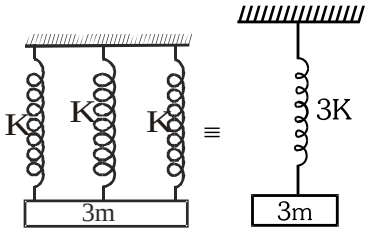
$\Rightarrow$ Frequency $n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{1000}{10}}$

$= \frac{5}{\pi} s^{-1}$

Beginner's Box - 4

1. watch will give correct time because it depends on spring action and does not depend on gravity.

2.



$\therefore$ time period = $2\pi\sqrt{\frac{3m}{3K}} = 2\pi\sqrt{\frac{m}{K}}$

3. (a) If length of spring is made half,

then $K' = 2K$ $\because T \propto \frac{1}{\sqrt{K}}$

$\therefore$ $\frac{T'}{T} = \sqrt{\frac{K}{K'}} = \sqrt{\frac{K}{2K}} \Rightarrow T' = \frac{T}{\sqrt{2}}$

(b) $\because$ $T \propto \sqrt{m}$

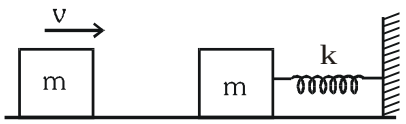
$\therefore$ $\frac{T'}{T} = \sqrt{\frac{m'}{m}} = \sqrt{\frac{m/2}{m}} \Rightarrow T' = \frac{T}{\sqrt{2}}$

4. If a force F produces an increase of x in the length of a spring, then the force constant of the spring $K = \frac{F}{x} = \frac{1 \times 9.8}{0.98}$ 10 N/m.

Period of oscillation

$T = 2\pi\sqrt{\frac{m}{K}} = 2 \times 3.14\sqrt{\frac{1}{10}} \simeq 2s$

5.



During collision, linear momentum remains conserve. $mv = (m + m)u$ $\Rightarrow$ $u = \frac{v}{2}$

Gained kinetic energy of second block will convert in potential energy of spring.

so $\frac{1}{2}KA^2 = \frac{1}{2}(2m)u^2 = \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2$

$\Rightarrow$ $KA^2 = \frac{mv^2}{2}$ $\Rightarrow$ $A = v\sqrt{\frac{m}{2K}}$

6. Maximum acceleration

$= \omega^2 A = \left(\frac{k}{m}\right)A = \left(\frac{1200}{3}\right)(2\times10^{-2}) = 8 \text{ ms}^{-2}$

7. Force constant $k = \frac{F}{x} = \frac{50 \times 9.8}{0.20} = 2450 \text{ Nm}^{-1}$

$\because$ $T = 2\pi\sqrt{\frac{m}{k}}$

$\Rightarrow$ $m = \frac{T^2k}{4\pi^2} = \frac{(0.6)^2 \times 2450 \times 7 \times 7}{4 \times 22 \times 22} = 22.32$ kg

$\therefore$ Weight of the body = mg

$= 22.32 \times 9.8 = 218.7$ N

8. (a) In the given arrangement, let the force constant of two springs be K, then

$K = k_1 + k_2 = 2k \qquad [\because\ k_1 = k_2 = k]$

$\therefore$ Time period

$$T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{m}{2k}} = 2\times 3.14\sqrt{\frac{3}{2\times 600}}$$

$= 0.314$ s

(b) Maximum speed of trolley,

$$v_{max} = A\omega = (A)\left(\frac{2\pi}{T}\right) = 5\times 10^{-2}\times\frac{2\times\pi}{0\cdot 314}$$

$= 1\ ms^{-1}$

Energy dissipated as heat by the time in which trolley comes to rest

$$= \frac{1}{2}KA^2 = \frac{1}{2}\times 1200\times(0.05)^2 = 1.5J$$

Beginner's Box - 5

1. Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}}$

$\Rightarrow \frac{\ell}{g} = \frac{1}{\pi^2}$ But $g = \pi^2$ therefore $\ell = 1$ m

2. The time period will increase at first, then decrease until the sphere is empty and will acquire its initial value again.

3. The system may be roughly regarded as a simple pendulum. When the girl stands up, the length of the simple pendulum is reduced so frequency of oscillation is increased.

4. In both cases effective acceleration due to gravity becomes zero. In absence of 'g_{eff}' there is no restoring force and pendulum does not oscillate.

5. Time period of simple pendulum

$$T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1.000}{9.800}} \approx 2s$$

Frequency $f = \frac{1}{T} = \frac{1}{2} = 0.5$ Hz

6. Time period $T = 2\pi\sqrt{\frac{L}{g}}$

$\Rightarrow T^2 = 4\pi^2\times\frac{L}{g}$

$\Rightarrow L = \frac{gT^2}{4\pi^2} = \frac{9\cdot 8\times 4}{4\times(3\cdot 14)^2} = 0\cdot 99$ m

7. Gravity disappears under free fall, so frequency is zero.

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Thermal Physics

SOLUTIONS

BEGINNER'S BOX-1

1. Convert all in same unit °C.

(a) $\frac{C}{5}=\frac{F-32}{9}$

$C=\frac{(50-32)\times5}{9}=\frac{18\times5}{9}=10\,°C$

(b) 50 K = 50 – 273 = – 223 °C

Increasing order ⇒ 50 K < 50 °F < 50 °C.

2. (a) X = Y = W all have same difference of maximum and minimum temperature.

(b) Relation will be

$\frac{X-(-20)}{70-(-20)}=\frac{W-30}{120-30}=\frac{Y-0}{90-0}$

$\frac{X+20}{90}=\frac{W-30}{90}=\frac{Y}{90}$

$\therefore$ X + 20 = W – 30 X + 20 = Y

W = X + 50 X = Y – 20

Now change all in X

First	50 °X		
Second	50 °W	⇒	0 °X
Third	50 °Y	⇒	30 °X

$\therefore$ In order of higher first 50 °X, 30 °X, 0 °X

i.e. 50 °X > 50 °Y > 50 °W

3. $\frac{C}{5}=\frac{F-32}{9}$

C = F

$\frac{F}{5}=\frac{F-32}{9}$

9F = 5F – 160

160 = –4F

F = –40

That temperature will be –40 °C or –40 °F.

4. % change in length = $\alpha\Delta t \times 100 = 1\%$

% change in area = $\beta\Delta t \times 100$

$= 2\alpha\Delta t \times 100$

$= 2 \times 1\%$

$= 2\%$

5. The diameter increases by an amount

$\Delta d = d_0\alpha\Delta T = 4.24 \times 1.7 \times 10^{-5} \times 200$

$= 1.44 \times 10^{-2}$ cm

6. $\Delta\ell = \ell\alpha\Delta T$

$\Delta\ell \propto \alpha$

So strip will bend towards the metal of lower thermal expansion co-efficient.

BEGINNER'S BOX-2

1. $K.E = \frac{1}{2}mv^2$

$= \frac{1}{2} \times 10 \times 10^{-3} \times 400 \times 400 = 800$ J

$= \frac{800}{4.186}$ cal $= 191.11$ cal $\simeq 200$ cal

2. $Q = ms\Delta\theta = 1000 \times 0.5 \times 100$

$= 50$ kcal

3. $Q = ms_w\Delta\theta$

$= 1 \times 1 \times 1 = 1$ cal

4. $Q = ms\Delta\theta$

$420 = 10 \times 10^{-3} \times 4200 \times \Delta\theta$

$\Delta\theta = 10°C$

5. $\frac{\frac{m_1s_1}{V_1}}{\frac{m_2s_2}{V_2}}=\frac{\rho_1s_1}{\rho_2s_2}=\frac{3}{4}\times\frac{4}{3}=1:1$

6. Heat released $= m_vs_v\Delta\theta + m_vL_v + m_ws_w\Delta\theta$

$= 1000 \times 0.5 \times 50 + 1000 \times 540 + 1000 \times 1 \times 50$

$= (25 + 540 + 50]$ kcal

$= 615$ kcal

7. $Q = ms\Delta\theta$

$= 200 \times 1 \times 20 = 4$ kcal

8. Loss in K.E = Heat energy

$\frac{1}{2}mv^2 = ms\Delta\theta$

$\frac{400\times400}{2}=500\times\Delta\theta$

$\Delta\theta = 160°C$

9. Let equilibrium temperature be θ

Heat gained by ice at –10°C = Heat loss by water at 100°C

$m_is_i \times 10 + m_iL_i + m_ws_w(\theta-0) = m'_ws_w(100-\theta)$

$10^3 \times 0.5 \times 10 + 10^3 \times 80 + 10^3 \times 1 \times \theta = 10^3 \times 1 \times (100-\theta)$

$5 + 80 + \theta = 100 - \theta$

$2\theta = 15$

$\theta = 7.5°C$ water.

10. Heat required to melt the ice completely
$Q_1 = m_i L_i = 540 \times 80$ cal
Heat rejected by 540 gm water at 80°C water to lower its temperature to 0°C water
$Q_2 = m_w s_w \Delta\theta = 540 \times 1 \times 80 = 540 \times 80$ cal
$Q_1 = Q_2$
$\therefore$ All ice will melt and equilibrium temperature will be 0°C.

BEGINNER'S BOX-3

2. (a) Thermal current = $\frac{\text{Temperature difference}}{R_H}$

where $\frac{1}{R_H} = \frac{K_1 A}{\ell} + \frac{K_2 A}{\ell} = \frac{(K_1 + K_2)a^2}{a}$

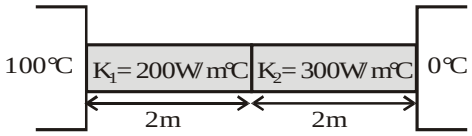
$=(60 + 40)(3 \times 10^{-2}) = 3\ WK^{-1}$

$\Rightarrow$ Thermal current $= 80 \times 3 = 240$ W

(b) Ratio of thermal currents $\frac{H_{Cu}}{H_{Al}} = \frac{K_{Cu}}{K_{Al}} = \frac{60}{40}$

$= 1.5$

3.



Let temperature of interface be T then

$\left(\frac{dQ}{dt}\right)_1 = \left(\frac{dQ}{dt}\right)_2$

$\Rightarrow K_1(100 - T) = K_2(T - 0)$

$\Rightarrow 200(100 - T) = 300(T)$

$\Rightarrow 200 = 5T \Rightarrow T = 40°C$

4. In steady state rate of flow of heat in the whole system will be same.

$\frac{KA(200-\theta_1)}{\ell} = \frac{2KA(\theta_1-\theta_2)}{\ell} = \frac{1.5KA(\theta_2-18)}{\ell}$

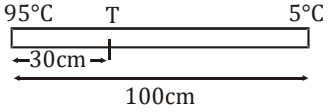
$\Rightarrow 200-\theta_1 = 2\theta_1-2\theta_2$ ...(1)

$2\theta_1-2\theta_2 = 1.5\theta_2 -27$ ...(2)

By solving (1) and (2), we get

$\Rightarrow \theta_1 = 116°C, \theta_2 = 74°C$

5.



$\frac{K(95-T)A}{30} = \frac{K(95-5)A}{100} \Rightarrow T = 68°C$

Temperature gradient

$= \frac{\Delta\theta}{\Delta x} = \frac{95-5}{100} = 0.9$ °C/cm

6. Heat current in BD = Heat current in BC

$\frac{90-20}{R_{BD}} = \frac{20-0}{R_{BC}}$

$70\ R_{BC} = 20 R_{BD}$

Temperature of A and B is same

So, no heat flow is AB.

$\frac{70\ell_{BC}}{KA} = \frac{20\ell_{BD}}{KA} \Rightarrow \frac{7}{2} = \frac{\ell_{BD}}{\ell_{BC}}$

BEGINNER'S BOX-4

2. Emitted power $P = \sigma A T^4$

According to question $\sigma A_1 T_1^4 = \sigma A_2 T_2^4$

$\Rightarrow \sigma(4\pi r_1^2)T_1^4 = \sigma(4\pi r_2^2)T_2^4 \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{r_2}{r_1}}$

3. $\frac{\theta_1 - \theta_2}{t} = K\left(\frac{\theta_1 + \theta_2}{2} - \theta_0\right)$

In first case, $\frac{80 - 70}{30} = K\left(\frac{80 + 70}{2} - \theta_0\right)$

$\frac{1}{3} = K(75 - \theta_0)$ ...(1)

In second case, $\frac{60 - 50}{70} = K\left(\frac{60 + 50}{2} - \theta_0\right)$

$\frac{1}{7} = K(55 - \theta_0)$ ...(2)

Divide equation (1) by equation (2)

$\frac{7}{3} = \frac{(75 - \theta_0)}{(55 - \theta_0)} \Rightarrow 385 - 7\theta_0 = 225 - 3\theta_0$

$\Rightarrow \theta_0 = \frac{160}{4} = 40°C$

4. According to newton's law of cooling

$\frac{\theta_1 - \theta_2}{t} = K\left[\left(\frac{\theta_1 + \theta_2}{2}\right) - \theta_0\right]$

$\Rightarrow \frac{60-40}{10} = K\left[\frac{60+40}{2} - 10\right]$ ...(i)

& $\frac{40-\theta}{10} = K\left[\frac{40+\theta}{2} - 10\right]$ ...(ii)

$\Rightarrow \frac{20}{40-\theta} = \frac{100-20}{40+\theta-20} \Rightarrow \theta = 28°C$

5. By using Wien's displacement law

$T = \frac{b}{\lambda_m} = \frac{2.89 \times 10^{-3}}{1.5 \times 10^{-6}} = 1927\ K \simeq 2000\ K$

BEGINNER'S BOX-5

1. Initial molar amount

$$\mu_1 = \frac{PV}{RT} = \frac{200\times10^3\times8.3\times10^{-3}}{8.3\times300} = \frac{2}{3}\text{ mole}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{\mu_1 T_1}{\mu_2 T_2}$$

$$\Rightarrow \frac{200}{100} = \frac{\frac{2}{3}\times300}{\mu_2\times600}$$

$$\Rightarrow \mu_2 = \frac{1}{6}\text{ mole}$$

$$\Rightarrow \text{leaked out moles} = \mu_1-\mu_2 = \frac{2}{3}-\frac{1}{6} = \frac{1}{2}$$

mole

2. By Mole conservation $\mu_1 + \mu_2 = \mu_1' + \mu_2'$

$$\frac{P_1V_1}{RT_1}+\frac{P_2V_2}{RT_2} = \frac{P_1'V_1'}{RT_1'}+\frac{P_2'V_2'}{RT_2'}$$

$$\Rightarrow \frac{105\text{kPa}\times V}{R\times300\text{K}}+\frac{105\text{kPa}\times V}{R\times300\text{K}}$$

$$= \frac{P_{mix}V}{R\times300\text{K}}+\frac{P_{mix}V}{R\times400\text{K}}$$

$$\Rightarrow P_{mix}=120\text{ kPa}$$

3. $N = \frac{PV}{kT} = \frac{(1.2\times10^7)\times(10^{-3})}{(1.38\times10^{-23})\times(400)} = 2.2\times10^{24}$

4. (a) Ideal gas equation ; PV = NkT(V, T $\rightarrow$ same)

$$\Rightarrow \frac{N_1}{N_2}=\frac{P_1}{P_2}\Rightarrow \frac{N_1}{N_2}=\frac{3}{2}$$

(b) From equation $\frac{P}{\rho}=\frac{RT}{M_w}\Rightarrow \rho=\frac{PM_w}{RT}$

(T $\rightarrow$ same)

In the mixture T = constant then

$$\Rightarrow \frac{\rho_1}{\rho_2}=\frac{P_1M_{w_1}}{P_2M_{w_2}}\Rightarrow \frac{\rho_1}{\rho_2}=\frac{3}{2}\times\frac{20.2}{32.0}=0.947$$

BEGINNER'S BOX-6

1. Absolute temperature, T = 27 + 273 = 300 K. By the law of equipartition of energy, the average kinetic energy of a gas molecule per degree of freedom is $\frac{1}{2}kT$.

(a) An oxygen molecule has 3 translational degrees of freedom. Hence its average kinetic energy of translation is

$$3\times\frac{1}{2}kT = 3\times\frac{1}{2}(1.38\times10^{-23})\times300$$

$$= 6.21\times10^{-21}\text{ J}$$

(b) Oxygen molecule is diatomic, hence it has 2 rotational degrees of freedom also, that is, it has a total of 5 degrees of freedom. Hence its total kinetic energy is

$$5\times\frac{1}{2}kT = 5\times\frac{1}{2}(1.38\times10^{-23})\times300$$

$$= 10.35\times10^{-21}\text{ J}$$

(c) One mole oxygen contains N_A molecules. Hence the total kinetic energy at 1 mole gas is

$$N_A\times\frac{5}{2}kT=(6.02\times10^{23})\times(10.35\times10^{-21})$$

$$= 6231\text{J}$$

2. Initial Temperature T_1 = 200 K

(a) $E = \frac{3}{2}kT\Rightarrow E\propto T\Rightarrow \frac{E_2}{E_1}=\frac{T_2}{T_1}$

$$\Rightarrow\frac{2E}{E}=\frac{T_2}{200}\Rightarrow T_2 = 400\text{ K} = 127^\circ\text{C}$$

(b) $v_{rms}\propto\sqrt{T}\Rightarrow\frac{v_{rms_2}}{v_{rms_1}}=\sqrt{\frac{T_2}{T_1}}$

$$\Rightarrow\frac{2v}{v}=\sqrt{\frac{T_2}{200}}\Rightarrow T_2 = 800\text{ K} = 527^\circ\text{C}$$

3. $\frac{R}{C_p}=0.4\Rightarrow C_P=\frac{5}{2}R$ then

(a) $C_v = C_P - R\Rightarrow C_v=\frac{3}{2}R$ (mono atomic)

$C_v=\frac{f}{2}R$ then degree of freedom f = 3

(b) $C_v=\frac{3}{2}R$ and $\gamma = 1+\frac{2}{f} = 1+\frac{2}{3}=\frac{5}{3}$

(c) $E=\frac{3}{2}RT=\frac{3}{2}\times R\times300 = 450R = 500\text{ cal}$

4. Average speed $v_{avg}=\frac{v_1+v_2+v_3}{3}$

Root mean square speed $=\sqrt{\frac{v_1^2+v_2^2+v_3^2}{3}}$

5. Root mean square speed

$= \sqrt{\frac{(1)^2+(4)^2+(8)^2+(16)^2}{4}} = 9.2\text{m/sec}$

6. (a) Average translational kinetic energy per molecule $E \propto T$

$\Rightarrow \frac{E_1}{E_2} = \frac{1}{1}$ ($\because$ T = same for both)

(b) $v_{rms} = \sqrt{\frac{3RT}{M_w}}$ ($\because$ T = same)

$\Rightarrow \frac{v_{rms_1}}{v_{rms_2}} = \sqrt{\frac{M_{w_2}}{M_{w_1}}}$

$\Rightarrow \frac{v_{rms_1}}{v_{rms_2}} = \sqrt{\frac{70.9}{39.9}} = \frac{1.33}{1} \simeq \frac{4}{3}$

BEGINNER'S BOX-7

1. (a) $W_{ABCA} = \frac{1}{2} \times (2V_0 - V_0) \times (3P_0 - P_0) = P_0V_0$

(b) Heat rejected in path CA = $-(W + \Delta U)$

$= -[P_0(V_0 - 2V_0) + \mu C_V \Delta T]$

where $\Delta T = T_A - T_C = \frac{P_0(V_0 - 2V_0)}{\mu R} = \frac{-P_0V_0}{\mu R}$

Therefore heat rejected

$= P_0V_0 + \frac{3}{2}P_0V_0 = \frac{5}{2}P_0V_0$

OR

Heat rejected in path CA in isobaric process

$= -\mu C_P \Delta T = -1 \times \left(\frac{5}{2}R\right)\left(-\frac{P_0V_0}{1 \times R}\right) = \frac{5}{2}P_0V_0$

Heat absorb in path AB = $\Delta U = \mu C_V \Delta T$

Therefore heat absorbed $= 1 \times \frac{3}{2}R(T_B - T_A)$

$= \frac{3}{2}R(3T_0 - T_0) = 3RT_0$

Here $P_0V_0 = RT_0$ then Heat absorbed

$= 3P_0V_0$

(c) Heat absorbed in BC path = $W + \Delta U$

$= P_0V_0 + \left(\frac{1}{2} \times 2P_0 \times V_0\right) + \mu C_V \Delta T$

$= 2P_0V_0 + 1 \times \frac{3}{2}R(T_C - T_B)$

But $T_C = 2T_0$ where T_0 = temp. of A & T_B

$= 3T_0$ also $P_0V_0 = RT_0$

Therefore heat absorbed

$= 2P_0V_0 + 1 \times \frac{3}{2}R(2T_0 - 3T_0)$

$= 2P_0V_0 - \frac{3}{2}P_0V_0 = \frac{P_0V_0}{2}$

2. In given process $W = -1/2 \times (3 - 1) \times (5 \times 10^5)$

$= -5 \times 10^5$ J

and $\Delta U = 0$ ($\because$ cyclic process)

$\therefore$ $Q = W + \Delta U = -5 \times 10^5$ J

3. The work done in a thermodynamic process is equal to the area enclosed between the P–V curve and the volume-axis.

Work done by the gas in the process A→B is

$W_1 = (6.0 - 1.0)$ litre $\times (12 \times 10^5)$ N/m^2

$= 5.0 \times 10^{-3}$ m$^3 \times (12 \times 10^5)$ N/m^2

$= 6000$ Nm $= 6000$ J

Work done by the process B → C is zero since volume remains constant.

Work done by the gas in process C → D is

$W_2 = -(6.0 - 1.0) \times 10^{-3} \times 2 \times 10^5$

$= -(5.0 \times 10^{-3}) \times (2 \times 10^5) = -1000$ J

Work done in the process D → A is also zero.

4. (a) $W = P\Delta V$

here P = 8 atm = 8×10^5 N/m^2,

$V_1 = 2$litre $= 2 \times 10^{-3}$ m^3, $V_2 = 5$litre

$= 5 \times 10^{-3}$ m^3

$W = 8 \times 10^5 \times (5-2) \times 10^{-3} = 24 \times 10^2$ J

(b) $\Delta U = \mu C_V \Delta T$; $C_V = \frac{3}{2}R$ (for monoatomic gas)

$= \mu \times \frac{3}{2}R(T_2 - T_1) = \frac{3}{2}P\Delta V$

[$\because P\Delta V = \mu R \Delta T$]

$= \frac{3}{2} \times 24 \times 10^2$ J $= 36 \times 10^2$ J

(c) $Q = W + \Delta U = 24 \times 10^2 + 36 \times 10^2$

$= 60 \times 10^2$ J

5. In given process

$W = \frac{1}{2} \times (700 - 500) \times 10^{-6} \times (200 - 100) \times 10^3$

$= 10J$

$\because \quad \Delta U = 0$ so $Q = \frac{W}{J}$

$\Rightarrow \quad J = \frac{W}{Q} = \frac{10}{2.4} = 4.16 J/cal$

BEGINNER'S BOX-8

1. For adiabatic process

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$\Rightarrow T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 300\,(3)^{1.5-1} = 519.6\ K$

For adiabatic process

$Q = 0$ so $\Delta U = -W = -\frac{\mu R(T_1 - T_2)}{\gamma - 1}$;

$\mu = \frac{32}{32} = 1$

$= -\frac{1 \times 8.31 \times (-219.6)}{(1.5-1)}$

$= 3649.7 \simeq 3650\ J$

2. (a) Heat energy given at constant volume.

$Q_{mix} = \mu_{H_2} C_{V_{H_2}} \Delta T + \mu_{He} C_{V_{He}} \Delta T \left[C_V = \frac{f}{2}R\right]$

$= \frac{1}{2} \times \frac{5}{2} \times 2 \times 50 + \frac{1}{4} \times \frac{3}{2} \times 2 \times 50$

$= 125 + 37.5 = 162.5$ cal

(b) Heat energy given at constant pressure.

$Q_{mix} = \mu_{H_2} C_{P_{H_2}} \Delta T + \mu_{He} C_{P_{He}} \Delta T$

$\left[C_P = \left(\frac{f}{2} + 1\right)R\right]$

$= \frac{1}{2} \times \frac{7}{2} \times 2 \times 50 + \frac{1}{4} \times \frac{5}{2} \times 2 \times 50$

$= 175 + 62.5 = 237.5$ cal

3. (A) $\rightarrow$ p, (B) $\rightarrow$ q , (C) $\rightarrow$ r

4. (A) $\rightarrow$ q , (B) $\rightarrow$ r , (C) $\rightarrow$ p , (D) $\rightarrow$ s

5. (A) $\rightarrow$ p,s, (B) $\rightarrow$ p,q,r,s, (C)$\rightarrow$s , (D)$\rightarrow$q,s, (E) $\rightarrow$ r,s

6. (a) AB is isothermal process hence $\Delta U_{AB} = 0$

$\Rightarrow Q_{AB} = W_{AB} = 700$ J

(b) BC is an adiabatic process hence

$Q_{BC} = 0$ & $W_{BC} = 400$ J

ABCA is cyclic process so

$(\Delta U)_{wholecycle} = 0 = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA}$

From First law of thermodynamics

$Q_{ABCA} = W_{ABCA}$

$Q_{AB} + Q_{BC} + Q_{CA} = W_{AB} + W_{BC} + W_{CA}$

$\Rightarrow \quad 700 + 0 - 100 = 700 + 400 + W_{CA}$

$W_{CA} = -500$ J so work is done on the gas is 500 J

7. (a) For process A to B (adiabatic)

$\therefore \quad TV^{\gamma-1} =$ constant

$\Rightarrow T_AV_A^{\gamma-1} = T_BV_B^{\gamma-1}$

$\Rightarrow 300(0.36)^{1.5-1} = T_B(1.44)^{1.5-1}$

$\Rightarrow T_B = 300\left(\frac{0.36}{1.44}\right)^{0.5} = 150\ K$

For process B to C (isobaric)

$V \propto T \Rightarrow \frac{V_B}{V_C} = \frac{T_B}{T_C}$

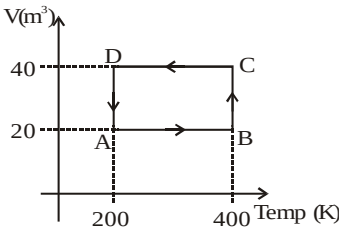
$\Rightarrow \frac{1.44}{0.36} = \frac{150}{T_C} \Rightarrow \quad T_C = 37.5\ K$

(b) $W_{AB} = \frac{\mu R(T_A - T_B)}{\gamma - 1}$

$= \frac{P_AV_A}{RT_A} \times R\frac{(300-150)}{(1.5-1)}$

$= \frac{2 \times 10^5 \times 0.36}{300} \times 300 = 72 \times 10^3\ J$

8.



Work done $= W_{AB} + W_{BC} + W_{CD} + W_{DA}$

$= 0 + 2.303\mu RT_B \log_{10}\left(\frac{V_C}{V_B}\right)$

$+0 + 2.303\mu RT_D \log_{10}\left(\frac{V_A}{V_D}\right)$

$= 2.303 \times \mu R\left[T_B \log_{10}\left(\frac{V_C}{V_B}\right) + T_D \log\left(\frac{V_A}{V_D}\right)\right]$

$= 2.303\times1\times8.314\left[400\log_{10}\left(\frac{40}{20}\right)+200\log\left(\frac{20}{40}\right)\right]$

$= 2.303\times8.314\left[400\log_{10}2-200\log_{10}2\right]$

$= 2.303 \times 8.314 \times 200 \times \log_{10}2$

$= 2.303 \times 8.314 \times 200 \times 0.3010$

$= 1152.65$ J

BEGINNER'S BOX-9

1. For an ideal engine

$\eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q_1}$

$\Rightarrow W = Q_1\left(1-\frac{T_2}{T_1}\right) = 100\left(1-\frac{300}{500}\right)$ cal

$= 40 \times 4.2$ J $= 168$ J

2. Coefficient of Performance (β)

$\beta = \frac{T_2}{T_1 - T_2}$ Where $T_2 = -23 + 273 = 250$ K,

$\beta = \frac{5}{3} \Rightarrow \frac{5}{3} = \frac{250}{T_1 - 250} \Rightarrow T_1 = 400$ K

3. (a) Efficiency

$\eta = \left(1-\frac{Q_2}{Q_1}\right)\times100 = \left(1-\frac{600}{1000}\right)\times 100 = 40\%$

(b) $\frac{Q_2}{Q_1} = \frac{T_2}{T_1} \Rightarrow \frac{600}{1000} = \frac{T_2}{400}$

$\Rightarrow T_2 = 240$ K

(c) $\eta = \frac{W}{Q_1} \Rightarrow \frac{40}{100} = \frac{W}{1000}$

$\Rightarrow W = 400$ J

4. $\because \eta = 1 - \frac{T_2}{T_1}$

$\therefore \frac{40}{100} = 1 - \frac{300}{T_1}$

$\Rightarrow T_1 = 500$ K

If efficiency increase up to 60% then

$\frac{60}{100} = 1 - \frac{300}{T_1'} \Rightarrow T_1' = 750$ K

Then, change in temperature of source

$\Delta T = 750 - 500 = 250$ K

5. $\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$

where $T_1 = 400$ K,

$T_2 = 300$K & $Q_1 = 120$ cal/s

$\Rightarrow \frac{Q_2}{120 - Q_2} = \frac{300}{100}$

$\Rightarrow Q_2 = 90$ cal/s

Therefore wattage

$\frac{W}{t} = \frac{(Q_1 - Q_2)\times4.2}{t} = \frac{30\times4.2\text{J}}{1\text{s}} = 126$ W

AIM: FREE EDUCATION TO ALL APUL

Properties of Matter and Fluid Mechanics

SOLUTIONS

BEGINNER'S BOX-1

1. Tangential stress

$= \frac{F}{A} = \frac{100\cos 30°}{5\times 2} = 5\sqrt{3} N/m^2$

Longitudinal stress $= \frac{F_\perp}{A} = \frac{100\sin 30°}{5\times 2} = 5\ N/m^2$

2. Shear strain $= \frac{r\theta}{\ell} = \frac{1\times 10^{-2}\times 0.8}{2} = 0.004$ rad

3. Maximum stress $= \frac{F}{A} = \frac{M(g+a)}{\pi r_{min}^2}$

$\therefore r_{min} = \sqrt{\frac{M(g+a)}{\pi \times \text{Max. stress}}} = \sqrt{\frac{900(9.8+2.2)}{\pi\times\frac{3}{\pi}\times 10^8}}$

$= 6 \times 10^{-3}$ m $= 6$ mm

4. Applied stress $= \frac{F}{A} = \frac{5\times 10^5}{4\times 10^{-4}}$

$= 12.5 \times 10^8\ N/m^2$

Compressive strength $= 1.7 \times 10^8\ N/m^2$

Applied stress is greater than compressive strength of bond so bone will break.

5. Stress $= \frac{F}{A} = \frac{mgL}{AL} = \frac{\rho VgL}{v} = \rho gL$

$L_{max} = \frac{\text{maximum stress}}{\rho g} = \frac{7.9\times 10^8}{7.9\times 10^3\times 10}$

$= 10^4$m $= 10$ km

6. In both case Y, A & F remain same so breaking stress will be also same.

BEGINNER'S BOX-2

1. Strain $= \frac{\Delta\ell}{L} = \frac{2L-L}{L} = 1$

Young modulus $Y = \frac{\text{stress}}{\text{strain}} = 2.0 \times 10^9\ N/m^2$

2. $Y = \frac{FL}{\pi r^2 \Delta\ell}$ $\Delta\ell \propto \frac{L}{r^2}$

$\frac{\Delta\ell_1}{\Delta\ell_2} = \left(\frac{L_1}{L_2}\right)\left(\frac{r_2}{r_1}\right)^2 = \frac{3}{1}\times\left(\frac{3}{1}\right)^2 = \frac{27}{1}$

3. $\Delta\ell = \frac{FL}{YA} \Rightarrow \Delta\ell \propto \frac{L}{r^2}$

4. (Bulk modulus) $K = \frac{\Delta P}{(-\Delta V/V)}$;

$\left(\frac{\Delta V}{V}\right) = \frac{345\times 10^6}{138\times 10^9} = \frac{1}{400}$

% Change in volume = 0.25%

5. (Bulk modulus) K =

$\frac{1}{\text{compressibility}} = \frac{1}{4\times 10^{-5}} = 2.5 \times 10^4$

$K = \frac{\Delta P}{(-\Delta V/V)}$

$\Rightarrow (-\Delta V) = \frac{100\times 100}{2.5\times 10^4} = 0.4\ cm^3$

6. Shear modulus $= \frac{\text{shear stress}}{\text{shear strain}} = \frac{F_t}{A}\frac{L}{\ell}$

$8 \times 10^{10} = \frac{4000\times 25\times 10^{-2}}{(25\times 10^{-2})^2\times\ell}$

$\Rightarrow \ell = 2 \times 10^{-7}$ m

7. Increase in energy $= \frac{1}{2} F \times \Delta\ell = \frac{1}{2} \times F \times \frac{FL}{AY}$

$= \frac{1}{2}\times(10\times 10)^2 \times \frac{0.1}{10^{-4}\times 10^{11}} = 5 \times 10^{-5}$ J

BEGINNER'S BOX-3

1. When equal volume of two metals are mixed

$\rho_{mix} = \frac{\rho_1+\rho_2}{2} = 4$

$\rho_1 + \rho_2 = 8$ ....(i)

When equal masses of the same two metals are mixed

$\rho_{mix} = \frac{2\rho_1\rho_2}{\rho_1+\rho_2} = 3$

$\rho_1\rho_2 = 12$ .....(ii)

from equation (i) & (ii)

$(8 - \rho_2)\ \rho_2 = 12$

$\Rightarrow \rho_2^2 - 8\rho_2 + 12 = 0$

$\Rightarrow (\rho_2 - 6)\ (\rho_2 - 2) = 0$

if $\rho_2 = 6\ kg/m^3$ then $\rho_1 = 2\ kg/m^3$

or $\rho_2 = 2 kg/m^3$ then $\rho_1 = 6 kg/m^3$

2. ρgh = constant $\Rightarrow$ gh = constant

$g_1h_1 = g_2h_2$

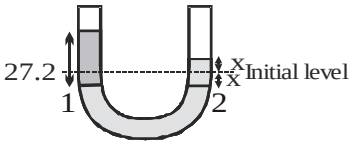
$\Rightarrow g \times 75 = (g - a)h$

$\Rightarrow h = \frac{10\times75}{9} = 83.33$ cm

3. $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_1}{D_1^2} = \frac{F_2}{D_2^2}$ ($\because$ D = diameter)

$\Rightarrow \frac{1}{(10)^2} = \frac{F_2}{(50)^2} \Rightarrow F_2 = 25N$

4.



$P_1 = P_2$

$P_0 + \rho_w g \times 27.2 = P_0 + \rho_{Hg} \times g \times 2x$

$1 \times 10 \times 27.2 = 13.6 \times 10 \times 2x$

x = 1 cm

5. R.D block = $\frac{w_A}{w_A - w_w} = \frac{15}{15-12} = 5$

R.D of liquid = $\frac{w_A - w_L}{w_A - w_w} = \frac{15-13}{15-12} = \frac{2}{3} = 0.67$

6. For floating condition

(i) $W = Th \Rightarrow \rho_s Vg = \rho_w V_{in} g$

$\rho_s \times V = \rho_w \times \frac{2}{3}V$ [$\because \rho_w = 10^3$ kg/m^3]

$\rho_s = 0.67 \times 10^3$ kg/m^3

(ii) $\rho_s V = \rho_{oil} \times 0.90\ V$

$\rho_{oil} = \frac{0.67\times10^3}{0.90} = 0.74 \times 10^3$ kg/m^3

7. In floating condition $W = Th \Rightarrow Mg = V_{in}\rho_w g$

$700 \times g = V_{in} \times 1 \times g \Rightarrow V_{in} = 700$ cm^3

outside volume $V_0 = V - V_{in}$

$= 1000 - 700 = 300$ cm^3

8. $W_{app} = (\rho_s - \rho_w) V_{in} g$

$W_{app} = (5 - 1) \times 5 \times 5 \times 5 \times g$ newton

$W_{app} = 4 \times 5 \times 5 \times 5$ g-f = 500 g-f

BEGINNER'S BOX-4

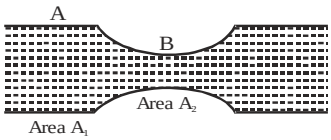
1. By mass conservation $A_1v_1 = A_2v_2 + A_3v_3$

$\Rightarrow v_3 = \frac{A_1v_1 - A_2v_2}{A_3} = \frac{(0.12)(3)-(0.12)(1.5)}{0.18}$

$= 1$ m/s

2. $v \simeq \sqrt{2gh} = \sqrt{2\times10\times3.2} = 8$ m/s

3.



$A_1v_1 = A_2v_2$ or $v_2 = \left[\frac{A_1}{A_2}\right]v_1$ ...(i)

From Bernoulli's equation for horizontal flow,

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

or $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho\left[\frac{A_1^2}{A_2^2}\right]v_1^2$ [from equation (i)]

or $P_1 - P_2 = \frac{1}{2}\rho v_1^2\left[\frac{A_1^2}{A_2^2} - 1\right]$

But $P_1 - P_2 = 600$ N/m^2

$\therefore \quad 600 = \frac{1}{2}\rho v_1^2\left[\frac{A_1^2}{A_2^2} - 1\right]$

$\therefore \quad v_1 = \sqrt{\frac{1200}{\rho}}\left[\frac{A_1^2}{A_2^2} - 1\right]^{-\frac{1}{2}}$ ...(ii)

If Q be the volume of liquid flowing per unit time then $Q = A_1v_1 = A_2v_2$. Putting values of A_1 and v_1

$Q = \frac{6}{\sqrt{10}}10^{-3}$ m^3/s

BEGINNER'S BOX-5

1. $F_{net} = F_1 - F_2 = (T_1 - T_2)\ell$

$= (80 - 60) \times 4 = 80$ dyne

2. $T = \frac{F}{2(2\pi R)} = \frac{8}{4\pi\times1}$ = N/m

3. $F_{ex} = 2T\ell = 2 \times 70 \times 20 = 2800$ dyne

4. $2T\ell = mg$

$\Rightarrow 2T\ell = (\pi r^2\ell)\rho g \Rightarrow r = \sqrt{\frac{2T}{\pi\rho g}}$

5. For film we have two free surface

$\Delta W = 2T\Delta A \Rightarrow T = \frac{\Delta W}{2\Delta A} = \frac{6\times10^{-4}}{60\times10^{-4}} = 0.1N/m$

6. Surface Energy = $T \times \Delta A = 75 \times 0.04 = 3$J

7. If volume of a soap bubble increasing by 700%

$\Rightarrow V_2 = V + \frac{700}{100}\times V = 8V$

$\therefore \quad V \propto r^3 ; \frac{8V}{V} = \left(\frac{r_2}{r_1}\right)^3$

$\Rightarrow \quad r_2 = 2r_1 = 2R$

$W = T\Delta A$

$\Rightarrow \quad W = T \times 2\,[4\pi(2R)^2 - 4\pi R^2]$

$\Rightarrow \quad W = 24\pi R^2 T$

8. $W = 4\pi R^2 T = 4\pi(2)^2 \times 60 = 960\pi$ erg

9. $R = n^{\frac{1}{3}} r = \left(10^6\right)^{\frac{1}{3}} r = 100\, r$

$$\% \text{ energy loss} = \left(\frac{SE_i - SE_f}{SE_i}\right) \times 100$$

$$= \left(1 - \frac{SE_f}{SE_i}\right) \times 100$$

$$= \left(1 - \frac{4\pi R^2 T}{n 4\pi r^2 T}\right) \times 100$$

$$= \left(1 - \frac{1}{n^{\frac{1}{3}}}\right) 100$$

$$= \left(1 - \frac{1}{100}\right) \times 100 = 99\%$$

10. Increase in energy $= 4\pi R^2 T\,(n^{\frac{1}{3}} - 1)$

$= 4\pi R^2 T\,(4-1) = 12\pi r^2 T$

BEGINNER'S BOX-6

1. Since angle of contact does not depend upon the angle of inclination. So, angle of contact is same (80°).

2. If liquid rises, meniscus will be concave. So, angle of contact will be acute (<90°).

3. $(P_{ex})_1 = 3(P_{ex})_2$

$$\frac{4T}{r_1} = 3\left(\frac{4T}{r_2}\right) = \frac{r_1}{r_2} = \frac{1}{3}$$

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1}{27}\right)$$

4. $P_{in} - P_{out} = \frac{2T}{r} \quad \Rightarrow P_{in} = P_{out} + \frac{2T}{r}$

$\Rightarrow P_{in} = P_0 + \rho gh + \frac{2T}{r}$

$= 10^5 + (0.8\times10^3\times10\times 25\times10^{-2}) + \frac{2\times25\times10^{-3}}{0.02\times10^{-3}}$

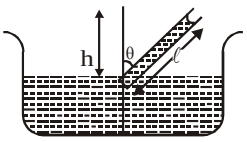
$= 10^5 + 2000 + 2500 = 1.045 \times 10^5$ Pa

5. $P_{ex} = P_{in} - P_{out} = \frac{4T}{r}$

$$\frac{(P_{ex})_1}{(P_{ex})_2} = \frac{1.01-1.00}{1.02-1.00} = \frac{1}{2} \Rightarrow \frac{r_1}{r_2} = \frac{2}{1}$$

$$\therefore \quad \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \frac{8}{1}$$

6.



h = constant

$\ell\cos\theta$ = constant

$\ell_1\cos\theta_1 = \ell_2\cos\theta_2$

$$\frac{\ell_1}{\ell_2} = \frac{\cos\theta_2}{\cos\theta_1} = \frac{\cos 60^\circ}{\cos 30^\circ}$$

$$= \frac{1}{\sqrt{3}}$$

7. $h = \frac{2T\cos\theta}{r\rho g}$

$h \propto \frac{1}{r}$

$$\frac{h_1}{h_2} = \frac{r_2}{r_1} \Rightarrow \frac{r_1}{r_2} = \frac{60}{40} = \frac{3}{2}$$

8. $h = \frac{2T\cos\theta}{rdg}$

$\Rightarrow r = \frac{2T\cos\theta}{hdg} = \frac{2\times75\times10^{-3}\times1}{0.015\times10^3\times10}$

$= 1 \times 10^{-3}$ m $= 1$ mm

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BEGINNER'S BOX-7

1. $F = \eta A\frac{dv}{dx} = 10^{-3} \times 10 \times \frac{2}{1} = 2 \times 10^{-2}$ N

2. $F = \eta A\frac{dv}{dx}$

$2 = \frac{0.01}{10} \times (1)^2 \times \frac{4}{\ell}$

$\ell = 2 \times 10^{-3}$ m = 2mm

3. $v_T = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta}$

$= \frac{2}{9} \times \frac{(0.01 \times 10^{-3})^2 \times (1000 - 1.2) \times 10}{1.8 \times 10^{-5}}$

$= 0.012$ m/s

5. $Q = \frac{dV}{dt} = \frac{\pi Pr^4}{8\eta L}$

$P \propto \frac{L}{r^4}$

$\Rightarrow \frac{P_1}{P_2} = \frac{L_1}{L_2} \times \left(\frac{r_2}{r_1}\right)^4 = \frac{\ell}{2\ell} \times \left(\frac{2r}{r}\right)^4 = \frac{8}{1}$

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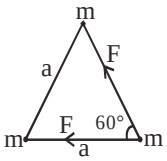
Gravitation

SOLUTIONS

Beginner's Box - 1

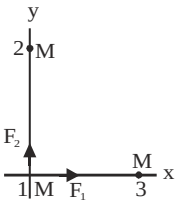
1. All forces having equal magnitude and act at the centre of square with equal angle to each other so net force on mass m_1 kept at the centre is zero.

2.



$F_{net} = \sqrt{3}\,F = \sqrt{3}\dfrac{Gm^2}{a^2}$

3. $F_1 = F_2 = \dfrac{G(1)(1)}{(0.2)^2} = \dfrac{6.67\times10^{-11}}{0.04} = 1.67\times10^{-9}$



$\vec{F}_{net} = F_1(\hat{i}) + F_2(\hat{j}) = F(\hat{i}+\hat{j}) = 1.67\times10^{-9}(\hat{i}+\hat{j})N$

Beginner's Box - 2

1. Field intensity $F = \dfrac{GMr}{R^3}$ (Inside the sphere)

$(F_1 \propto r_1, F_2 \propto r_2)$

$\dfrac{F_1}{F_2} = \dfrac{r_1}{r_2}$ if $r_1 < R$ & $r_2 < R$

Field intensity $F \propto \dfrac{1}{r^2}$ (Out side the sphere)

$\therefore \dfrac{F_1}{F_2} = \dfrac{r_2^2}{r_1^2}$ if $r_1 > R$ and $r_2 > R$

2. $t = \sqrt{\dfrac{2h}{g}} = 1$ sec; $t' = \sqrt{\dfrac{2h}{g'}} = \sqrt{6}\sqrt{\dfrac{2h}{g}} = \sqrt{6}$ sec

3. $g = \dfrac{GM}{R^2}$ $R_{earth} = 2\,R_{mars}$

$M_{earth} = 10\,M_{mars}$

$g_{mars} = \dfrac{4}{10}g_{earth}$ and $W_{mars} = \dfrac{4}{10}W_{earth} = 80N.$

4. $g' = g\left(1-\dfrac{2h}{R}\right)$; $\dfrac{\Delta g}{g} = \dfrac{2h}{R}$

$1 = 2\,\dfrac{h}{R} \Rightarrow \dfrac{h}{R} = \dfrac{1}{2}$;

$g' = g\left(1-\dfrac{d}{R}\right)$; $= \dfrac{\Delta g'}{g} = \dfrac{d}{R} = \dfrac{h}{R}$,

g decreases by 0.5%

5. $\dfrac{\Delta g}{g} = -\dfrac{2h}{R_e} \Rightarrow \dfrac{1}{100} = \dfrac{2h}{6400} \Rightarrow h = 32$ km

6. $\dfrac{\Delta g}{g} = -\dfrac{2h}{R_e} \Rightarrow \dfrac{\Delta g}{g}\times100 = \dfrac{2\times16}{6400}\times100 = 0.5\%$

7. $g' = \dfrac{g}{\left(1+\dfrac{h}{R_e}\right)^2}$

$\therefore \quad h = \dfrac{R_e}{2} \quad \Rightarrow g' = \dfrac{4g}{9} = 4.44\ m/s^2$

8. $g_{height} = g/4 = \dfrac{g}{\left(1+\dfrac{h}{R_e}\right)^2}$

$\Rightarrow h = R_e = 6400$ km

9. $g\left(1-\dfrac{2h}{R}\right) = g\left(1-\dfrac{d}{R}\right)$

$2h = d$

$h = d/2 = 50$ km

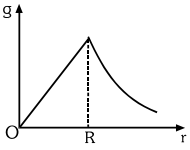
10. $g_{depth} = g_{surface}\left(1-\dfrac{d}{R_e}\right)$

$\Rightarrow \dfrac{7}{10}g = g\left(1-\dfrac{d}{R_e}\right) \Rightarrow d = \dfrac{3}{10}\times 6400$ km

$= 1920$ km

11. $\dfrac{g}{4} = g\left(1-\dfrac{d}{R_e}\right) \Rightarrow d = \dfrac{3}{4}R$

12.



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Beginner's Box - 3

1. $PE_i = -\frac{GM_e m}{R} = -mgR$;

$PE_f = -\frac{GM_e m}{2R} = -\frac{mgR}{2}$

Increase in PE is $\frac{mgR}{2}$

2. $\int dV = -\int I.dr$, $\int_{V_0}^{V} dV = \int_{r_0}^{r} \frac{K}{r} dr$

$|V|_{V_0}^{V} = K|\log r|_{r_0}^{r}$

By substituting the value c in equation

$V - V_0 = K \log\left(\frac{r}{r_0}\right)$

$V = K \log\left(\frac{r}{r_0}\right) + V_0$

3. $V = \frac{-GM_1}{r/2} - \frac{GM_2}{r/2} = \frac{-2G}{r}(M_1 + M_2)$

$V = -2200 \times 6.67 \times 10^{-11}$ J/kg

4. $I_g = -\frac{GM}{R^2}$, $V = -\frac{GM}{R}$,

$V = I_g R = 6 \times 8 \times 10^6 = 4.8 \times 10^7$

5. $\int dV = -\int I_g.dx$; $\int_{v}^{0} dV = -\int_{x}^{\infty} \frac{K}{x^3} dx$

$0 - V = K\left[+\frac{1}{2x^2}\right]_x^{\infty} \Rightarrow V = +\frac{K}{2x^2} \Rightarrow V = \frac{K}{2x^2}$

6. Equilibrium position of the neutral point from mass 'm' is :-

$= \left(\frac{\sqrt{m}}{\sqrt{m}+\sqrt{M}}\right)d$

$V_1 = \frac{-Gm_1}{r_1}$; $V_2 = \frac{-Gm_2}{r_2}$

$V_1 = \frac{-Gm}{\sqrt{m}d}(\sqrt{M}+\sqrt{m})$; $V_2 = \frac{-GM}{\sqrt{M}d}(\sqrt{M}+\sqrt{m})$

$V_1 = \frac{-G}{d}\sqrt{m}(\sqrt{M}+\sqrt{m})$;

$V_2 = \frac{-G}{d}\sqrt{M}(\sqrt{M}+\sqrt{m})$

$V = \frac{-G}{d}(\sqrt{M}+\sqrt{m})^2$

7. To escape from the earth total energy of the body should be zero KE+ PE = 0

$\frac{1}{2}mv^2 - \frac{GMm}{5R_e} = 0 \Rightarrow KE_{min} = \frac{mgR_e}{5}$

8. $PE = PE_{12} + PE_{23} + PE_{31} = -$

$\frac{Gm^2}{2R} - \frac{Gm^2}{2R} - \frac{Gm^2}{2R} \Rightarrow \boxed{PE = -\frac{3Gm^2}{2R}}$

Beginner's Box - 4

1. $T \propto r^{3/2}$

$\frac{T_e}{T_m} = \left(\frac{r_e}{r_m}\right)^{3/2} \Rightarrow T_m = \left(\frac{2}{5}\right)^{3/2}$ year

2. $T \propto r^{3/2}$

$\left(\frac{T_1}{T_2}\right) = \left(\frac{r_1}{r_2}\right)^{3/2}$ $\therefore r_2 = 2r_1$

$T_2 = 2\sqrt{2}$ year

3. $F \propto \frac{1}{r^m}$; $F = \frac{C}{r^m}$

This force will provide the required centripetal force

Therefore

$m\omega^2 r = \frac{C}{r^m}$; $\omega^2 = \frac{C}{mr^{m+1}}$

$T = \frac{2\pi}{\omega} \Rightarrow T \propto r^{(m+1)/2}$

4. By applying conservation of angular momentum

$m\omega_1 r_1^2 = m\omega_2 r_2^2$

5. For moon time period $T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$

$\Rightarrow$ mass of the earth $M = \frac{4\pi^2 r^3}{T^2 G}$

$\Rightarrow M = \frac{4\times(3.14)^2 \times(4\times10^8)^3}{(28\times24\times60\times60)^2 \times 6.67\times10^{-11}}$

$= 6.47 \times 10^{24}$ kg.

6. $\because$ angular momentum $L = mr_P v_P = mr_A v_A$

$\therefore \frac{v_P}{v_A} = \frac{r_A}{r_P}$ Since $r_A > r_P, v_P > v_A$.

The area SBAC bounded by the ellipse and the radius vectors SB and SC is larger than SBPC in Fig. From Kepler's second law, equal areas are swept in equal times. Hence the planet will take a longer time to traverse BAC than CPB.

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7. All quantities vary over an orbit except angular momentum and total energy.

8. $T \propto r^{3/2}$

$$\frac{T_s}{T_m}=\left(\frac{r_s}{r_m}\right)^{3/2}=\left(\frac{1}{2}\right)^{3/2} \Rightarrow T_s = 9.7 \text{ days}$$

9. Let the radius of the planet be R then time period

$$T=\frac{2\pi R}{v}=\frac{2\pi R}{\sqrt{\frac{GM}{R}}}=\frac{2\pi R^{3/2}}{\sqrt{GM}}$$

$\because$ mass of planet $M = \frac{4}{3}\pi R^3 \rho$

$$\therefore \quad T=\frac{2\pi R^{3/2}}{\sqrt{G.\frac{4}{3}\pi R^3.\rho}}=\sqrt{\frac{3\pi}{G\rho}}$$

10. total mechanical energy $E = -\frac{GMm}{2r}$

$$\therefore \quad \frac{E_1}{E_2}=\frac{m_1}{m_2}\times\frac{r_2}{r_1}=\frac{3}{1}\times\frac{4r}{r}=\frac{12}{1}$$

11. Since total energy of the satellite in orbit of radius r is $E = -\frac{GMm}{2(R+h)}$. Therefore energy of amount equal to E has to be expanded to take the satellite out of the gravitational pull of the earth.

$$\therefore \quad E=\frac{6.67\times10^{-11}\times6\times10^{24}\times200}{2\left(6.4\times10^{6}+400\times10^{3}\right)}$$

$$= \frac{6.67\times6}{6.8}\times10^{9} = 5.89 \times 10^9 \text{ J}$$

12. Kinetic energy increases, but potential energy decreases, and the sum decreases due to dissipation against friction.

13. $$\frac{K.E}{P.E}=\frac{\frac{GMm}{2R}}{-\frac{GMm}{R}}=-\frac{1}{2}$$

14. $W = \frac{GMm}{R^2} = 10$

$$W_{true}=\frac{GMm}{(7R)^2}=\frac{GMm}{49R^2}=\frac{10}{49}$$

$\simeq 0.2$ N $\Rightarrow$ $W_{app} = 0$

15. Here h = 2R

$$\boxed{r=R+h}$$

Necessary centripetal force is provided by gravitational force

$$\Rightarrow \quad \frac{GMm}{(3R)^2}=\frac{mv^2}{(3R)} \Rightarrow v=\sqrt{\frac{GM}{3R}}$$

By conservation of mechanical energy

$U_i + K_i = U_f + K_f$.

$$\Rightarrow \quad -\frac{GMm}{R}+K_i=-\frac{GMm}{3R}+\frac{1}{2}m\left(\sqrt{\frac{GM}{3R}}\right)^2$$

$$K_i=\frac{1}{2}\frac{GM}{3R}+\frac{GMm}{R}-\frac{GMm}{3R}=\frac{5GMm}{6R}$$

Rotational Motion

SOLUTIONS

Beginner's Box - 1

1. (a) Angular speed $\omega = \frac{v}{r} = \frac{36}{0.30} = 120$ rad/s

(b) (i) $\because \omega^2 = \omega_0^2 + 2\alpha\theta$

$\therefore 0 = (120)^2 + 2\alpha(40\times 2\pi) \Rightarrow \alpha = -\frac{90}{\pi}$ rad/s^2

(ii) distance covered during braking

= circumference of wheel × number of revolution

$= 2\pi r \times 40 = 2\pi \times 0.30 \times 40 = 24\pi$m

2. $\int_2^{\omega} d\omega = \int_0^t \alpha dt \Rightarrow \omega - 2 = (t^4 - t^3 + t^2)t_0$

$\therefore \omega - 2 = t^4 - t^3 + t^2$

$\therefore \omega = 2 + t^2 - t^3 + t^4$ rad/s

$$\int_1^{\theta} d\theta = \int_0^t \omega dt \Rightarrow \theta - 1 = \left(2t + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5}\right)_0^t$$

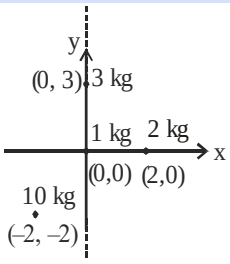
$$\theta = 1 + 2t + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5} \text{ rad}/s^2$$

4. $\omega = \omega_0 + \alpha t$

$2\pi \times \frac{600}{60} = \alpha \times 2 \Rightarrow 10\pi$ rad/s^2

Beginner's Box - 2

1.



$I = 1 \times (0)^2 + 2 \times (2)^2 + 3 \times (0)^2 + 10(2)^2$

$= 48$ kg-m^2

2. $I_{ring} = I_{disc}$

$$\frac{MR_1^2}{2} = \frac{MR_2^2}{4} \Rightarrow \frac{R_1}{R_2} = \frac{1}{\sqrt{2}}$$

3. $$\frac{I_1}{I_2} = \frac{M_1R_1^2}{M_2R_2^2} \Rightarrow \frac{4}{1} = \frac{M_1}{M_2} \times \left(\frac{4}{1}\right)^2 \Rightarrow \frac{M_1}{M_2} = \frac{1}{4}$$

4. $$I = m_1\left(\frac{L}{2}\right)^2 + m_2\left(\frac{L}{2}\right)^2$$

$= 1 \times (1)^2 + 2 \times (1)^2 = 3$ kg-m^2

5. $$I = \frac{5}{4}MR^2 \Rightarrow MR^2 = \frac{4}{5}I$$

$$I' = \frac{3}{2}MR^2 = \frac{3}{2} \times \frac{4}{5}I = \frac{6}{5}I$$

6. Moment of inertia of hollow cylinder will be larger as compare to disc because most of its mass is located away from the axis of rotation as compare to that of disc.

7. Spokes do not carry much mass. Most of the mass is located at the rim. This gives cycle wheel more moment of inertia for same mass.

8. $M_A = \pi r^2 t\rho$ and $M_B = \pi(4r)^2 \frac{t}{4}\rho \Rightarrow M_B = 4M_A$

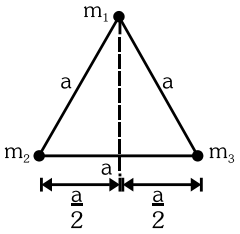
$$\therefore \frac{I_A}{I_B} = \frac{M_A r^2}{M_B \times 16r^2} = \frac{1}{64} \Rightarrow I_B = 64I_4$$

9. Mass of plate

= (Mass per unit area) (Area) = $\mu\ell^2$

Moment of inertia I = $\frac{M\ell^2}{6} = \frac{\mu\ell^2 \times \ell^2}{6} = \frac{\mu\ell^4}{6}$

10.



Moment of inertia

$$I = m_1 \times 0 + m_2 \times \left(\frac{a}{2}\right)^2 + m_3 \times \left(\frac{a}{2}\right)^2$$

$$= \frac{a^2}{4}(m_2 + m_3)$$

11. Moment of inertia $I = MR^2 + \frac{MR^2}{2}$

$$= \frac{2MR^2 + MR^2}{2} = \frac{3}{2}MR^2$$

12. It will be again = $\frac{2}{5}MR^2$, because the axis in question is also diameter of the sphere.

Beginner's Box - 3

1. $\vec{\tau}=\vec{r}\times\vec{F}$

$$\vec{\tau}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & 3\\ 1 & -2 & 5\end{vmatrix}$$

$\vec{\tau}=\hat{i}(5-(-6))-\hat{j}(10-3)+\hat{k}(-4-1)$

$\vec{\tau}=11\hat{i}-7\hat{j}-5\hat{k}\,N-m$

2. About point B moment of inertia is less than A so it is easier to rotate about point B.

3. $\tau = I\alpha$

$$2\times\frac{20}{100}=\frac{2}{2}\times\left(\frac{20}{100}\right)^2\times\alpha$$

$\alpha = 10\ rad/sec^2$

$a_t = r\alpha = \frac{20}{100}\times 10 = 2\ m/s^2$

4. $\because\ r_1F_1 = r_2F_2$

$\because\ 1.6 \times 1 = 0.4 \times F_2$

$\because\ F_2 = 4\ N$

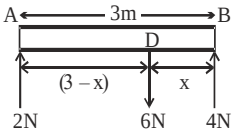
5. Net force on rod $F_{net} = 0$

$\tau_{net} = -F\times 20 + F\times 40 - F\times 60 + F\times 80$

$= 40\ F$

so rod experinces a torque

6.

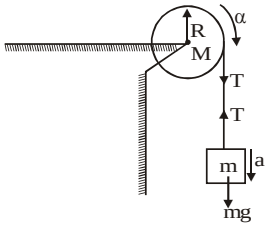


for equilibrium net torque about end B = 0

$-2(3) + 6x = 0;\ x = 1m$

7. For block : $mg - T = ma$ .....(i)

For disc (pulley) $TR = I\alpha = \frac{MR^2}{2}\alpha$



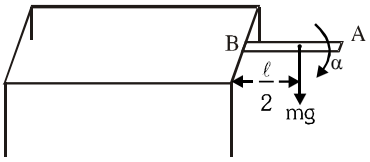
But $\alpha = \frac{a}{R}$ so $T = \frac{Ma}{2}$ ...(ii)

Therefore

$$\frac{mg-T}{T}=\frac{2m}{M}\Rightarrow\frac{mg}{T}=\frac{2m}{M}+1$$

$$\therefore T=\frac{mg}{\left(\frac{2m}{M}+1\right)}=\frac{(1.2\times10)}{\left(\frac{2\times1.2}{2.4}+1\right)}=6N$$

8.



For angular motion of the stick

$$\frac{1}{100}=mg\left(\frac{\ell}{2}\right)=Ia\,.$$

moment of intertia of stick about B is

$$I=\frac{m\ell^2}{3}\Rightarrow mg\left(\frac{\ell}{2}\right)=\left(\frac{m\ell^2}{3}\right)\alpha\Rightarrow\alpha=\frac{3g}{2\ell}$$

Acceleration of centre of mass

$$=a\left(\frac{\ell}{2}\right)=\left(\frac{3g}{2\ell}\right)\left(\frac{\ell}{2}\right)=\frac{3}{4}g$$

Beginner's Box - 4

1. $I = m_{earth}r^2 = 6\times10^{24}\times(1.5\times10^8\times10^3)^2$

$$\omega=\frac{2\pi}{T}=\frac{2\pi}{3.14\times10^7}$$

Angular momentum = $I\omega$

$= 2.7 \times 10^{40}\ kg\ m^2/s$

2. The angular momentum of particle about origin will be zero because given straight line passes through origin, so line of action of velocity (momentum) also passes through this point.

3. Moment of inertia about diametric axis

$I = \frac{1}{4}MR^2 = \frac{1}{4}(0{\cdot}2)\ (0{\cdot}04)^2 = 8\times10^{-5}\ kg\text{–}m^2$

$\therefore$ kinetic energy $= \frac{1}{2}I\omega^2$

$= \frac{1}{2}\times(8\times10^{-5})(10)^2 = 4\times10^{-3}\ J$

Angular momentum $L = I\omega = 8\times10^{-4}$ J-s

Beginner's Box - 5

1. $\because \tau = 0 \Rightarrow L = \text{constant}$

$\therefore \quad I_1\omega_1 = I_2\omega_2$

$\therefore \quad MK_1^2\omega_1 = MK_2^2\omega_2 \Rightarrow \frac{K_1}{K_2} = \sqrt{\frac{\omega_2}{\omega_1}}$

2. $\because \quad L = I\omega$

$\therefore \frac{L_1}{L_2} = \frac{I_1}{I_2} \times \frac{\omega_1}{\omega_2}$

$\therefore \frac{L}{L_2} = \frac{mr^2}{m(r/2)^2} \times \frac{\omega}{\omega} = \frac{4}{1}$

$\therefore L_2 = \frac{L}{4}$

3. $KE = \frac{1}{2}I\omega^2 = \frac{1}{2} \times \frac{m\ell^2}{3} \times (2\pi f)^2$

$= \frac{2}{3} m\ell^2\pi^2 f^2$

4. $WD = \Delta KE = \frac{1}{2}I(\omega_2^2 - \omega_1^2)$

$= \frac{1}{2} \times I\,[(5\omega)^2 - \omega^2] = 12\ I\omega^2$

$= 12 \times 10 \times \left(\frac{10 \times 2\pi}{60}\right)^2 = 131.6\ J$

5. $KE = \frac{1}{2}I\omega^2$

$K.E. = \frac{1}{2} \times \frac{2}{5} \times 1 \times \left(\frac{3}{100}\right)^2 \times (50)^2 = 0.45\ J$

6. When sand is poured on the a rotating disc its moment of inertia gets increased. Since no external torque acts on the system, hence angular momentum must conserved. As $L = I\omega$ so angular velocity will decrease.

7. $K.E. = \frac{1}{2} I\omega^2$

$\therefore\ 600 = \frac{1}{2} \times 3.0 \times \left(\frac{2\pi}{T}\right)^2$

$\therefore \left(\frac{2\pi}{T}\right)^2 = 400$

$\therefore \frac{2\pi}{T} = 20$

$\therefore\ T = 0.314$ second

8. (a) Angular acceleration $\alpha = \frac{\omega_2^2 - \omega_1^2}{2\theta}$

$= \frac{(100)^2 - (10)^2}{2 \times 100} = 49.5\ rad/s^2$

Torque $\tau = I\alpha = 1000 \times 49.5$

$= 4.95 \times 10^4$ N–m

(b) Increase in KE $= \frac{1}{2} I\,(\omega_2^2 - \omega_1^2)$

$= \frac{1}{2} \times 1000 \times [(100)^2 - (10)^2]$

$= 4.95 \times 10^6\ J$

Beginner's Box - 6

1. $\because v = \sqrt{\frac{2gs\sin\theta}{1 + K^2/R^2}}$

$\therefore v = \sqrt{\frac{2 \times 10 \times 10 \times \sin 30^\circ}{1+1}} = 7m/s$

2. $KE = \frac{1}{2} I\omega^2$

$I = \frac{2KE}{\omega^2} = \frac{2 \times 360}{30 \times 30}$

$I = 0.8\ kg\text{-}m^2$

3. $v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$

both disc rolling down from same height so,

$\frac{v_1}{v_2} = 1$

4. Translational K.E. $= \frac{1}{2} MV^2$

Total K.E. $= \frac{1}{2}MV^2 + \frac{1}{2}I\omega^2$

$= \frac{1}{2}MV^2 + \frac{1}{2} \times \frac{2}{3}MR^2 \times \frac{V^2}{R^2} = \frac{5}{6}MV^2$

Fractional part of translational kinetic energy

$= \frac{\frac{1}{2}Mv^2}{\frac{5}{6}Mv^2} \times 100 = 60\%$

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Collisions and Centre of Mass

SOLUTIONS

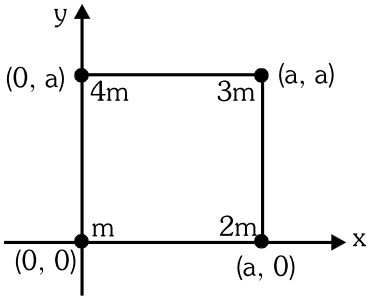
Beginner's Box - 1

1. $x = \frac{3\times 0 + 4\times 2 + 8\times 1}{3+4+8}$ m $= \frac{16}{15}$ m $= 1{\cdot}1$ m

$y = \frac{3\times 0 + 4\times 1 + 8\times 2}{3+4+8}$ m $= \frac{20}{15}$ m $= 1{\cdot}3$ m

so co-ordinates (1·1m, 1·3 m)

2.



$X_{CM} = \frac{m\times 0 + 2m\times a + 3m\times a + 4m\times 0}{m+2m+3m+4m} = \frac{5}{10}a = \frac{a}{2}$

$Y_{CM} = \frac{m\times 0 + 2m\times 0 + 3m\times a + 4m\times a}{m+2m+3m+4m} = \frac{7}{10}a$

so co-ordinates $\left(\frac{a}{2}, \frac{7}{10}a\right)$

3. Length of rod $= |\vec{r}_2 - \vec{r}_1| = \sqrt{(4-2)^2 + (2-5)^2}$

$= \sqrt{13}$m

centre of mass $\vec{r}_{CM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$

$= \frac{3}{5}(2\hat{i}+5\hat{j}) + \frac{2}{5}(4\hat{i}+2\hat{j}) = \left(\frac{14}{5}\hat{i} + \frac{19}{5}\hat{j}\right)$m

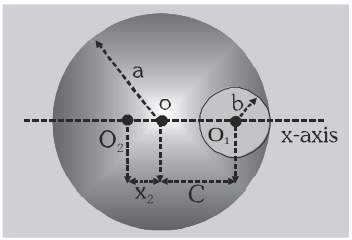
4. (a) at O (b) III quadrant (c) on OY' axis
(d) at O (e) iv quadrant (f) at O

5. Let the circular disc of radius 'a' is made up of the circular section of radius b as a remainder. Further let the line of symmetry joining the centres O and O_1 be the x-axis with O as origin. The centre of mass of the disc of radius 'a' will be given by

$$X_{CM} = \frac{m_1x_1 + m_2x_2}{m_1+m_2} \qquad ...(1)$$

while Y_{CM} and Z_{CM} will be zero (as for all points on x-axis y and z = 0).

If σ is the density of the material of disc.



$m_1 = \pi b^2\sigma$ and $x_1 = c$

$m_2 = \pi(a^2 - b^2)\sigma$ and $x_2 = ?$

$M = m_1 + m_2 = \pi a^2\sigma$ and $X_{CM} = 0$

So substituting these values in equation (1)

$$0 = \frac{\pi b^2\sigma c + \pi(a^2-b^2)\sigma x_2}{\pi a^2\sigma}$$

$$x_2 = \frac{-cb^2}{(a^2-b^2)} \qquad ...(2)$$

i.e., centre of mass of the remainder (say O_2) is at a distance $cb^2/(a^2 - b^2)$ to the left of O on the line joining the centres O and O_1.

Beginner's Box - 2

1. $\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$

$= \frac{1\times 2\hat{i} + 2\times(2\cos 30\hat{i} - 2\sin 30\hat{j})}{3}$

$= \left(\frac{2+2\sqrt{3}}{3}\right)\hat{i} - \frac{2}{3}\hat{j}$

2. $\vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$

$= \frac{10(2\hat{i}-7\hat{j}+3\hat{k}) + 2(-10\hat{i}+35\hat{j}-3\hat{k})}{12}$

$= \frac{24}{12}\hat{k} = 2\hat{k}$ m/s

3. $v_{cm} = \frac{m_1v_1 + m_2v_2}{m_1+m_2} = \frac{5(14)+2(0)}{5+2}$

$= \frac{70}{7} = 10$ m/s

4. $\vec{a}_{CM} = \dfrac{m_1\vec{a}_1 + m_2\vec{a}_2 + m_3\vec{a}_3}{m_1 + m_2 + m_3}$

$= \dfrac{(3\hat{i} - 2\hat{j} + 2\hat{k}) + (-\hat{i} + 2\hat{j} - \hat{k}) + (\hat{i} + \hat{j} + \hat{k})}{1 + 2 + 3}$

$= \dfrac{3\hat{i} + \hat{j} + 2\hat{k}}{6}$

$|\vec{a}_{CM}| = \dfrac{\sqrt{14}}{6}\, m/s^2$

5. Let x = shift of plank to the right

$\Delta\vec{x} = \dfrac{m_A\Delta\vec{x}_A + m_B\Delta\vec{x}_B + m_c\Delta\vec{x}_c + m_p\Delta\vec{x}_p}{m_A + m_B + m_C + m_P}$

$0 = \dfrac{40(x+4) + 50x + 60(x-4) + 90x}{40 + 50 + 60 + 90}$

$\Rightarrow x = \dfrac{1}{3}$ m (Right)

6. $\Delta\vec{x} = \dfrac{m_1.\Delta\vec{x}_1 + m_2.\Delta\vec{x}_2}{m_1 + m_2}$

$\Rightarrow 0 = \dfrac{m_1 a + m_2\Delta x_2}{m_1 + m_2}$

$\Rightarrow \Delta x_2 = -\dfrac{m_1 a}{m_2}$

7. CM remains at rest if initially it is at rest.

8. $\Delta\vec{y} = \dfrac{m_1\Delta\vec{y}_1 + m_2\Delta\vec{y}_2}{m_1 + m_2}$

$\Rightarrow 0 = \dfrac{m}{4}(15) + \dfrac{3m}{4}(y_2) \Rightarrow y_2 = -5$ cm

Beginner's Box - 3

1. $v_1 = \dfrac{m_1 - m_2}{m_1 + m_2}u_1$

$\Rightarrow \quad \dfrac{u}{4} = \dfrac{2-m}{2+m} \times u$

$\Rightarrow \quad (2 + m) = 4(2 - m)$

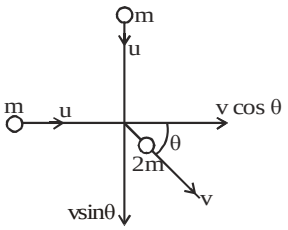
$\Rightarrow \quad m = 1.2$ kg

2. For e = 1 and same mass velocity will exchange after collision.

3. $m_1v_1 + m_2v_2 = (m_1 + m_2)\, v_{system}$

$v_{system} = \dfrac{mv + 2m \times 0}{m + 2m} = \dfrac{v}{3}$

4.



COLM in x-axis

$m \times 45\sqrt{2} = 2mv\cos\theta$

$v\cos\theta = \dfrac{45}{\sqrt{2}}$ ....(1)

COLM in y-axis

$m \times 45\sqrt{2} = 2\, mv \sin\theta$

$v\sin\theta = \dfrac{45}{\sqrt{2}}$ ....(2)

from equation (1) & (2)

$v^2\cos^2\theta + v^2\sin^2\theta = \left(\dfrac{45}{\sqrt{2}}\right)^2 + \left(\dfrac{45}{\sqrt{2}}\right)^2$

$v = 45$ m/s

5. $v_{system} = \dfrac{m_1v_1 - m_2v_2}{m_1 + m_2} = \dfrac{2\times3 - 1\times4}{3} = \dfrac{2}{3}$

6. m = 1kg, h = 20m, e = 0.5

(i) $v_2 = e^2u = \left(\dfrac{1}{2}\right)^2 \times \sqrt{2g\times20} = 5m/s$

(ii) $h_2 = e^4h = \left(\dfrac{1}{2}\right)^4 \times 20 = \dfrac{20}{16} = \dfrac{5}{4}$ m

(iii) $v_{avg} = \dfrac{\text{total distance}}{\text{total time}}$

$= \sqrt{\dfrac{gh}{2}}\left[\dfrac{1+e^2}{(1+e)^2}\right] = \dfrac{50}{9}\, m/s$

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7. $u = 40$ m/s, $e = 0.5 = \frac{1}{2}$

travelled total distance

$S = 2h + 2h_1 + 2h_2 + \ldots\ldots + \infty$

$S = \left(\frac{2}{1-e^2}\right)h = \left(\frac{2}{1-\frac{1}{4}}\right)\left(\frac{40\times 40}{2\times 10}\right) = 213.33$ m

Total time

$T = 2t + 2t_1 + 2t_2 + \ldots\ldots\ldots$

$\Rightarrow \quad T = \left(\frac{1}{1-e}\right)2t$

$= \left(\frac{1}{1-\frac{1}{2}}\right)2\times\left(\frac{40}{10}\right) = 16$ sec.

8. $S = h + 2h_1 + 2h_2 + \ldots\ldots$

$e = 0.2 = \frac{1}{5}$

$S = h\left(\frac{1+e^2}{1-e^2}\right) = h\left[\frac{1+\frac{1}{25}}{1-\frac{1}{25}}\right] = \frac{13}{12}h$

9. $v_1 = \frac{m_1 - em_2}{m_1+m_2}u_1 + \frac{(1+e)m_2}{m_1+m_2}u_2$

$v_2 = \frac{(1+e)m_1}{m_1+m_2}u_1 + \frac{m_2 - em_1}{m_1+m_2}u_2$

$u_1 = 6$ m/s $\qquad u_2 = -6$ m/s

$m_1 = m_2 \qquad e = \frac{1}{3}$

$\therefore \quad v_1 = \frac{(1-e)}{2}u_1 + \frac{(1+e)}{2}u_2$

$\Rightarrow \quad v_1 = -2$ m/s $\qquad \Rightarrow \quad v_2 = 2$ m/s

10. In the one dimensional elastic collision with one body at rest, the body moving initially comes to rest & the one which was at rest earlier starts moving with the velocity that first body had before collision.

so, if m & v_0 be the mass & velocity of body,

the change in momentum = $mv_0 \Rightarrow \int F dt =$

$mv_0 \Rightarrow \int F dt = mv_0 \Rightarrow F = \frac{mv_0}{\Delta t} = 2N$

11. Using relative motion, the time of collision is

$t = \frac{h}{20+0} = 1$s

using equation $v = u + at$, for $t = 1$

velocity of particle A & B is $10\hat{j}$ and $-10\hat{j}$ respectively at the time of collision.

By conservation of momentum for collision

$3\left(-10\hat{j}\right) + 1\left(10\hat{j}\right) = 4\vec{v}_f \Rightarrow v_f = 5$ m/s

$KE = \frac{1}{2}(4)(5)^2 = 50$J

12. (a) $\vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2} = \frac{2(5\hat{i}) + 3(-2\hat{i})}{2+3}$

$= 0.8\,\hat{i}$ m/s

(b) By momentum conservation law

$(2)\,(5\hat{i}) + 3(-2\hat{i}) = (2)\,(-1.6\hat{i}) + 3(\vec{v}_2)$

$\Rightarrow$ velocity of 2nd particle $\vec{v}_2 = 2.4\hat{i}$ m/s

(c) $e = \frac{|\vec{v}_2 - \vec{v}_1|}{|\vec{u}_1 - \vec{u}_2|} = \frac{2.4+1.6}{5+2} = \frac{4}{7}$

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Circular Motion

SOLUTIONS

BEGINNER'S BOX-1

1. Given $\omega = \theta^2 + 2\theta$

$$\frac{d\omega}{d\theta} = 2\theta + 2$$

$$\alpha = \omega\frac{d\omega}{d\theta} = (\theta^2 + 2\theta)(2\theta + 2)$$

at $\theta = 1$

$\alpha = 12$ rad/sec^2

2. $\ell = 6$cm, v = ?

$$\omega = \frac{2\pi}{60} = \frac{\pi}{30} \text{ rad/s.}$$

$$\text{So } v = \omega\ell = \frac{\pi}{30}\times 6 = \frac{\pi}{5} \text{ cm/s} = 2\pi \text{ mm/s}$$

$$\text{Magnitude of difference} = \sqrt{2}\frac{\pi}{5} \text{ cm/s}$$

$= 2\sqrt{2}\,\pi$ mm/s

3. v = 2t

At t = 3, v = 6m/s; r = 6m

$$\text{So } a_{cp} = \frac{v^2}{r} = 6\text{m/s}^2$$

4. Since the particles maintain straight line at centre

so their relative angular displacement as well as relative angular velocity both are zero. Hence

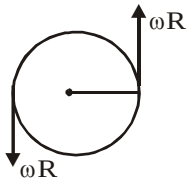
$\omega_1 : \omega_2 = 1 : 1$

6. $$\omega = \frac{14\times 2\pi}{25}$$

$\therefore$ magnitude of acceleration $= |\vec{a}_R|$

$$|\vec{a}_R| = \omega^2 r = \left(\frac{14\times 2\pi}{25}\right)^2 \times \frac{80}{100} \simeq 9.9 \text{ m/s}^2$$

7. ω = constant, $a_T = 0$



$$\left[\omega = \frac{2\pi}{T}, \frac{T}{2} = \frac{\pi}{\omega}\right]$$

$$a_{av} = \frac{2\omega R}{\pi/\omega} = \frac{2\omega^2 R}{\pi}$$

$a_{inst} = \omega^2 R$

$$\text{So ratio} = \frac{a_{av}}{a_{inst}} = \frac{2}{\pi}$$

8. $\vec{a_t} = 6\hat{i}$

$\vec{\alpha}\times\vec{R} = 6\hat{i}$

$\vec{\alpha}\times 2\hat{j} = 6\hat{i}$

$\vec{\alpha} = -3\hat{k}$ rad/s^2

$\vec{a_r} = -8\hat{j}$

$\vec{\omega}\times\vec{v} = -8\hat{j}$

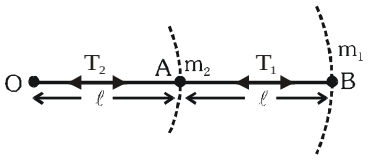
$\vec{\omega}\times\vec{v} = -8\hat{j}$

$\vec{\omega}\times(\omega R)\hat{i} = -8\hat{j}$

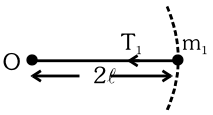
$\Rightarrow \vec{\omega} = -2\hat{k}$ rad/s

BEGINNER'S BOX-2

1. Suppose OA = AB = ℓ



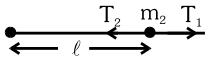
For mass m_1



$T_1 = m_1\omega^2(2\ell)$

For mass m_2

$T_2 - T_1 = m_2\omega^2\ell$



$\Rightarrow T_2 = m_2\omega^2\ell + m_1\omega^2 2\ell$

$$\frac{T_1}{T_2} = \frac{m_1\omega^2(2\ell)}{m_2\omega^2\ell + m_1\omega^2 2\ell} = \frac{2m_1}{m_2 + 2m_1}$$

2. Here $\sin\theta = \frac{1.28}{8} = 0.16$

since θ = is very small

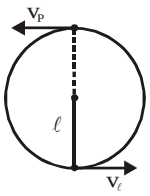
$\sin\theta \approx \theta \approx \tan\theta$

So $\tan\theta = 0.16$ & R = 40 m

Most appropriate velocity = $\sqrt{Rg\tan\theta}$ = 8m/s

3. $\because \quad \frac{T_{max}}{T_{min}} = 3 \quad \therefore \quad \frac{\frac{mv_\ell^2}{\ell} + mg}{\frac{mv_p^2}{\ell} - mg} = 3$

$\Rightarrow \quad \frac{v_\ell^2 + g\ell}{v_p^2 - g\ell} = 3$



velocity at minimum position

By COME

$v_\ell^2 = v_p^2 + 4g\ell \Rightarrow \frac{v_P^2 + 5g\ell}{v_P^2 - g\ell} = 3 \Rightarrow 2v_P^2 = 8g\ell$

$\Rightarrow v_P = \sqrt{4g\ell} = \sqrt{4\times10\times10} = 20$ m/s,

$v_\ell = \sqrt{8g\ell} = \sqrt{8\times10\times10} = 20\sqrt{2}$ m/s

4. (a) $v_D^2 = v_C^2 - 2gR = 5gR$

$\Rightarrow v_D = \sqrt{5gR}$

(b) $mg + N_D = \frac{mv_D^2}{R}$

$\Rightarrow N_D = \frac{m(5gR)}{R} - mg = 4mg$

(c) by energy conservation between point A & C

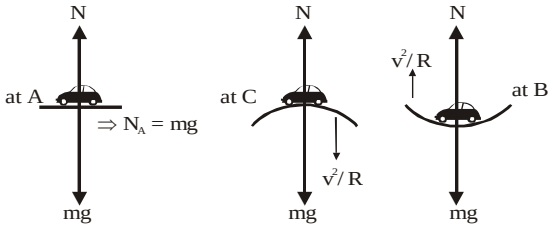
$mgH = \frac{1}{2}mv_C^2 + mgR$

$= \frac{1}{2}mv_D^2 + mg2R$

$= \frac{1}{2}m(5gR) + mg2R$

$= \frac{9}{2}mgR \Rightarrow H = \frac{9}{2}R$

5.



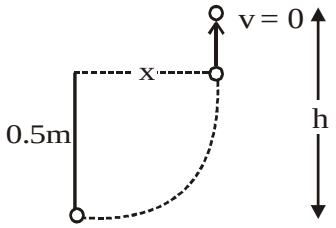
At C, $N = mg - mv^2/R$ & at B, $N = mg + mv^2/R$

$\therefore f_{max} = \mu_S N \rightarrow$ maximum for B

6. $T = mg + \frac{mv^2}{\ell}$

m = 0.5 kg, ℓ = 0.5 m & T = 45 N

Height upto which the stone will rise h = $\frac{v^2}{2g} = 2m$



So height upto which the stone will rise after the string breaks = h – 0.5 = 1.5 m

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Work, Energy & Power

SOLUTIONS

BEGINNER'S BOX-1

1. $W = Fd\cos\theta \Rightarrow \cos\theta = \dfrac{W}{Fd}$

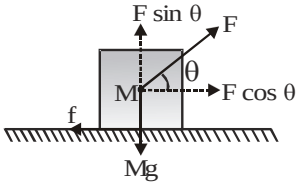
$W = 25$ J, $F = 5$ N & $d = 10$ m

so, $\cos\theta = \dfrac{25}{5\times10} = \dfrac{1}{2} \Rightarrow \theta = 60°$

2. Work has to be done against gravity in second case.

3. From the figure $F\sin\theta + N = Mg$

$\therefore$ $N = Mg - F\sin\theta$



$F\cos\theta = f = \mu N = \mu[Mg - F\sin\theta]$

$F(\cos\theta + \mu\sin\theta) = \mu Mg$

$\therefore F = \dfrac{\mu Mg}{\cos\theta + \mu\sin\theta}$ = force required to pull an object

Work done in pulling an object

$W = Fd = \dfrac{\mu Mgd\cos\theta}{\cos\theta + \mu\sin\theta}$

4. $W = \int_{x_1}^{x_2} F\,dx = \int_{x_1}^{x_2} 2x\,dx$

$W = x_2^2 - x_1^2$

5. $W = \int_0^2 F\,dx = \int_0^2 (10 + 0.5x)\,dx$

$W = 10[x]_0^2 + 0.5\left[\dfrac{x^2}{2}\right]_0^2$

$W = 21$ J

6. $v = ax^{3/2}$

$\Rightarrow \dfrac{dv}{dx} = \dfrac{3}{2}ax^{1/2}$

So, $a = \dfrac{v\,dv}{dx} = \dfrac{3}{2}a^2x^2$

$F = ma = \dfrac{3}{4}a^2x^2$

$W = \int_0^2 F\,dx = \dfrac{3}{4}a^2\int_0^2 x^2\,dx$

$W = 50$ J

7. $W = \vec{F}.\vec{x}$

$\vec{F} = 4\hat{i} + \hat{j} + 3\hat{k}$ N & $\vec{x} = \vec{r}_2 - \vec{r}_1$

$\vec{x} = 11\hat{i} + 11\hat{j} + 15\hat{k}$

So, $W = 100$ J

8. $W_1 = \int_0^a F_x\,dx = \int_0^a Kx\,dx = \dfrac{Ka^2}{2}$

$W_2 = \int_0^a Fy\,dy = \int_0^a Ky\,dy = \dfrac{Ka^2}{2}$

$W = W_1 + W_2 = \dfrac{Ka^2}{2} + \dfrac{Ka^2}{2} = Ka^2$

9. Resultant of three given forces –

$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$

$= 3\hat{i} + 9\hat{j} + 3\hat{k} + \left[-\dfrac{10}{7}\hat{i} + \dfrac{45}{7}\hat{k}\right] +$

$[22\hat{i} + 11\hat{j} + 66\hat{k}]$

$= \left[\dfrac{165}{7}\hat{i} + 20\hat{j} + \dfrac{528}{7}\hat{k}\right]$ units

Displacement $\vec{r} = \vec{r}_2 - \vec{r}_1$

$= [11\hat{i} + 6\hat{j} + 8\hat{k}] - [4\hat{i} - 1\hat{j} + 1\hat{k}]$

$= (7\hat{i} + 7\hat{j} + 7\hat{k})$ units

work done $= \vec{F}\,.\,\vec{r}$

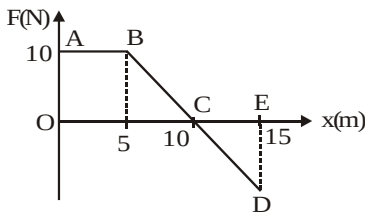
$= \left[\dfrac{165}{7}\hat{i} + 20\hat{j} + \dfrac{528}{7}\hat{k}\right] \cdot [7\hat{i} + 7\hat{j} + 7\hat{k}]$

$= 165 + 140 + 528 = 833$ unit

10. In rectilinear motion work done by a force equals to area between the force-position graph and the position axis

(a) $W_{0\to10}$ = Area of trapazium OABC = 75 J

(b) $W_{10\to15}$ = –Area of triangle CDE = –25 J



(c) $W_{0\to15}$ = Area of trapazium OABC – Area of triangle CDE = 50 J

BEGINNER'S BOX-2

1. $K = \frac{p^2}{2m} \Rightarrow K \propto p^2$

$\frac{K_f}{K_i} = \frac{p_f^2}{p_i^2} = (1.5)^2 = 2.25$

$K_f = 2.25\ K_i$

$\frac{\Delta K}{K_i} \times 100 = \frac{K_f - K_i}{K_i} \times 100 = 125\%$

So, kinetic energy increases by 125%

2. Let p = x so K = 3x

Now $K = \frac{1}{2}mv^2 = \frac{1}{2}p.v$

$\Rightarrow v = \frac{2K}{p} = 6$ units

3. According to work energy theorem

$W_{gravity} + W_{air} = \Delta KE \Rightarrow mgh + W_{air} = \frac{1}{2}mv^2 - 0$

$10 \times [10] \times [20] + W_{air} = 500$

Work done by air on object $W_{air} = -1500$ J

4. $\because x = \frac{t^3}{3} \quad \therefore v = \frac{dx}{dt} = t^2,$

Velocity at t = 0, u=0 and at t=1s v=1 m/s

Using work energy theorem :

$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}1(1)^2 = 0.5$ J

BEGINNER'S BOX-3

1. (A) – p, r, s, (B) – t

2. **For (A) :** $p = \sqrt{2mK}$ if K ↑ then p ↑

For (B) : It is not necessary that height continuously decrease.

For (C) : W = ΔK, if ΔK = positive then W = positive

For (D) : The resultant force on the particle must be at an angle less than 90° all times

3. Mechanical energy = kinetic energy + potential energy

$E = K + U(x)$ where $K = \frac{1}{2}mv^2$

If K = 0 then E = U(x)

If F = 0 then $F = -\frac{dU(x)}{dx} = 0 \Rightarrow \frac{dU(x)}{dx} = 0$

4. (A) – q, (B) – r, (C) – p

5. $W_C + W_{nc} + W_{ext} = \Delta K$

$mgh - fs + 0 = 0 \Rightarrow mgh - \mu mgs = 0$

$s = \frac{h}{\mu} = \frac{1}{0.2} = 5$m

BEGINNER'S BOX-4

1. $E = \frac{1}{2}Kx^2$

If E = constant then $x \propto \frac{1}{\sqrt{K}}$

So $\frac{F_1}{F_2} = \frac{K_1}{K_2}.\frac{x_1}{x_2} = \frac{K_1}{K_2}\sqrt{\frac{K_2}{K_1}}$

$\frac{F_1}{F_2} = \sqrt{\frac{K_1}{K_2}}$

2. In 1st pm compression is x_m

Using law of conservation of mechanical energy

$mg(h + x_m) = \frac{1}{2}kx_m^2$

$20(4 + x_m) = \frac{1}{2} \times 1960\ x_m^2$

$980\ x_m^2 - 20\ x_m - 80 = 0$

$49x_m^2 - x_m - 4 = 0$

4. By applying work energy theorem

$\frac{1}{2}m\frac{v^2}{4} - \frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$\frac{-3mv^2}{8} = \frac{-1}{2}kx^2;\ k = \frac{3mv^2}{4x^2}$

5. From COME $\Rightarrow K_1 + U_1 = K_2 + U_2$

$0 + \frac{1}{2}k_1x^2 + \frac{1}{2}k_2x^2$

$= \frac{1}{2}mv^2 + \frac{1}{2}k_1\left(\frac{x}{2}\right)^2 + \frac{1}{2}k_2\left(\frac{x}{2}\right)^2$

$\frac{1}{2}(k_1 + k_2)x^2 = \frac{1}{2}mv^2 + \frac{1}{8}(k_1 + k_2)x^2$

$v = \sqrt{\frac{3}{4}\frac{(k_1 + k_2)x^2}{m}}$

AIM: FREE EDUCATION TO ALL APUL

6. Output power of motor

$= \frac{mgh}{t} = \frac{(30\times1000)\times9.8\times40}{15\times60}$

$\because$ % efficiency $= \frac{\text{Output power of motor}}{\text{Power consumed by motor}}$

$\Rightarrow 30 = \frac{30\times1000\times9.8\times40}{15\times60\times P}$

$\Rightarrow P = \frac{9.8\times1000\times40}{15\times60} = 43.55 \times 10^3\,W$

$= 43.6$ kW

7. Output power of pump

$P = \frac{mgh}{t} = \frac{100\times10\times10}{5}$

$P_{output} = 2$ kW

therefore, $P_{input} = \frac{P_{output}}{\eta} = \frac{2}{0.6} = 3.33$ kW

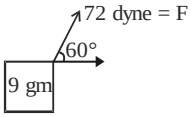
AIM: FREE EDUCATION TO ALL APUL

Laws of Motion and Friction

SOLUTIONS

Beginner's Box - 1

1.



In horizontal direction

$\vec{F} = m\vec{a}$

$F\cos 60° = 9a$

$72 \times \frac{1}{2} = 9a;\ a = 4\ cm/s^2$

2. $F = ma$

$-50 = 20 \times a$

$a = \frac{-5}{2} ms^{-2}$

$v = u + at$

$15 = \frac{5}{2} t;\ t = 6\ sec$

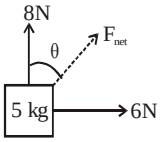
3. $v = u + at$ $\qquad 3.5 = 2 + 25\ a$

$a = \frac{3}{50} ms^{-2}$

$F = ma$ $\qquad F = \frac{3\times3}{50} = 0.18\ N$

In the direction of motion of body

4.



$F_{net} = \sqrt{8^2 + 6^2} = 10\ N$

$F_{net} = ma \Rightarrow 10 = 5a$

$a = 2\ m/s^2$, $\cos\theta = \frac{4}{5}$, $\theta = 37°$ from 8 N

5. $v_i = \sqrt{2\times9.8\times10} = 14 m/s$

$v_f = \sqrt{2\times4.9\times10} = 7\sqrt{2}\ m/s$

v_i = velocity before striking the ground

v_f = velocity after striking the ground

$\vec{I} = \Delta\vec{P} = \vec{P}_f - \vec{P}_i$

$I = 1\times[7\sqrt{2} - (-14)]$

$I = 14 + 7\sqrt{2} = 23.898\ Ns$

$\langle F \rangle = \frac{I}{T} = 238.98\ N$

6. $F = n \times mV$

$200 = n \times \frac{50}{1000} \times 1000$

$n = 4$ bullets/sec

$= (4 \times 60) = 240$ bullets/min

7. Force applied by bullet on gun = Rate of change of momentum of bullet

60 sec $\rightarrow$ 400 bullet

1 bullet = $\frac{60}{400}$ sec.

$t = \frac{60}{400}$ s, m = mass of bullet

$F = \frac{m(v-u)}{t} = \frac{35}{1000} \times \frac{400}{60} \times 400 = 93.33N$

Force required = Force on gun= 93.33 N

8. $I = F.t$; $I = 10 \times 3 \times 10^{-6} = 3 \times 10^{-5}$ N-s

$I = \Delta P \Rightarrow 3 \times 10^{-5} = 5 \times 10^{-3}\ \Delta v$

$\Delta v = \frac{3}{5} \times 10^{-2} = 6 \times 10^{-3}$ m/s

9. $F = ma$

$\frac{6}{10} = \frac{1}{4} \times a$

$2.4 = a$ (Retardation)

$v = u + at$

$u = at$

$12 = \frac{24}{10} \times t \Rightarrow t = 5sec$

$I = F \times t = 3$ N-s

10. $v_f = v_i = 15$ m/s $\qquad I = m\ (\vec{v}_f - \vec{v}_i)$

$I = m\sqrt{v_f^2 + (v_i)^2}$

$I = \frac{15}{100} \times 15\sqrt{2} = 3.1815N - s$

11. Impulse = $\int F.dt$ =

Area of F – t graph b/w 4 μs to 16 μs

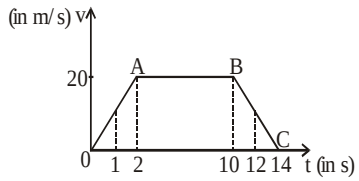
$= \frac{1}{2}(200+800)\times2\times10^{-6} + \frac{1}{2}\times800\times10\times10^{-6}$

$= 10^{-3} + 4 \times 10^{-3}$

$= 5 \times 10^{-3}$ Ns

Beginner's Box - 2

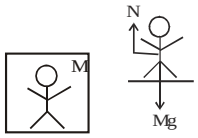
1.



Acceleration of lift from O to A = 10 m/s² = g

Acceleration of lift from B to C = –5 m/s² = $-\frac{g}{2}$

Weight of man = normal force exerted by the ground



(i) At t = 1 sec

a = 10 m/s²

N – Mg = Mg

N = 2Mg

W = 2Mg = 20 M

(ii) at t = 8 sec.

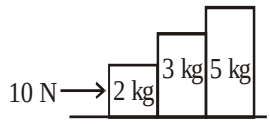
a = 0 N = Mg

N = Mg = 10 M

(iii) at t = 12

a = – 5 m/s², $Mg - N = \frac{Mg}{2}$; N = 5M

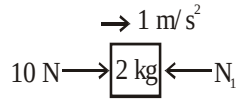
2.



acceleration will be same for all the 3 blocks so it can be considered a single block of mass 10 kg (2 + 3 + 5)

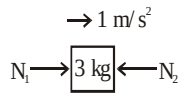
F = ma; 10 = 10a; a = 1 m/s²

(i)



$10 - N_1 = 2$; $N_1 = 8$ N

(ii)

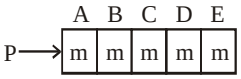


$N_1 - N_2 = 3$

$8 - N_2 = 3$

$N_2 = 5$ N

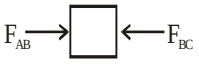
3.



$P = 5ma$; $a = \frac{P}{5m}$

$P - F_{AB} = \frac{P}{5m} \times m$ $\qquad$ $F_{DE} = \frac{P}{5m} \times m$

$\frac{4P}{5} = F_{AB}$ $\qquad$ $F_{DE} = \frac{P}{5}$

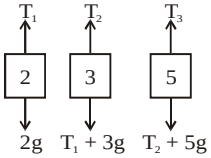


$F_{AB} - F_{BC} = P/5 \Rightarrow F_{BC} = \frac{4P}{5} - \frac{P}{5} = \frac{3P}{5}$

$F_{CD} = \frac{P}{5m} \times 2m = \frac{2P}{5}$

Beginner's Box - 3

1.



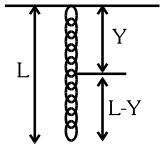
$T_1 = 2g$

$T_2 = T_1 + 3g \Rightarrow T_2 = 5g$

$T_3 = T_2 + 5g \Rightarrow T_3 = 10g$

$T_1 : T_2 : T_3 = 2 : 5 : 10$

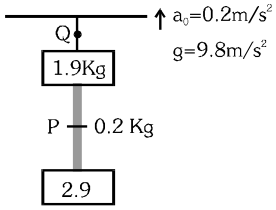
2.



$T = \frac{M}{L}(L - Y)g$

$$\begin{pmatrix} L\text{length} & \rightarrow & \text{mass}\,M \\ 1\,\text{Length} & \rightarrow & \frac{M}{L} \\ \text{Mass of}\,(L-Y) & = & \frac{M}{L}(L-Y) \\ \text{length} & & \end{pmatrix}$$

3.

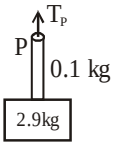


$T_Q - (1.9 + 0.2 + 2.9)g = (1.9 + 0.2 + 2.9)\,a_0$

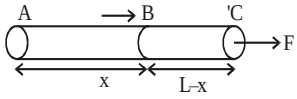
$T_Q = 5 \times 10$

$T_Q = 50$ N

$T_P - 3g = 3a_0$; $T_P = 30$ N



4.



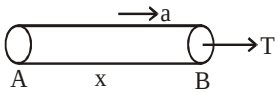
$\text{mass} = \frac{Mx}{L}$

$F = ma$

$a = \frac{F}{M}$

$T = \frac{Mx}{L} \times \left(\frac{F}{M}\right)$

FBD of AB



$T = \frac{Fx}{L}$

5. $F_1 - F_2 = Ma$

$a = \frac{F_1 - F_2}{M}$



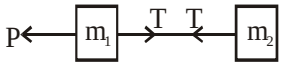
$F_1 - T = \frac{Mx}{L} \times \frac{(F_1 - F_2)}{M}$

$T = F_1 + (F_2 - F_1)\frac{x}{L}$

6. $-T + Mg = \frac{M}{2}g$

$T = \frac{Mg}{2}$; $T = 25$ N

7.



$P = (m_1 + m_2)a$

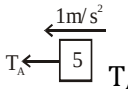
$a = \frac{P}{m_1 + m_2}$

$T = m_2 a$

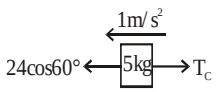
$T = \frac{Pm_2}{(m_1 + m_2)}$

8. $24 \cos 60° = (5 + 5 + 2)a$

$a = 1\ m/s^2$



$T_A = 5 \times a \Rightarrow T_A = 5N$



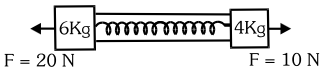
$24 \cos 60° - T_C = 5 \times a$

$T_C = 7$ N

$T_B - T_A = 1a$

$T_B = 6N$

9.



$20 - 10 = (6 + 4)a$

$a = 1\ m/s^2$

$20 - T = 6 \times 1$

$T = 14$ N

Reading = 14 N

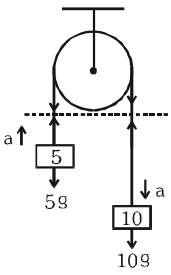
Beginner's Box - 4

1. $a = \frac{(10-5)g}{10+5}$

$a = \frac{g}{3}$

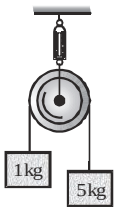
$T_s = \frac{2 \times 5 \times 10}{5+10} g$

$= \frac{20}{3} g$



$T_P = 2T_s = \frac{40}{3} g$

2.



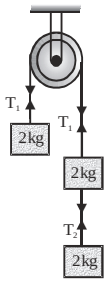
$F = 2T$; $T = \frac{2 \times 5 \times 1}{5+1} g = \left(\frac{10}{6} g\right) N$

$F = \left(\frac{10}{3} g\right) N$ Reading = $\frac{10}{3}$ kg-wt

3. $a_{block} \frac{(2+2-2)}{2+2+2} g = g/3$

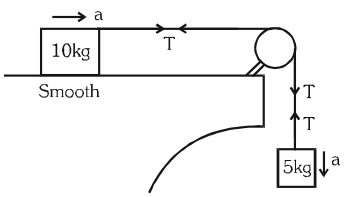
$T_1 - 2g = \frac{2g}{3}$

$T_1 = \frac{8g}{3}$



$2g - T_2 = \frac{2g}{3}$; $T_2 = \frac{4g}{3}$ N

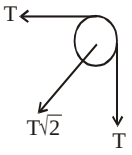
4.



$a = \frac{5g}{10+5} = \frac{g}{3}$

$T = 10 \times \frac{g}{3}$

$T = \frac{10g}{3}$

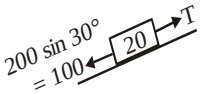


Thrust of pulley = $T_P = \sqrt{T^2 + T^2}$

$T_P = \frac{10\sqrt{2}g}{3}$

5. $\therefore$ 100 – T = 20a ...(i)

$\therefore$ T – 50 = 5a ...(ii)





(i) + (ii)

$\Rightarrow$ 50 = 20a + 5a

$\Rightarrow$ $a = \frac{50}{25} = 2\ m/s^2$

Substituting in (ii)

T – 50 = 5 × 2

T = 50 + 10 = 60 N

6. F = 2T = 100 N

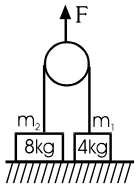
T = 50 N

T – 40 = 4a

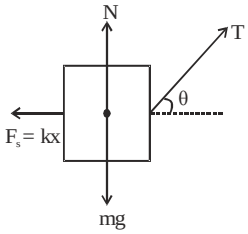
50 – 40 = 4a

$a = \frac{10}{4} = 2.5\ m/s^2$

a of 8 kg block = 0



7. (a)



(b) N should be zero

T sinθ = mg

T = mg cosecθ

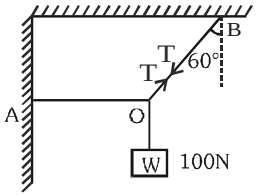
(c) T cos θ = kx

$mg = \frac{\cos\theta}{\sin\theta} kx$

$x = \frac{mg\cot\theta}{k}$

Beginner's Box - 5

1.



T_{OW} = 100 N

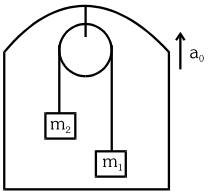
T_{OB} cos 60 = T_{OW}

T_{OB} = 200 N

$T_{OA} = T_{OB}$ sin 60°

T_{OA} = 100N

2. In the frame of lift



a = acc. in lift frame

$a = \frac{(m_1 - m_2)}{m_1 + m_2} g_{eff}$ & $T = \frac{2m_1m_2}{m_1 + m_2} g_{eff}$

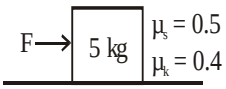
$g_{eff} = g + a_0$

$a = \frac{(m_1 - m_2)}{m_1 + m_2}(g + a_0)$

$T = \frac{2m_1m_2}{m_1 + m_2}(g + a_0)$

Beginner's Box - 6

1. Limiting Friction = 25 N



(i) when F = 15 N.

f = F = 15 N

(ii) when F = 25 N ($\because f_L = \mu_s mg$

f = F = 25 N = 0.5 × 5 × 10 = 25 N)

(iii) when F = 35 N

f = μ_k × mg ($\therefore$ slipping takes place) = 20 N

2. F sin 30° + N = Mg

μN = F cos 30°

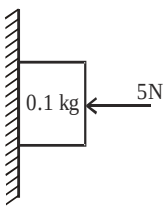
$N = Mg - \frac{F}{2}$

$\frac{1}{\sqrt{3}}(Mg - \frac{F}{2}) = \frac{\sqrt{3}F}{2}$

Mg = 2F

F = 25 N

3.



$f_L = \mu N$

= 0.5 × 5

= 2.5 N

F = mg

F = 1N

(< f_L= ie 2.5 N)

So, f = F = 1 N

4. $v^2 = u^2 + 2as$

$8 \times 8 = 2 \times 40 \times a$

$a = 0.8\ m/s^2$

$\because \quad \mu N = ma$

$\&\quad N = mg$

$\mu = \frac{a}{g} \Rightarrow \mu = 0.0816$

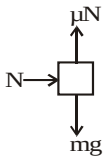
5. $a = \left(\frac{30 - 20\mu g}{20 + 3}\right)$ $\qquad a = \frac{22}{23}\ m/s^2$

$30 - T = 3a$

$30 - T = \frac{66}{23}$

$T = \frac{626}{23} = 27.13\ N$

6.



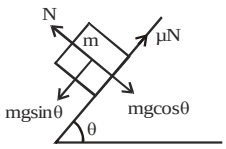
$\mu N = mg$

$N = ma$

$a = \frac{g}{\mu}$

Beginner's Box - 7

1.



$N = mg\cos\theta$

$\mu N = mg\sin\theta$

$\tan\theta = \mu$

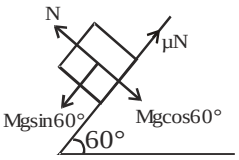
$\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = 30°$

2. Angle of Repose $\theta = 30°$

$\mu = \tan\theta$

$\mu = \frac{1}{\sqrt{3}}$



$N = Mg\cos 60°$

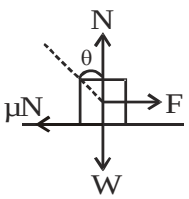
$Mg\sin 60° - \mu N = Ma$

$Mg\sin 60° - \frac{1}{\sqrt{3}} Mg\cos 60° = Ma$

$\left(\frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{3}}\right) g = a$

$a = \frac{g}{\sqrt{3}} = \frac{10}{\sqrt{3}}\ m/s^2$

3.



$\tan\theta = \frac{\mu N}{N}$

$\tan\theta = \mu$

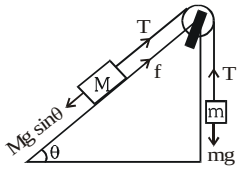
$N = W$

$\mu N = F$

$W\tan\theta = F$

4. For min value, friction will be in upward direction along the plane

Net pulling force = 0



$Mg\sin\theta = T + \mu Mg\cos\theta$

$T = mg$

$m_{min} = M(\sin\theta - \mu\cos\theta)$

For max value, friction will act in downward direction

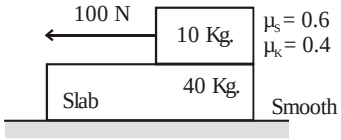
$Mg\sin\theta + \mu Mg\cos\theta = T$

$T = mg$

$m_{max} = M(\sin\theta + \mu\cos\theta)$

Beginner's Box - 8

1.



max. acceleration for 40 kg

$= \frac{F_{max}}{40} = \frac{60}{40} = 1.5 m/s^2$

If they were to move together

$\text{Acceleration} = \frac{100}{10 + 40} = 2 m/s^2$

So, They will not move together and slippings will take place

acceleration of 40 kg

$= \frac{\mu_k \times 9.8 \times 10}{40} = 0.98\ m/s^2$

For 10 kg mass :-

$100 - 0.4 \times 10 \times 9.8 = 10a$

$a = 6.08\ m/s^2$

2. The blocks will not move as 30 N is less than maximum friction available between B & Table So Friction force between A & B = 0 (No relative slipping)

3. $F = Ma$

$F = 65\ N$

$ma_{max} = f_r = 65 \times 0.2 \times 9.8$

$a_{max} = 1.96\ m/s^2$

AIM: FREE EDUCATION TO ALL APUL

Kinematics

SOLUTIONS

BEGINNER'S BOX-1

1. t = 2 min 20 sec = 120 + 20 = 140 sec

after each complete rotation, displacement becomes zero.

Number of rotation = $\frac{140}{40}=3+\frac{1}{2}$

$\therefore$ displacement in $\frac{1}{2}$ rotation = 2r

(displacement for 3 rotation = 0)

Distance in $3+\frac{1}{2}$ rotation = $3\times 2\pi r+\pi r=7\pi r$

2. Since displacement is independent of path, the displacement is the shortest distance between initial position & final position. So the displacement for each girl will be 400 m. For girl 'B' the displacement will be equal to actual path length

3. Distance = 4 + 3 + 12 = 19 m

displacement = $\sqrt{4^2+3^2+12^2}$ = 13 m

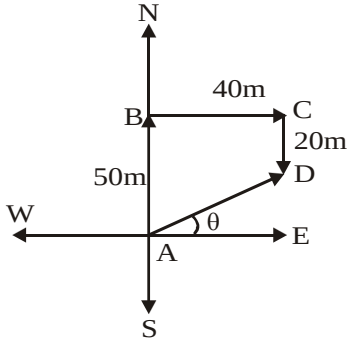
4. Distance = path length = $\frac{1}{2}$ × circumference

$=\frac{1}{2}\times 2\pi R=\pi R=(40\pi)$ m,

Displacement $=\left|\vec{d}\right|$ = 2r = 80m

5. (A) Total distance travelled = 50m + 40m + 20m = 110m

(B) Let east direction is $\hat{i}$ and north direction is $\hat{j}$ then $\vec{AB}=50\hat{j},\vec{BC}=40\hat{i},\vec{CD}=-20\hat{j}$



According to law of polygon

$\vec{AD}=\vec{AB}+\vec{BC}+\vec{CD}$

$\vec{AD}=50\hat{j}+40\hat{i}-20\hat{j}=40\hat{i}+30\hat{j}$

$|\vec{AD}|=\sqrt{40^2+30^2}=50$m

Now, if angle of $\vec{AD}$ with east towards north is θ, then

$\tan\theta=\frac{30}{40}=\frac{3}{4}$, so $\theta=\tan^{-1}\left(\frac{3}{4}\right)=37°$

So direction of displacement is E 37° N (37° N of E)

BEGINNER'S BOX-2

1. Air distance (shortest distance)

= displacement = 260 km

Road distance = 320 km

(i) Average speed of bus = $\frac{\text{Total distance}}{\text{total time}}$

$=\frac{320}{8}$

= 40 km/h

(ii) Average velocity of bus

$=\frac{\text{Total displacement}}{\text{Time}}=\frac{260}{8}$

= 32.5 km/h

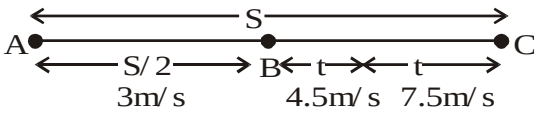
(iii) Average speed of plane

$=\frac{260}{\frac{1}{4}}$ = 1040 km/h

(iv) Average velocity of aeroplane = $\frac{260}{1/4}$

=1040km/h

2.



Average speed from B to C

$(v_{Avg})_{B\to C}=\frac{4.5\times t+7.5\times t}{2t}$

$(v_{Avg})_{B\to C}$ = 6 m/s

Average speed from A to C

$$(v_{Avg})_{A \to C} = \frac{\frac{S}{2}+\frac{S}{2}}{\frac{S}{2\times3}+\frac{S}{2\times6}}$$

$(v_{Avg})_{A \to C} = 4$ m/s

3. **Between 6:00 A.M. to 6:30 A.M.**

In this time interval the tip of minute hand moves from (12 mark) to (6 mark).

$\therefore$ displacement

= 2 × (length of minute hand)

= 2 × 4.5 cm = 9 cm

and time taken (t) = 30 min = 30 × 60 = 1800s

$\therefore$ average velocity

$$= \frac{S}{t} = \frac{9}{1800} = 5\times10^{-3}\,cm/s$$

direction of average velocity is from 12 mark to 6 mark on the clock panel.

Between 6:00 A.M. to 6:30 P.M.

In this time interval also minute hand will move from (12 mark) to (6 mark).

$\therefore$ displacement

= 2 × (length of minute hand)

= 2 × 4.5 cm = 9 cm

and time taken (t) = 12 hrs and 30 minute

= (12 × 60 × 60)s + (30 × 60)s

= 43200 s + 1800 s = 45000 s

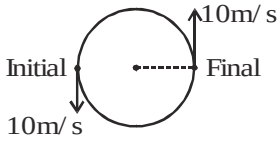
$\therefore$ average velocity

$$= \frac{s}{t} = \frac{9}{45000} = 2\times10^{-4}\,cm/s$$

direction of average velocity is from 12 mark to 6 mark on the clock panel.

4. Change in speed

$\Delta v = 10 - 10 = 0$ Change in velocity



$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$

$= 10\hat{j} - (-10\hat{j}) = 20\hat{j}$

Magnitude $|\Delta\vec{v}| = 20$ m/s

5. (a) The distance travelled during time 0 to 5.0s is, $S = (2.5)(5.0)^2 = 62.5$ m

The average speed during this time is

$$v_{avg} = \frac{\text{distance travelled}}{\text{time taken}} = \frac{62.5}{5} = 12.5 \text{ m/s}$$

(b) Given $S = (2.5)t^2$

$$\text{Instantaneous speed} = \frac{dS}{dt} = \frac{d}{dt}(2.5t^2) =$$

$(2.5)(2t) = 5t$

At t = 5.0 s the speed is

$[v]_{t=5S} = (5.0) \times (5.0) = 25$ m/s

6. Average velocity

$$= \frac{\text{Net displacement}}{\text{total time}} = \frac{AO+OB}{\text{time}}$$

$$= \frac{1+1}{1} = 2 \text{ m/s}$$

7. (a) Average speed = $\frac{\text{Total path length}}{\text{Total time}}$

$$= \frac{23}{(28/60)} = 49.3 \text{ kmh}^{-1}$$

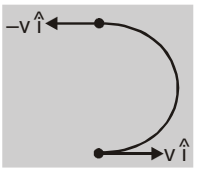
(b) Average velocity = $\frac{\text{Displacement}}{\text{Time}}$

$$= \frac{\text{Shortest distance}}{\text{Time}} = \frac{10}{28/60} = 21.4 \text{ km/h}$$

(c) Both are Not equal

BEGINNER'S BOX-3

1. $t = \frac{\pi R}{v} = \frac{\pi\times5}{5} = \pi$ sec



$\Delta\vec{v} = (-5\hat{i}) - (5\hat{i}) = -10\hat{i}$

$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{t} = \frac{-10\hat{i}}{\pi} \quad \rightarrow \quad |\vec{a}_{avg}| = \frac{10}{\pi} \text{ m/s}^2$$

2. $x = 7t^2 - 2t + 5$

(a) $v = \frac{dx}{dt} = 14t - 2$

at t = 5, $v = 14 \times 5 - 2 = 68$

(b) $a = \frac{dv}{dt} = 14$

(c) Average velocity

$= \frac{\text{displacement}}{\text{Time}} = \frac{x_5 - x_0}{5-0}$

$x_5 = 7(5)^2 - 2(5) + 5 = 170$

$x_0 = 7(0)^2 - 2(0) + 5 = 5$

$v_{Avg} = \frac{170-5}{5} = 33$

(d) Average acceleration

$= \frac{\text{Change in velocity}}{\text{Time interval}} = \frac{v_5 - v_0}{5-0}$

$v_5 = 14 \times 5 - 2 = 68$

$v_0 = 14 \times 0 - 2 = -2$

$a_{Avg} = \frac{68-(-2)}{5-0} = 14$

BEGINNER'S BOX-4

1. Let $a = a_0$ & $u = 0$

So $S_1 = 0 + \frac{1}{2} a_0 (5)^2 = \frac{25}{2} a_0$

distance covered in 15 sec.

$S_1 + S_2 = \frac{1}{2} a_0 (15)^2 = \frac{225}{2} a_0$

$S_2 = \frac{225}{2} a_0 - \frac{25}{2} a_0 = 100 a_0$

$\frac{S_1}{S_2} = \frac{\frac{25}{2} a_0}{100 a_0} = \frac{1}{8} \Rightarrow S_1 = \frac{S_2}{8}$

2. $v^2 = u^2 + 2aL$ (L is length of train)

$2aL = v^2 - u^2$

For mid point

also $v_m^2 = u^2 + 2a\frac{L}{2} = u^2 + \frac{1}{2}(v^2 - u^2)$

$v_m^2 = \frac{2u^2 + v^2 - u^2}{2} = \frac{u^2 + v^2}{2}$; $v_m = \sqrt{\frac{u^2 + v^2}{2}}$

3. By using $v^2 = u^2 + 2as$

We have $\left(u - \frac{u}{n}\right)^2 = u^2 + 2(-a)s$

$\left(\frac{2}{n} - \frac{1}{n^2}\right) u^2 = 2as$

Let no. of planks required be N

Then,

$0^2 = u^2 + 2(-a)Ns$

Therefore, $\left(\frac{2}{n} - \frac{1}{n^2}\right) u^2 = \frac{u^2}{N}$

$N = \frac{n^2}{2n-1}$

4. initial velocity $u = 126 \times \frac{5}{18} = 35$ m/s,

$s = 200$ m

From equation of motion $v^2 = u^2 + 2as$

$\Rightarrow 0 = (35)^2 + 2a \times 200 \Rightarrow a = -3.06$ m/s^2

$\therefore$ retardation is 3.06 m/s^2

$v = u + at$; $0 = 35 - 3.06\,t$

$t = 11.4$ sec.

5. During the reaction time, the car will move with constant speed.

So, $S_1 = ut$

The distance covered by the car after brakes are applied $S_2 = \frac{0^2 - u^2}{-2a} = \frac{u^2}{2a}$

So Total distance travelled $= S_1 + S_2 = ut + \frac{u^2}{2a}$

6. $S_{n^{th}} = u + \frac{1}{2} a(2n-1)$

$1.2 = 0 + \frac{1}{2} \times a \times 11$;

$a = \frac{2.4}{11}$ m/s^2; $a = 0.218$ m/s^2

BEGINNER'S BOX-5

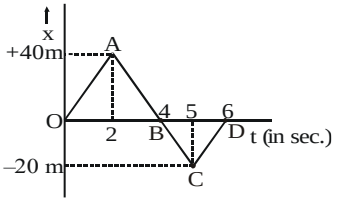
1. Velocity = Slope of s – t curve

So $\frac{v_A}{v_B} = \frac{\text{Slope of A}}{\text{Slope of B}} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{3}$

2. (i) Total distance travelled

$= x_{OA} + x_{AB} + x_{BC} + x_{CD}$

$= 40 + 40 + 20 + 20 = 120$ m

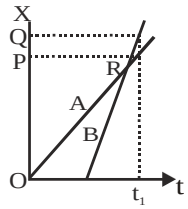


(ii) Displacement = final position – initial position $= x_D - x_O = 0 - 0 = 0$

(iii) Average speed $= \frac{\text{Total distance}}{\text{Total time}} = \frac{120}{6} =$ 20 m/s

(iv) Average velocity $= \frac{\text{displacement}}{\text{Time}} = \frac{0}{6} = 0$ m/s

3.



(a) A lives closer to the school than B (as P is close to O than Q)

(b) A starts earlier than B (A starts when time t = 0)

(c) B walks faster than A (slope of B is greater than slope of A)

(d) A and B reach home at the same time t_1 (as shown by dotted lines in the graph)

(e) B overtakes A on the road once (as graphs for A and B intersect only at one point R.)

4. (i) Total distance covered = Area under v – t curve

$$\frac{1}{2} = (2 \times 2) + 2(4-2) + \frac{1}{2}(10+2) \times 1$$

$$\frac{1}{2} \times 5 \times 10 = 37\text{m}$$

(ii) $\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{37}{10} = 3.7\,\text{m/s}$

(iii) Acceleration = Slope of v – t curve

So maximum acceleration will be in the part where the slope will be maximum i.e. BC

$$a_{max} = a_{BC} = \frac{10-2}{1} = 8\text{m/s}^2$$

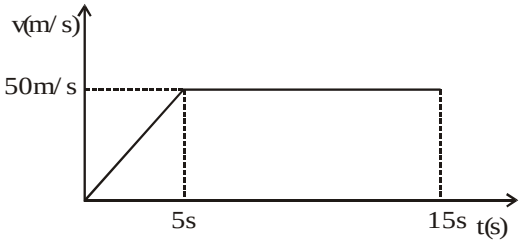
(iv) Retardation = Slope of CD (As it is negative)

$$= \left|\frac{0-10}{5}\right| = |-2| = 2\text{m/s}^2$$

5. Velocity after 5 seconds = 0 + 10 × 5

= 50m/s

v – t curve for the situations is as follows :



Total distance travelled = Area under v-t curve

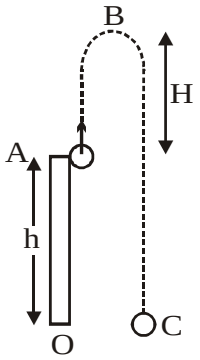
$$= \frac{1}{2}(50 \times 5) + 50 \times (15-5) = 625\text{m}$$

BEGINNER'S BOX-6

1. Time taken to reach the highest point = $\frac{u}{g}$

$$t = \frac{10}{10} = 1\text{s}$$

2. Time taken to reach B



$$t_{AB} = \frac{u}{g} = \frac{10}{10} = 1\text{ sec.}$$

t_{AC} = 5sec. (given)

so t_{BC} = 5 – 1 = 4 sec.

$$S_{BC} = \frac{1}{2}gt^2 = \frac{1}{2}(10)(4)^2$$

= 80 m

$$S_{AB} = \frac{u^2}{2g} = \frac{(10)^2}{20} = 5\text{m}$$

(a) Height of tower h = 80 – 5 = 75 m

(b) $v_c = 0 + gt_{BC}$ = 40 m/s

(c) Distance travelled = 5 + 80 = 85 m

(d) Average speed = $\frac{85}{5}$ = 17 m/s

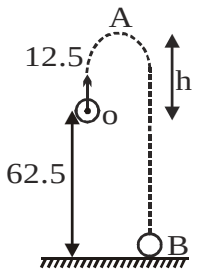
Average velocity = $\frac{75}{5}$ = 15 m/s

3. Velocity of balloon after 10 sec. = 0 + 1.25 × 10

= 12.5 m/s

distance travelled = $\frac{1}{2}at^2 = \frac{1}{2}(1.25) \times 10^2$

= 62.5 m



(i) $h = \frac{u^2}{2g} = \frac{(12.5)^2}{20} = 7.8125\,\text{m}$

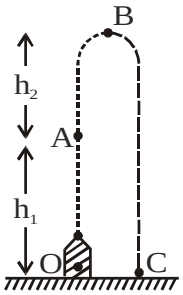
height from ground=62.5+7.8125=70.31 m

(ii) $t_{OA} = \frac{u}{g} = \frac{12.5}{10} = 1.25$ sec.

$t_{AB} = \sqrt{\frac{2h_{AB}}{g}} = 3.75$ sec.

Total time = 5 sec.

4. Point O → Point of projection



Point A → Point up to which fuel is consumed.

Point B → Highest point of path

$h_1 = 0 \times 60 + \frac{1}{2} \times 10 \times (60)^2$

$= 18$ km

$v_A = 0 + 10 \times 60 = 600$ m/s

$O = v_A^2 - 2gh_2 \Rightarrow h_2 = \frac{v_A^2}{2g} = \frac{(600)^2}{20} = 18$ km.

maximum height from ground = 18 + 18 = 36 km.

time taken from A to B :→ $O = 600 - gt \Rightarrow t = 60$ sec.

time taken in coming down to earth –

$36000 = \frac{1}{2} gt^2 \Rightarrow t = 60\sqrt{2}$ sec.

$\therefore$ Total time = $60 + 60 + \sqrt{2}\,60 = 60(2 + \sqrt{2})$ s.

$= (2 + \sqrt{2})$ min.

5. Let it takes n seconds to fall from tower then

total distance $s = \frac{1}{2} g\,(n)^2$

distance travelled in n^{th} second

$s_{n^{th}} = 0 + \frac{g}{2}(2n - 1) \Rightarrow s_{n^{th}} = \frac{9}{25} s$

$\Rightarrow \frac{g}{2}(2n - 1) = \frac{9}{25}\frac{g}{2}(n^2) \Rightarrow 9n^2 = 50n - 25$

$\Rightarrow n = \frac{50 + \sqrt{50^2 - 4 \times 9 \times (+25)}}{2 \times 9} = 5$ sec.

$H = \frac{1}{2} g(5)^2 = 125$ m

6. Let the particle covers its total journey in n seconds then for last 2 sec.

$S_n - S_{n-2} = 40$ m

$\frac{1}{2} g(n)^2 - \frac{1}{2} g(n-2)^2 = 40$

$\Rightarrow (4n - 4) = 8 \Rightarrow n = 3$

height of tower $= \frac{1}{2} g(3)^2 = 45$ m

7. (a) Acceleration is directed downwards

(b) At highest point v = 0, a = g (downwards)

(c) For upward motion (Reference point to be the highest point, taking downward direction positive)

x → Positive

→ negative

a → positive

For downward motion

x → Positive

v → Positive

a → Positive

(d) $H = \frac{v^2}{2g} = 44.1$ m, $T = \frac{2v}{g} = 6$ sec.

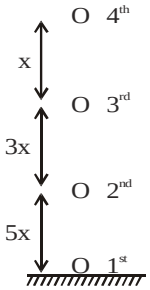
8. Let time of fall be 'n'

$\frac{1}{2} g(2n - 1) = \frac{1}{2} g\,(3)^2$

$\Rightarrow n = 5$ sec.

Height of tower $= \frac{1}{2} g(5)^2 = 125$ m.

9. By galileo's law of odd number



$x + 3x + 5x = 9m$

$x = 1$ m

Position of 2nd drop from top

$x + 3x = 4x = 4m$

Position of 3rd drop from top

$x = 1$ m

BEGINNER'S BOX-7

1. (a) Time taken by the ball to reach the highest point

$$t = \frac{T}{2} = \frac{u\sin\theta}{g} = \frac{20}{10} \times \sin 30°$$

(b) The maximum height attained

$$= \frac{u^2 \sin^2\theta}{2g} = \frac{(20)^2 \times \sin^2 30°}{2\times 10} = 5\text{m}$$

(c) The horizontal range

$$= \frac{u^2 \sin 2\theta}{g} = \frac{(20)^2 \times \sin(2\times 30°)}{10} = 34.64\text{m}$$

(d) The time of flight

$$= \frac{2u\sin\theta}{g} = \frac{2\times 20\times \sin 30°}{10} = 2\text{s}$$

2. Let u be the velocity of projection of the ball. The ball will cover maximum horizontal distance when angle of projection with horizontal, $\theta = 45°$. Then $R_{max} = \frac{u^2}{g} = 100$ m

If ball is projected vertically upwards ($\theta = 90^0$ from ground) then H attains maximum value,

$$H_{max} = \frac{u^2}{2g} = \frac{R_{max}}{2}$$

$\therefore$ the height to which cricketer can throw the ball is $= \frac{R_{max}}{2} = \frac{100}{2} = 50$ m.

3. $\frac{H_1}{H_2} = \frac{\sin^2\alpha}{\sin^2(90°-\alpha)} = \tan^2\alpha$

$\frac{R_1}{R_2} = \frac{\sin 2\alpha}{\sin 2(90°-\alpha)} = 1$

4. For the first ball, angle of projection = θ, velocity of projection, u = 40 m/s.

Let h be the maximum height attained by it.

As maximum height attained $= \frac{u^2\sin^2\theta}{2g}$

$$\therefore \quad h = \frac{(40)^2\sin^2\theta}{2\times 10} \qquad .....(1)$$

For second ball, Angle of projection = $(90° - \theta)$.

velocity of projection, u = 40 m/s

Maximum height reached = (h + 50) m

$$\therefore \quad h + 50 = \frac{u^2\sin^2(90°-\theta)}{2g} = \frac{(40)^2\cos^2\theta}{2\times 10}$$

.....(2)

By adding (1) and (2),

$$2h + 50 = \frac{(40)^2}{2\times 10} \times (\sin^2\theta + \cos^2\theta) = \frac{(40)^2}{2\times 10} = 80$$

$\Rightarrow$ 2h = 80 – 50 = 30 $\Rightarrow$ h = 15 m

Height of the first ball, h = 15 m & Height of the second ball = h + 50 = 15 + 50 = 65 m

5. $R = \frac{u^2\sin(2\times 15°)}{g} = 1.5$ or $\frac{u^2}{g} \times \frac{1}{2} = 1.5$

or $\frac{u^2}{g} = 3$ km

Horizontal range for angle of projection 45° is

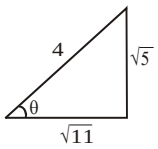
$$R' = \frac{u^2}{g} \times \sin(2\times 45°) = \frac{u^2}{g}\sin 90° = \frac{u^2}{g} = 3\text{km}$$

6. $H_m = \frac{u^2\sin^2\theta_0}{2g}$; $R = \frac{u^2}{g}\sin 2\theta_0 = \frac{2u^2}{g}$ $\sin\theta_0.\cos\theta_0$

$$\frac{H_m}{R} = \frac{1}{4}\tan\theta_0 \Rightarrow \theta_0 = \tan^{-1}\left(\frac{4H_m}{R}\right)$$

7. Since their time of flight is same so their vertical velocity will be same, maximum heights attain by them will be same

8. $H \le 25$ m



$$\Rightarrow \frac{u^2}{2g}\sin^2\theta \le 25 \text{ m}$$

$$\Rightarrow \sin^2\theta \le \frac{25}{80} = \frac{5}{16}$$

$$\Rightarrow \sin\theta \le \frac{\sqrt{5}}{4}$$

For maximum range $\sin\theta = \sqrt{\frac{5}{4}}$

$$\tan\theta = \sqrt{\frac{5}{11}} \Rightarrow R = \frac{4H}{\tan\theta} = \frac{4\times 25}{\sqrt{5/11}} = 148.32 \text{ m}$$

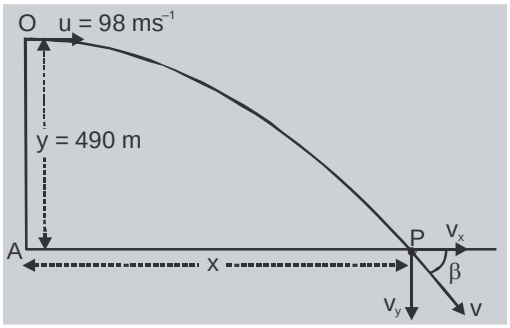
BEGINNER'S BOX-8

1. (i) The projectile is fired from the top O of a hill with velocity u = 98 m/s along the horizontal OX. It reaches the target P vertical distance

OA = y = 490 m

As $y = \frac{1}{2} gt^2$

$\therefore 490 = \frac{1}{2} \times 9.8\, t^2$



or $t = \sqrt{100} = 10\,s.$

(ii) Distance of the target from the hill is

AP = x = horizontal velocity × time

= 98 × 10 = 980m.

(iii) The horizontal components of velocity v of the projectile at point P is $v_x = u = 98$ m/s

and vertical component

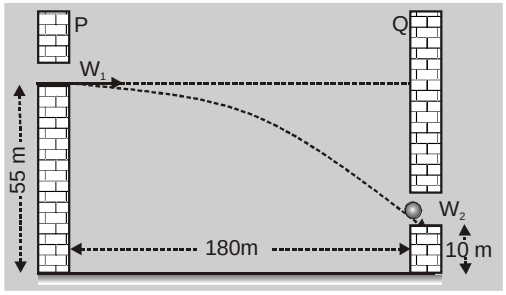
$v_y = u_y + gt = 0 + 9.8 \times 10 = 98$ m/s

If the resultant velocity v makes angle β with the horizontal, then

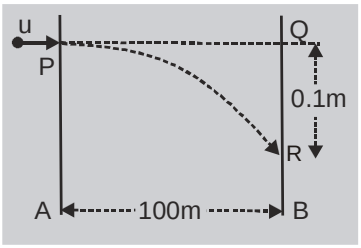
$$\tan\beta = \frac{v_y}{v_x} = \frac{98}{98} = 1 \quad \text{or} \quad \beta = 45°$$

2. Equation of trajectory $y = \frac{gx^2}{2u^2}$

$$45 = \frac{10\times(180)^2}{2u^2} \Rightarrow u = 60 \text{ m/s}$$



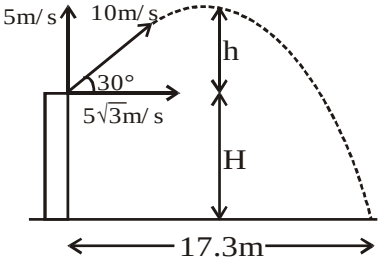
3. Assume that the bullet hits the screen A with velocity u and pierces the screen B after time t.



equation of trajectory $y = \frac{gx^2}{2u^2}$

$$(10 \times 10^{-2}) = \frac{9.8\times(100)^2}{2u^2} \Rightarrow u = 700 \text{ m/s}$$

4. Total flight time



$R = u_x T$

$T = \frac{R}{u_x}$

$T = \frac{17.3}{5(1.73)} = 2\text{sec.}$

Time to reach maximum height $= \frac{5}{g} = 0.5$ sec.

$h = \frac{1}{2} g(0.5)^2 = 1.25$ m

$H + h = \frac{1}{2} g(1.5)^2 = 11.25$ m

So H = 10 m

BEGINNER'S BOX-9

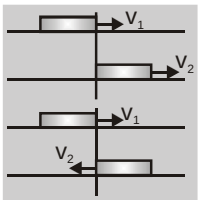
1. In case of overtake

$v_1 - v_2 = \frac{100}{40} = 2.5$

In case of crossing

$v_1 + v_2 = \frac{100}{20} = 5$

Adding, $2v_1 = 7.5$



$\rightarrow v_1 = 3.75$ m/s and $v_2 = 1.25$ m/s

2. Let s = distance between the trains initially
Velocity of train B w.r.t. A is zero, acceleration of train B w.r.t. A is = 1 m/s²

distance $s_{rel} = u_{rel}t + \frac{1}{2}a_{rel}t^2 = 0 + \frac{1}{2} \times (50)^2$

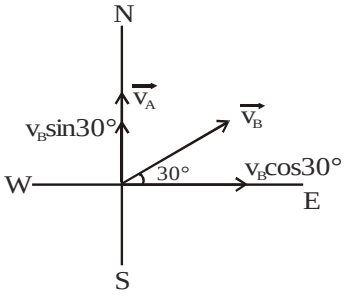
$= 1250m$

3. $T = \frac{2u_{rel}}{a_{rel}} = \frac{2\times 49}{9.8} = 10\ s$

since the lift is not accelerating, time of flight will remain same.

BEGINNER'S BOX-10

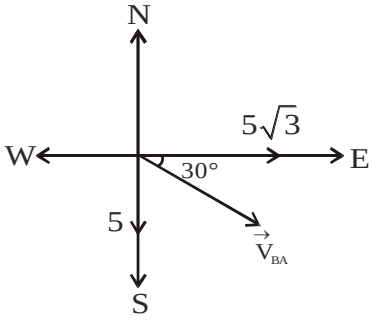
1.



$\overrightarrow{v_A} = 10\hat{j}$

$\overrightarrow{v_B} = 5\sqrt{3}\hat{i} + 5\hat{j}$

$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A} = 5\sqrt{3}\hat{i} - 5\hat{j}$

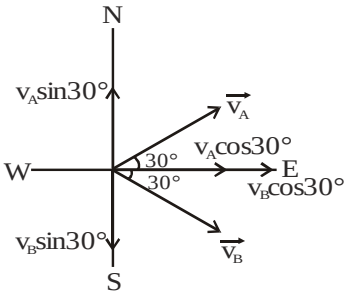


angle with east direction

$\tan\theta = -\frac{5}{5\sqrt{3}} = -\frac{1}{\sqrt{3}}$

$\theta = 30°$ i.e. E – 30° – S

2.



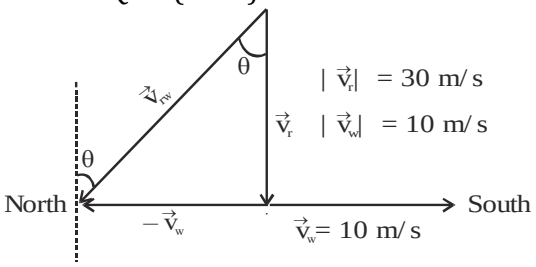
$\overrightarrow{v_{AB}} = \overrightarrow{v_A} - \overrightarrow{v_B}$

$\overrightarrow{v_{AB}} = \left(5\sqrt{3}\hat{i} + 5\hat{j}\right) - \left(5\sqrt{3}\hat{i} - 5\hat{j}\right)$

$= 10\hat{j}$ m/s i.e. 10 m/s in north direction

3. Same as Q.32 (Ex.-2)

4.



From the diagram

$\tan\theta = \frac{10}{30} = \frac{1}{3}$

So $\theta = \tan^{-1}\left(\frac{1}{3}\right)$ from vertical towards south

5. $\overrightarrow{v_m} = 2\hat{i} + 3\hat{j}$; $\overrightarrow{v_{rm}} = -4\hat{j}$

$\overrightarrow{v_r} = \overrightarrow{v_{rm}} + \overrightarrow{v_m} = 2\hat{i} - \hat{j}$

Now velocity of man $\overrightarrow{v_{m'}} = -2\hat{i} - 3\hat{j}$

$\overrightarrow{v_{rm'}} = \overrightarrow{v_r} - \overrightarrow{v_{m'}} = 4\hat{i} + 2\hat{j}$

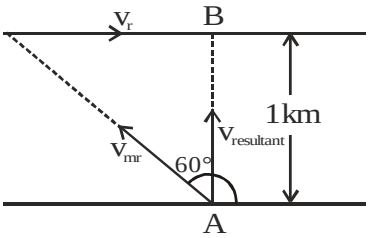
$\left|\overrightarrow{v_{rm'}}\right| = \sqrt{20}$ m/s

6. For bodies to collide the vertical velocities should be same

$\Rightarrow\ u_1 \sin 60° = u_2 \Rightarrow \frac{u_1}{u_2} = \frac{2}{\sqrt{3}}$

BEGINNER'S BOX-11

1. (a) $\sin 60° = \frac{v_r}{v_m}$



$\Rightarrow v_r = \frac{2\sqrt{3}}{2} = \sqrt{3}$ m/s

(b) $t = \frac{1000}{2\cos 60°} = 1000$ sec.

2. For shortest path

(a) $\sin\theta = \frac{v_r}{v_{mr}} = \frac{1}{2} \Rightarrow \theta = 30°$

So direction from downstream = 90° + 30° = 120°

(b) crossing time $T = \frac{d}{v_{mr}\cos\theta} = \frac{2}{10\cos 30°}$

$T = \frac{2}{5\sqrt{3}}$ h

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3. (a) Speed of the child running in the direction of motion of the belt
= 9 + 4 = 13km/h

(b) Speed of the child in opposite direction of the motion of the belt
= 9 – 4 = 5km/h

(c) Since both the parents and child are on the moving belt. So their relative position and speed remain unchanged.

speed of child = $9 \times \frac{5}{18}$ = 2.5 m/s,

time taken = $\frac{\text{distance}}{\text{speed}} = \frac{50}{2.5}$ = 20s

If the motion is viewed by one of the parents answer to (a) and (b) are altered while answer to (c) remain unchanged.

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Basic Mathematics Used in Physics and Vectors

SOLUTIONS

BEGINNER'S BOX-1

1. (i) $\tan(-30°) = -\tan 30° = -\frac{1}{\sqrt{3}}$

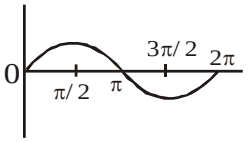
(ii) $\sin(120°)=\sin(180° - 60°)= \sin 60° = \frac{\sqrt{3}}{2}$

(iii) $\sin(135°)=\sin(90° + 45°) = \cos 45° = \frac{1}{\sqrt{2}}$

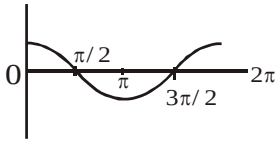
(iv) $\cos(150°) = \cos(180° - 30°) = -\cos(30°)$

$= -\frac{\sqrt{3}}{2}$

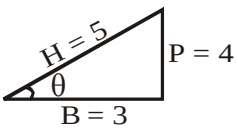
(v) $\sin(270°) = (-1)$ (from graph)



(vi) $\cos 270° = 0$



2. $\sec\theta = \frac{5}{3}$



we know that, $\sec\theta = \frac{H}{B} = \frac{5}{3}$

$\cos\theta = \frac{3}{5}$;

we know 3, 4, 5 are pythogorous triplets

$\therefore$ P = 4

i.e. H : B : P = 5 : 3 : 4

$\therefore$ $\sin\theta = \frac{4}{5}$ $\quad$ $\text{cosec}\,\theta = \frac{5}{4}$

$\cos\theta = \frac{3}{5}$ $\quad$ $\sec\theta = \frac{5}{3}$

$\tan\theta = \frac{4}{3}$ $\quad$ $\cot\theta = \frac{3}{4}$

BEGINNER'S BOX-2

1. Let P(8, –4) and Q(0, a) be two points, distance PQ is 10.

$\therefore$ According to distance formula

$PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(0-8)^2+(a+4)^2}$

$\Rightarrow PQ = \sqrt{64+(a^2+16+8a)}$

$\Rightarrow 10 = \sqrt{80+a^2+8a}$

$\Rightarrow a^2 + 8a + 80 = 100 \Rightarrow a^2 + 8a = 20$

$\Rightarrow a^2 + 10a - 2a - 20 = 0$

$\Rightarrow (a - 2)(a + 10) = 0 \Rightarrow (a = 2)$ and $(a = -10)$

$\therefore a = 2$

2. $P(x_1, y_1, z_1) = (0, -1, +1)$

and $Q(x_2, y_2, z_2) = (3, 3, 13)$

using distance formula

$PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

$= \sqrt{9+16+144} = \sqrt{169} = 13$

3. Slope $m = \frac{\Delta y}{\Delta x} = \frac{y_2-y_1}{x_2-x_1} = \tan\theta$

$m = \frac{\Delta y}{\Delta x} = \frac{(0-2)}{(3-0)} = \frac{-2}{3}$

BEGINNER'S BOX-3

1. (i) $y = x^{7/2}$

$\frac{dy}{dx} = \frac{d}{dx}(x^{7/2}) = \frac{7}{2}x^{\frac{7}{2}-1} = \frac{7}{2}x^{5/2}$

(ii) $y = x^{-3}$ $\quad\therefore\quad$ $\frac{dy}{dx} = -3x^{-4}$

(iii) $y = x$ $\quad\therefore\quad$ $\frac{dy}{dx} = 1$

(iv) $y = x^5 + x^3 + 4x^{1/2} + 7$

$\frac{dy}{dx} = 5x^4 + 3x^2 + 2x^{-1/2} + 0$

$= 5x^4 + 3x^2 + 2x^{-1/2}$

(v) $y = 5x^4 + 6x^{3/2} + 9x$

$\frac{dy}{dx} = 20x^3 + 9x^{3/2-1} + 9$

$= 20x^3 + 9x^{1/2} + 9$

(vi) $y = ax^2 + bx + c \Rightarrow \frac{dy}{dx} = 2ax + b$

(vii) $y = 3x^5 - 3x - \frac{1}{x}$

$\frac{dy}{dx} = 15x^4 - 3 + \frac{1}{x^2}$

2. $s = t^2 + 5t + 3 \Rightarrow \frac{ds}{dt} = 2t + 5$

3. $s = ut + \frac{1}{2}at^2 \Rightarrow \frac{ds}{dt} = u + at$

4. $A = 3t^2 + 7\ cm^2$

$\frac{dA}{dt} = \frac{d}{dt}(3t^2 + 7) = 6t$

at $t = 5$, $\frac{dA}{dt} = 30 cm^2/s$

5. $A = \pi r^2$

$\frac{dA}{dr} = \frac{d}{dr}(\pi r^2) = 2\pi r$

6. (i) $\frac{d}{dx}(x - 1)(2x + 5)$

$= (x - 1)\frac{d}{dx}(2x + 5) + (2x + 5)\frac{d}{dx}(x - 1)$

$= (x - 1)(2) + (2x + 5)(1) = 2x - 2 + 2x + 5$

$= 4x+3$

(ii) $\frac{d}{dx}\frac{1}{(2x+1)} = \frac{0-2}{(2x+1)^2} = \frac{-2}{(2x+1)^2}$

(iii) $\frac{d}{dx}\left(\frac{3x+4}{4x+5}\right) = \frac{(4x+5)(3)-(3x+4)(4)}{(4x+5)^2}$

$= \frac{-1}{(4x+5)^2}$

(iv) $\frac{d}{dx}\left(\frac{x^2}{x^3+1}\right) = \frac{(x^3+1)(2x)-x^2(3x^2)}{(x^3+1)^2}$

$= \frac{2x^4+2x-3x^4}{(x^3+1)^2} = \frac{2x-x^4}{(x^3+1)^2}$

BEGINNER'S BOX-4

1. (i) $\int x^{15}dx = \frac{x^{15+1}}{15+1} + C = \frac{x^{16}}{16} + C$

(ii) $\int x^{-3/2}dx = \frac{x^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} + C = -2x^{-\frac{1}{2}} + C$

(iii) $\int(3x^{-7} + x^{-1})dx = \frac{3x^{-6}}{-6} + \log_e x + C$

$= \frac{-x^{-6}}{2} + \log_e x + C$

(iv) $\int\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx$

$= \int\left(x + \frac{1}{x} + 2\right)dx = \frac{x^2}{2} + \log_e x + 2x + C$

(v) $\int\left(x + \frac{1}{x}\right)dx = \frac{x^2}{2} + \log_e x + C$

(vi) $\int\left(\frac{a}{x^2} + \frac{b}{x}\right)dx = \int(ax^{-2} + bx^{-1})dx$

$= \frac{ax^{-2+1}}{-2+1} + b\log_e x + C = -\frac{a}{x} + b\log_e x + C$

BEGINNER'S BOX-5

1. (i) $\int_R^\infty \frac{GMm}{x^2}dx = GMm\int_R^\infty x^{-2}dx$

$= GMm\left[\frac{x^{-1}}{-1}\right]_R^\infty = \frac{GMm}{R}$

(ii) $\int_{r_1}^{r_2} \frac{-kq_1q_2}{x^2}dx$

$= -kq_1q_2\left[\frac{x^{-1}}{-1}\right]_{r_1}^{r_2} = kq_1q_2\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$

(iii) $\int_u^v Mv\,dv = M\left[\frac{v^2}{2}\right]_u^v = \frac{1}{2}M(v^2 - u^2)$

(iv) $\int_0^\infty x^{-1/2}dx = \left[\frac{x^{1/2}}{\frac{1}{2}}\right]_0^\infty = \infty$

(v) $\int_0^{\pi/2} \sin x\,dx = [-\cos x]_0^{\pi/2} = 1$

(vi) $\int_0^{\pi/2} \cos x\,dx = [\sin x]_0^{\pi/2} = 1$

(vii) $\int_{-\pi/2}^{\pi/2} \cos x\,dx = [\sin x]_{-\pi/2}^{\pi/2} = 2$

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BEGINNER'S BOX-6

1. (i) $10x^2 - 27x + 5 = 0$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{27 \pm \sqrt{(-27)^2-4(10)(5)}}{20}$$

$$= \frac{27 \pm 23}{20} \Rightarrow x = \frac{5}{2}, \frac{1}{5}$$

(ii) $(pq)x^2 - (p^2 + q^2)x + (pq) = 0$

$$x = \frac{(p^2+q^2) \pm \sqrt{(p^2+q^2)^2-4(pq)(pq)}}{2pq}$$

$$= \frac{(p^2+q^2) \pm (p^2-q^2)}{2pq} \Rightarrow x = \frac{p}{q}, \frac{q}{p}$$

BEGINNER'S BOX-7

1. $1 + 2 + 3 + \ldots\ldots\ldots\ldots + 50$

$$s_n = \frac{n}{2}[2a + (n-1)d] = \frac{50}{2}[2 + (49)1]$$

$$= 25[51] = 1275$$

2. $1^2 + 2^2 + 3^2 \ldots\ldots\ldots\ldots + 10^2$

$$s = \frac{n(n+1)(2n+1)}{6} = \frac{(10)(11)(21)}{6} = 385$$

3. $$s_\infty = \frac{a}{1-r} = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}$$

4. $$F_{net} = GMm\left[\frac{1}{r^2} + \frac{1}{2r^2} + \frac{1}{4r^2} + \ldots\ldots\infty\right]$$

$$= GMm\left[\frac{1/r^2}{1-\frac{1}{2}}\right] = \frac{2GMm}{r^2}$$

BEGINNER'S BOX-8

1. Let the magnitudes of forces be A and B.

then according to question $A - B = 10$

$\Rightarrow A = 10 + B$ .....(1)

and $\sqrt{A^2+B^2} = 50$

$\Rightarrow A^2 + B^2 = 2500$ .....(2)

Solving equation number (1) & (2)

$A = 40N$ and $B = 30N$

Therefore the magnitudes of forces are 40N & 30N

2. Unit vector along $(\vec{A}-\vec{B})$ $= \frac{\vec{A}-\vec{B}}{|\vec{A}-\vec{B}|}$

$$= \frac{(3\hat{i}+2\hat{j})-(2\hat{i}+3\hat{j}-\hat{k})}{|(3\hat{i}+2\hat{j})-(2\hat{i}+3\hat{j}-\hat{k})|} = \frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$$

3. $\because \quad |\hat{A}+\hat{B}| = \sqrt{2} \Rightarrow 2 \times 1 \times \cos\frac{\theta}{2} = \sqrt{2}$

$$\Rightarrow \cos\frac{\theta}{2} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\theta}{2} = 45° \Rightarrow \theta = 90°$$

$$|\hat{A}-\hat{B}| = 2 \times 1 \times \sin\frac{\theta}{2} = 2 \times \sin 45° = \sqrt{2}$$

BEGINNER'S BOX-9

1. $\because$ $A_x = 4$, $A_y = 6$ so $A_x + B_x = 10$ and $A_y + B_y = 9$

(a) $B_x = 10 - 4 = 6m$ and $B_y = 9 - 6 = 3m$

(b) length $= \sqrt{B_x^2+B_y^2} = \sqrt{36+9} = \sqrt{45}m$

(c) $\theta = \left(\frac{B_y}{B_x}\right)\tan^{-1} = \tan^{-1}\left(\frac{3}{6}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

2. Let α, β & γ are the angles of vector $(5\hat{i}+2\hat{j}+6\hat{k})$ from x, y & z-axis respectively.

then $$\cos\alpha = \frac{A_x}{A} = \frac{5}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{5}{\sqrt{65}},$$

$$\cos\beta = \frac{A_y}{A} = \frac{2}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{2}{\sqrt{65}}$$

$$\cos\gamma = \frac{A_z}{A} = \frac{6}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{6}{\sqrt{65}}$$

The sum of squares of directional cosines of this vector

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{5^2+2^2+6^2}{65} = \frac{65}{65} = 1$$

3. Resultant $(\vec{R})$ of

$\vec{A}$ & $\vec{B}$ $= \vec{A} + \vec{B} = (6\hat{i} - 6\hat{j} + 5\hat{k}) + (\hat{i} + 2\hat{j} - \hat{k})$

$= 7\hat{i} - 4\hat{j} + 4\hat{k}$

$\therefore$ unit vector parallel to resultant $(\hat{R})$

$$= \frac{\vec{R}}{|\vec{R}|} = \frac{7\hat{i}-4\hat{j}+4\hat{k}}{|7\hat{i}-4\hat{j}+4\hat{k}|} = \frac{7\hat{i}-4\hat{j}+4\hat{k}}{9}$$

BEGINNER'S BOX-10

1. $\because a = b = 1$ so $a^2 + b^2 + 2ab\cos\theta = (\sqrt{3})^2 = 3$

$\Rightarrow 1 + 1 + 2\cos\theta = 3 \Rightarrow \cos\theta = \frac{1}{2}$

$(\vec{a}-\vec{b}).(2\vec{a}+\vec{b}) = 2a^2 - b^2 + ab\cos\theta - 2ab\cos\theta$

$= 2 - 1 - \frac{1}{2} = \frac{1}{2}$

2. $\because$ $\vec{A}$ & $\vec{B}$ are perpendicular to each other

$\therefore \vec{A}.\vec{B} = 0$

$\Rightarrow \left(4\hat{i}-2\hat{j}+4\hat{k}\right).\left(-4\hat{i}+2\hat{j}+\alpha\hat{k}\right) = 0$

$\Rightarrow -16 - 4 + 4\alpha = 0 \Rightarrow \alpha = 5$

3. $\because (\hat{a}+2\hat{b})$ is perpendicular to $(5\hat{a}-4\hat{b})$

$\therefore (\hat{a}+2\hat{b})\cdot(5\hat{a}-4\hat{b}) = 0$

$\Rightarrow 5\hat{a}\cdot\hat{a} - 4\hat{a}.\hat{b} + 10\hat{b}.\hat{a} - 8\hat{b}.\hat{b} = 0$

$\Rightarrow 5 + 6\hat{a}\cdot\hat{b} - 8 = 0 \Rightarrow 6\hat{a}\cdot\hat{b} = 3$

$\Rightarrow \hat{a}\cdot\hat{b} = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

4. (a) Let the angle between $\vec{A}$ and $\vec{B}$ is θ, then

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB}$$

$$= \frac{\left(2\hat{i}-2\hat{j}-\hat{k}\right)\cdot\left(\hat{i}+\hat{j}\right)}{\left|2\hat{i}+2\hat{j}-\hat{k}\right|\left|\hat{i}+\hat{j}\right|} = \frac{0}{3\sqrt{2}} = 0 \Rightarrow \theta = 90°$$

(b) Resultant vector

$\left(\vec{R}\right) = \vec{A}+\vec{B} = \left(2\hat{i}-2\hat{j}-\hat{k}\right)+\left(\hat{i}+\hat{j}\right) = 3\hat{i}-\hat{j}-\hat{k}$

$\therefore$ projection of resultant on x-axis = 3

5. Vector component of $\vec{a}$ in the direction of $\vec{b}$

$$= (a\cos\theta)\hat{b} = \left(\frac{ab\cos\theta}{b}\right)\frac{\vec{b}}{b} = \frac{\vec{a}.\vec{b}}{|\vec{b}|^2}\vec{b}$$

$$= \frac{(2\hat{i}+3\hat{j}).(\hat{i}+\hat{j})}{[\sqrt{1^2+1^2}]^2}(\hat{i}+\hat{j})$$

$$= \frac{2\times1+3\times1}{2}(\hat{i}+\hat{j}) = \frac{5}{2}(\hat{i}+\hat{j})$$

BEGINNER'S BOX-11

1. $\sqrt{3}\left|\vec{A}\times\vec{B}\right| = \vec{A}.\vec{B}$

$\sqrt{3}\,AB\sin\theta = AB\cos\theta$

$\tan\theta = 1/\sqrt{3}$

$\theta = 30°$

2. As $|\vec{A}\times\vec{B}|^2 = |\vec{A}|^2|\vec{B}|^2 - (\vec{A}.\vec{B})^2$

$\therefore 8^2 = 2^2\times5^2 - (\vec{A}.\vec{B})^2$

$\therefore (\vec{A}.\vec{B})^2 = 100 - 64 = 36$

$\therefore \vec{A}.\vec{B} = \pm 6$

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Basic Mathematics Used in Physics and Vectors

SOLUTIONS

BEGINNER'S BOX-1

1. (i) $\tan(-30°) = -\tan 30° = -\frac{1}{\sqrt{3}}$

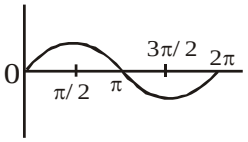
(ii) $\sin(120°)=\sin(180° - 60°)= \sin 60° = \frac{\sqrt{3}}{2}$

(iii) $\sin(135°)=\sin(90° + 45°) = \cos 45° = \frac{1}{\sqrt{2}}$

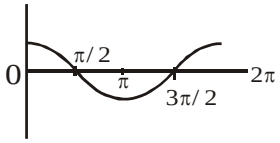
(iv) $\cos(150°) = \cos(180° - 30°) = -\cos(30°)$

$= -\frac{\sqrt{3}}{2}$

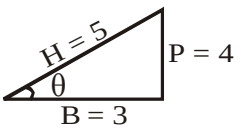
(v) $\sin(270°) = (-1)$ (from graph)



(vi) $\cos 270° = 0$



2. $\sec\theta = \frac{5}{3}$



we know that, $\sec\theta = \frac{H}{B} = \frac{5}{3}$

$\cos\theta = \frac{3}{5}$;

we know 3, 4, 5 are pythogorous triplets

$\therefore$ P = 4

i.e. H : B : P = 5 : 3 : 4

$\therefore$ $\sin\theta = \frac{4}{5}$ $\quad$ $\text{cosec}\,\theta = \frac{5}{4}$

$\cos\theta = \frac{3}{5}$ $\quad$ $\sec\theta = \frac{5}{3}$

$\tan\theta = \frac{4}{3}$ $\quad$ $\cot\theta = \frac{3}{4}$

BEGINNER'S BOX-2

1. Let P(8, –4) and Q(0, a) be two points, distance PQ is 10.

$\therefore$ According to distance formula

$PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(0-8)^2+(a+4)^2}$

$\Rightarrow PQ = \sqrt{64+(a^2+16+8a)}$

$\Rightarrow 10 = \sqrt{80+a^2+8a}$

$\Rightarrow a^2 + 8a + 80 = 100 \Rightarrow a^2 + 8a = 20$

$\Rightarrow a^2 + 10a - 2a - 20 = 0$

$\Rightarrow (a - 2)(a + 10) = 0 \Rightarrow (a = 2)$ and $(a = -10)$

$\therefore a = 2$

2. $P(x_1, y_1, z_1) = (0, -1, +1)$

and $Q(x_2, y_2, z_2) = (3, 3, 13)$

using distance formula

$PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

$= \sqrt{9+16+144} = \sqrt{169} = 13$

3. Slope $m = \frac{\Delta y}{\Delta x} = \frac{y_2-y_1}{x_2-x_1} = \tan\theta$

$m = \frac{\Delta y}{\Delta x} = \frac{(0-2)}{(3-0)} = \frac{-2}{3}$

BEGINNER'S BOX-3

1. (i) $y = x^{7/2}$

$\frac{dy}{dx} = \frac{d}{dx}(x^{7/2}) = \frac{7}{2}x^{\frac{7}{2}-1} = \frac{7}{2}x^{5/2}$

(ii) $y = x^{-3}$ $\quad\therefore\quad$ $\frac{dy}{dx} = -3x^{-4}$

(iii) $y = x$ $\quad\therefore\quad$ $\frac{dy}{dx} = 1$

(iv) $y = x^5 + x^3 + 4x^{1/2} + 7$

$\frac{dy}{dx} = 5x^4 + 3x^2 + 2x^{-1/2} + 0$

$= 5x^4 + 3x^2 + 2x^{-1/2}$

(v) $y = 5x^4 + 6x^{3/2} + 9x$

$\frac{dy}{dx} = 20x^3 + 9x^{3/2-1} + 9$

$= 20x^3 + 9x^{1/2} + 9$

(vi) $y = ax^2 + bx + c \Rightarrow \frac{dy}{dx} = 2ax + b$

(vii) $y = 3x^5 - 3x - \frac{1}{x}$

$\frac{dy}{dx} = 15x^4 - 3 + \frac{1}{x^2}$

2. $s = t^2 + 5t + 3 \Rightarrow \frac{ds}{dt} = 2t + 5$

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4. $A = 3t^2 + 7\ cm^2$

$\frac{dA}{dt} = \frac{d}{dt}(3t^2 + 7) = 6t$

at $t = 5$, $\frac{dA}{dt} = 30 cm^2/s$

5. $A = \pi r^2$

$\frac{dA}{dr} = \frac{d}{dr}(\pi r^2) = 2\pi r$

6. (i) $\frac{d}{dx}(x - 1)(2x + 5)$

$= (x - 1)\frac{d}{dx}(2x + 5) + (2x + 5)\frac{d}{dx}(x - 1)$

$= (x - 1)(2) + (2x + 5)(1) = 2x - 2 + 2x + 5$

$= 4x+3$

(ii) $\frac{d}{dx}\frac{1}{(2x+1)} = \frac{0-2}{(2x+1)^2} = \frac{-2}{(2x+1)^2}$

(iii) $\frac{d}{dx}\left(\frac{3x+4}{4x+5}\right) = \frac{(4x+5)(3)-(3x+4)(4)}{(4x+5)^2}$

$= \frac{-1}{(4x+5)^2}$

(iv) $\frac{d}{dx}\left(\frac{x^2}{x^3+1}\right) = \frac{(x^3+1)(2x)-x^2(3x^2)}{(x^3+1)^2}$

$= \frac{2x^4+2x-3x^4}{(x^3+1)^2} = \frac{2x-x^4}{(x^3+1)^2}$

BEGINNER'S BOX-4

1. (i) $\int x^{15}dx = \frac{x^{15+1}}{15+1} + C = \frac{x^{16}}{16} + C$

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$= \int\left(x + \frac{1}{x} + 2\right)dx = \frac{x^2}{2} + \log_e x + 2x + C$

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(vi) $\int\left(\frac{a}{x^2} + \frac{b}{x}\right)dx = \int(ax^{-2} + bx^{-1})dx$

$= \frac{ax^{-2+1}}{-2+1} + b\log_e x + C = -\frac{a}{x} + b\log_e x + C$

BEGINNER'S BOX-5

1. (i) $\int_R^\infty \frac{GMm}{x^2}dx = GMm\int_R^\infty x^{-2}dx$

$= GMm\left[\frac{x^{-1}}{-1}\right]_R^\infty = \frac{GMm}{R}$

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$= -kq_1q_2\left[\frac{x^{-1}}{-1}\right]_{r_1}^{r_2} = kq_1q_2\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$

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1. $1 + 2 + 3 + \ldots\ldots\ldots\ldots + 50$

$$s_n = \frac{n}{2}[2a + (n-1)d] = \frac{50}{2}[2 + (49)1]$$

$$= 25[51] = 1275$$

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$$s = \frac{n(n+1)(2n+1)}{6} = \frac{(10)(11)(21)}{6} = 385$$

3. $$s_\infty = \frac{a}{1-r} = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}$$

4. $$F_{net} = GMm\left[\frac{1}{r^2} + \frac{1}{2r^2} + \frac{1}{4r^2} + \ldots\ldots\infty\right]$$

$$= GMm\left[\frac{1/r^2}{1-\frac{1}{2}}\right] = \frac{2GMm}{r^2}$$

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1. Let the magnitudes of forces be A and B.

then according to question $A - B = 10$

$\Rightarrow A = 10 + B$ .....(1)

and $\sqrt{A^2+B^2} = 50$

$\Rightarrow A^2 + B^2 = 2500$ .....(2)

Solving equation number (1) & (2)

$A = 40N$ and $B = 30N$

Therefore the magnitudes of forces are 40N & 30N

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$$= \frac{(3\hat{i}+2\hat{j})-(2\hat{i}+3\hat{j}-\hat{k})}{|(3\hat{i}+2\hat{j})-(2\hat{i}+3\hat{j}-\hat{k})|} = \frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$$

3. $\because \quad |\hat{A}+\hat{B}| = \sqrt{2} \Rightarrow 2 \times 1 \times \cos\frac{\theta}{2} = \sqrt{2}$

$$\Rightarrow \cos\frac{\theta}{2} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\theta}{2} = 45° \Rightarrow \theta = 90°$$

$$|\hat{A}-\hat{B}| = 2 \times 1 \times \sin\frac{\theta}{2} = 2 \times \sin 45° = \sqrt{2}$$

BEGINNER'S BOX-9

1. $\because$ $A_x = 4$, $A_y = 6$ so $A_x + B_x = 10$ and $A_y + B_y = 9$

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(c) $\theta = \left(\frac{B_y}{B_x}\right)\tan^{-1} = \tan^{-1}\left(\frac{3}{6}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

2. Let α, β & γ are the angles of vector $(5\hat{i}+2\hat{j}+6\hat{k})$ from x, y & z-axis respectively.

then $$\cos\alpha = \frac{A_x}{A} = \frac{5}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{5}{\sqrt{65}},$$

$$\cos\beta = \frac{A_y}{A} = \frac{2}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{2}{\sqrt{65}}$$

$$\cos\gamma = \frac{A_z}{A} = \frac{6}{|5\hat{i}+2\hat{j}+6\hat{k}|} = \frac{6}{\sqrt{65}}$$

The sum of squares of directional cosines of this vector

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{5^2+2^2+6^2}{65} = \frac{65}{65} = 1$$

3. Resultant $(\vec{R})$ of

$\vec{A}$ & $\vec{B}$ $= \vec{A} + \vec{B} = (6\hat{i} - 6\hat{j} + 5\hat{k}) + (\hat{i} + 2\hat{j} - \hat{k})$

$= 7\hat{i} - 4\hat{j} + 4\hat{k}$

$\therefore$ unit vector parallel to resultant $(\hat{R})$

$$= \frac{\vec{R}}{|\vec{R}|} = \frac{7\hat{i}-4\hat{j}+4\hat{k}}{|7\hat{i}-4\hat{j}+4\hat{k}|} = \frac{7\hat{i}-4\hat{j}+4\hat{k}}{9}$$

BEGINNER'S BOX-10

1. $\because a = b = 1$ so $a^2 + b^2 + 2ab\cos\theta = (\sqrt{3})^2 = 3$

$\Rightarrow 1 + 1 + 2\cos\theta = 3 \Rightarrow \cos\theta = \frac{1}{2}$

$(\vec{a}-\vec{b}).(2\vec{a}+\vec{b}) = 2a^2 - b^2 + ab\cos\theta - 2ab\cos\theta$

$= 2 - 1 - \frac{1}{2} = \frac{1}{2}$

2. $\because$ $\vec{A}$ & $\vec{B}$ are perpendicular to each other

$\therefore \vec{A}.\vec{B} = 0$

$\Rightarrow \left(4\hat{i}-2\hat{j}+4\hat{k}\right).\left(-4\hat{i}+2\hat{j}+\alpha\hat{k}\right) = 0$

$\Rightarrow -16 - 4 + 4\alpha = 0 \Rightarrow \alpha = 5$

3. $\because (\hat{a}+2\hat{b})$ is perpendicular to $(5\hat{a}-4\hat{b})$

$\therefore (\hat{a}+2\hat{b})\cdot(5\hat{a}-4\hat{b}) = 0$

$\Rightarrow 5\hat{a}\cdot\hat{a} - 4\hat{a}.\hat{b} + 10\hat{b}.\hat{a} - 8\hat{b}.\hat{b} = 0$

$\Rightarrow 5 + 6\hat{a}\cdot\hat{b} - 8 = 0 \Rightarrow 6\hat{a}\cdot\hat{b} = 3$

$\Rightarrow \hat{a}\cdot\hat{b} = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

4. (a) Let the angle between $\vec{A}$ and $\vec{B}$ is θ, then

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB}$$

$$= \frac{\left(2\hat{i}-2\hat{j}-\hat{k}\right)\cdot\left(\hat{i}+\hat{j}\right)}{\left|2\hat{i}+2\hat{j}-\hat{k}\right|\left|\hat{i}+\hat{j}\right|} = \frac{0}{3\sqrt{2}} = 0 \Rightarrow \theta = 90°$$

(b) Resultant vector

$\left(\vec{R}\right) = \vec{A}+\vec{B} = \left(2\hat{i}-2\hat{j}-\hat{k}\right)+\left(\hat{i}+\hat{j}\right) = 3\hat{i}-\hat{j}-\hat{k}$

$\therefore$ projection of resultant on x-axis = 3

5. Vector component of $\vec{a}$ in the direction of $\vec{b}$

$$= (a\cos\theta)\hat{b} = \left(\frac{ab\cos\theta}{b}\right)\frac{\vec{b}}{b} = \frac{\vec{a}.\vec{b}}{|\vec{b}|^2}\vec{b}$$

$$= \frac{(2\hat{i}+3\hat{j}).(\hat{i}+\hat{j})}{[\sqrt{1^2+1^2}]^2}(\hat{i}+\hat{j})$$

$$= \frac{2\times1+3\times1}{2}(\hat{i}+\hat{j}) = \frac{5}{2}(\hat{i}+\hat{j})$$

BEGINNER'S BOX-11

1. $\sqrt{3}\left|\vec{A}\times\vec{B}\right| = \vec{A}.\vec{B}$

$\sqrt{3}\,AB\sin\theta = AB\cos\theta$

$\tan\theta = 1/\sqrt{3}$

$\theta = 30°$

2. As $|\vec{A}\times\vec{B}|^2 = |\vec{A}|^2|\vec{B}|^2 - (\vec{A}.\vec{B})^2$

$\therefore 8^2 = 2^2\times5^2 - (\vec{A}.\vec{B})^2$

$\therefore (\vec{A}.\vec{B})^2 = 100 - 64 = 36$

$\therefore \vec{A}.\vec{B} = \pm 6$